

**Topological Data Analysis of Wearable Sensor Signals:
Development and Evaluation of Recall-Oriented
Subject-General and Subject-Personalized Models**

M.Sc. THESIS

Kenan Evren BOYABATLI

Department of Mathematical Engineering

Mathematical Engineering Programme

Thesis Advisor: Prof. Dr. Atabey KAYGUN

JUNE 2026

**Topological Data Analysis of Wearable Sensor Signals:
Development and Evaluation of Recall-Oriented
Subject-General and Subject-Personalized Models**

M.Sc. THESIS

**Kenan Evren BOYABATLI
(509231205)**

Department of Mathematical Engineering

Mathematical Engineering Programme

Thesis Advisor: Prof. Dr. Atabey KAYGUN

JUNE 2026

**Giyilebilir Sensör Sinyalleriyle Topolojik Veri Analizi:
Düşme Vakalarını Kaçırmamayı Önceleyen Genel ve Kişisel
Modellerin Geliştirilmesi ve Değerlendirilmesi**

YÜKSEK LİSANS TEZİ

**Kenan Evren BOYABATLI
(509231205)**

Matematik Mühendisliği Anabilim Dalı

Matematik Mühendisliği Programı

Tez Danışmanı: Prof. Dr. Atabey KAYGUN

HAZİRAN 2026

Kenan Evren BOYABATLI, a M.Sc. student of ITU Graduate School student ID 509231205 successfully defended the thesis entitled “Topological Data Analysis of Wearable Sensor Signals: Development and Evaluation of Recall-Oriented Subject-General and Subject-Personalized Models”, which he/she prepared after fulfilling the requirements specified in the associated legislations, before the jury whose signatures are below.

Thesis Advisor : Prof. Dr. Atabey KAYGUN
İstanbul Teknik Üniversitesi

Jury Members : Prof. Dr. Placeholder One
Istanbul Technical University

Prof. Dr. Placeholder Two
Istanbul Technical University

Assoc. Prof. Dr. Placeholder Three
Istanbul Technical University

Date of Submission : June 2026
Date of Defense : June 2026

FOREWORD

I would like to express my sincere gratitude to my advisor, Prof. Dr. Atabey KAYGUN, for his guidance and support throughout this thesis.

Computing resources used in this work were provided by the National Center for High Performance Computing of Turkey (UHeM) under grant number 4025462026.

June 2026

Kenan Evren BOYABATLI

June 2026

Kenan Evren BOYABATLI

TABLE OF CONTENTS

FOREWORD	vii
TABLE OF CONTENTS	ix
ABBREVIATIONS	xv
LIST OF TABLES	xviii
LIST OF FIGURES	xxii
SUMMARY	xxiii
ÖZET	xxv
1. Introduction	1
1.1 Clinical and Societal Motivation	1
1.2 Limitations of Existing Approaches	2
Threshold-based detectors.	2
Statistical and spectral feature methods.	3
Deep learning approaches.	3
The personalization gap.	4
1.3 Topological Data Analysis for Wearable Signals	4
1.4 The Threshold Decision as a First-Class Component	6
1.5 Contributions	7
1.6 Thesis Organisation	8
Notation	9
2. Related Work	11
2.1 Threshold-Based Fall Detectors	11
2.1.1 Method family	11
2.1.2 Mathematical limitations	12
2.1.3 Comparison with the present work	12
2.2 Statistical Feature Methods	13
2.2.1 Method family	13
2.2.2 Mathematical limitations	13
2.2.3 Comparison with the present work	14
2.3 Deep Learning Approaches	15
2.3.1 Method family	15
2.3.2 Engineering and scientific limitations	16
2.3.3 Comparison with the present work	17
2.4 Topological Data Analysis for Time-Series Classification	17
2.4.1 Persistent homology in applied settings	17

2.4.2 TDA for activity recognition and wearable signals	18
2.5 Positioning of the Present Work.....	19
3. Methodology	21
3.1 Formal Problem Setting	22
3.1.1 Resampled discrete signal	22
3.1.2 Window operator	22
3.1.3 The hyperparameter tuple	23
3.1.4 Representation map and classifier.....	23
3.1.5 Performance functional	24
3.2 Signal Representation.....	24
3.2.1 Orientation invariance: the signal magnitude vector	25
3.2.2 Common sampling rate	25
Spectral sufficiency.	25
Integer divisibility.	26
Computational tractability.	26
3.2.3 Window extraction.....	26
3.2.4 Standard scaling.....	27
3.3 Phase-Space Reconstruction via Delay Embedding.....	27
3.3.1 Definition	27
3.3.2 The Takens embedding theorem	28
3.3.3 Geometric interpretation for fall detection.....	29
3.3.4 Filtration scale control	30
3.4 Topological Preliminaries	31
3.4.1 Topological spaces and homotopy	31
3.4.2 Abstract and geometric simplicial complexes.....	32
3.5 Simplicial Homology	32
3.5.1 Chains and the boundary operator	32
3.5.2 Homology groups and Betti numbers	33
3.5.3 Functoriality.....	34
3.6 Persistent Homology.....	35
3.6.1 Filtrations and complex families	35
Vietoris–Rips complex.	35
Alpha complex.	35
Sparse Rips complex.	36
3.6.2 Persistence modules	37
3.6.3 The structure theorem and persistence diagrams.....	37
3.6.4 Distances and stability	38
3.6.5 Computational reduction	40

3.7 Feature Vectorisation	41
3.7.1 Persistence-based statistics	41
3.7.2 Signal-level features	42
3.7.3 Complete feature vector	42
3.8 Classification Models	42
3.8.1 Logistic regression	42
3.8.2 Support vector machine	43
3.8.3 Platt score calibration	44
3.8.4 Class-weighted empirical risk	44
3.9 Bayesian Hyperparameter Optimisation	45
3.9.1 The optimisation problem	45
3.9.2 Sequential model-based optimisation	46
3.9.3 Tree-structured Parzen estimator	46
3.9.4 Comparison with alternative search strategies	47
Grid search.	47
Random search.	48
Gaussian-process Bayesian optimisation.	48
TPE (adopted strategy).	48
3.9.5 Search space	49
3.9.6 Outer LOSO and inner cross-validation	49
3.10 Evaluation Framework	51
3.10.1 Performance metrics	51
3.10.2 Decision threshold	52
3.10.3 Evaluation protocols	52
Naive protocol.	52
General protocol (leave-one-subject-out).	52
Personal protocol.	53
Decision-threshold sweep.	53
3.11 Statistical Inference	53
3.11.1 Repetition-level confidence intervals	54
3.11.2 Subject-level paired tests	54
3.11.3 Bootstrap confidence intervals	55
3.11.4 Permutation tests	55
3.11.5 Distributional analysis of personalisation	56
3.11.6 Cross-applied hyperparameters	56
3.11.7 Methodological correction over headline maxima	57
3.12 Computational Infrastructure	57
4. Experiments	59

4.1 Datasets.....	60
4.1.1 MobiFall	60
4.1.2 SisFall	61
4.1.3 FallAllD (FAD_40Hz).....	62
Original FallAllD dataset.	62
Selection of the 40 Hz derived variant.	62
4.2 Data Processing and Preprocessing.....	63
4.3 Feature Description	65
4.3.1 Persistence-Based Features.....	65
4.3.2 Signal-Level Features	66
4.4 Bayesian Hyperparameter Optimisation.....	67
4.5 Classifiers and Training Setup.....	69
4.6 Results.....	70
4.6.1 Optimal Hyperparameter Configurations	70
4.6.2 MobiFall Results	70
4.6.3 SisFall Results	71
4.6.4 FAD_40Hz Results	72
4.6.5 Cross-Dataset Performance Summary.....	73
4.6.6 Effect of Decision Threshold Optimisation	73
Computational cost of threshold personalisation. . .	74
Statistical analysis of threshold heterogeneity. . . .	74
5. Results and Discussion	77
5.1 H1 — Representation Effectiveness.....	77
5.1.1 F_2 outcomes	77
5.1.2 Naive vs. General protocol gap	79
5.1.3 Mathematical basis	79
5.1.4 Limitation: absence of baseline comparison	79
5.2 H2 — Dataset-Specific Topological Geometry	80
5.2.1 Distinct optimal configurations	80
5.2.2 Interpretation of configuration differences.....	80
Window length.	80
Complex type.	81
Distance measure.	81
5.2.3 Sensitivity of F_2 to λ	81
5.3 H3 — Representation Dominates Classifier Choice.....	82
5.3.1 Classifier gap	82
5.3.2 λ -tuning variation	83
5.3.3 Mathematical interpretation	83

5.4 H4 — Heterogeneous Threshold Personalisation	84
5.4.1 Per-subject sweep gain distribution	84
Distribution shape.	84
Near-zero gainers.	85
Heterogeneity metric.	85
Mathematical explanation.	86
Engineering contribution.	86
5.4.2 Personal training protocol.....	86
5.5 Ablation Study: Feature-Block Contributions.....	88
5.5.1 Within-topology contribution: H0 dominates H1.....	88
5.5.2 Topology versus signal: a dataset-dependent division of labour	89
5.5.3 Classifier invariance.....	90
5.5.4 Scope of the ablation.....	90
5.6 Cross-Dataset Synthesis	90
5.6.1 Performance ranking and subject pool.....	90
5.6.2 Classifier preference.....	91
5.6.3 Computational profile	91
5.7 Deployment Pathway.....	92
5.7.1 Sensor context	92
5.7.2 CPU-only operation	92
5.7.3 Prospective threshold personalisation	93
5.7.4 Sensor placement and orientation robustness.....	94
5.8 Limitations	94
1. No baseline comparison.	94
2. Cross-application drops not computed.	94
3. No ablation study.	95
4. Fixed sensor placement.	95
5. Retrospective threshold sweep.	95
5.9 Future Work	95
5.9.1 Online threshold adaptation.....	95
5.9.2 Embedded real-time benchmark.....	96
5.9.3 Multi-placement sensor fusion.....	96
5.9.4 Cross-dataset transfer of λ^*	96
5.9.5 Extension to other wearable time series	96
5.9.6 One-class fall detection from ADL-only supervision	97
5.9.7 Few-shot personalisation beyond threshold.....	99
6. Conclusion	101
6.1 Mathematical Perspective	101

Representation effectiveness (H1).	101
Dataset-specific topological geometry (H2).	102
Representation dominates classifier choice (H3). . .	102
6.2 Statistical Perspective	103
Threshold personalisation as a distributional phe- nomenon (H4).	103
Evaluation integrity under LOSO.	103
6.3 Engineering Perspective	104
Inference-time personalisation without retraining. .	104
CPU-only deployment profile.	105
Failure mode analysis and minimum calibration set size.	105
Path to consumer deployment.	106
6.4 Limitations and Open Questions.....	107
6.5 Closing Remarks.....	107
REFERENCES	109
CURRICULUM VITAE	113

ABBREVIATIONS

ADL	: Activities of Daily Living
CI	: Confidence Interval
CNN	: Convolutional Neural Network
CPU	: Central Processing Unit
CV	: Cross-Validation
EI	: Expected Improvement
FAD	: FallAllD 40 Hz Derived Variant
FN	: False Negative
FP	: False Positive
GP	: Gaussian Process
GPU	: Graphics Processing Unit
GUDHI	: Geometry Understanding in Higher Dimensions
HAR	: Human Activity Recognition
HPC	: High Performance Computing
KKT	: Karush–Kuhn–Tucker
LOSO	: Leave-One-Subject-Out
LR	: Logistic Regression
LSTM	: Long Short-Term Memory
PD	: Persistence Diagram
PH	: Persistent Homology
RBF	: Radial Basis Function
SMBO	: Sequential Model-Based Optimisation
SMV	: Signal Magnitude Vector
SVM	: Support Vector Machine
TDA	: Topological Data Analysis
TN	: True Negative
TP	: True Positive
TPE	: Tree-structured Parzen Estimator

LIST OF TABLES

2.1	Structural comparison of fall detection and TDA-for-wearables approaches. ✓ = property satisfied; × = property absent; ~ = partially satisfied.....	20
4.1	Summary of the three datasets as used in the experiments.....	60
4.2	Ten persistence-based statistics computed per homological dimension. Each row is computed twice: once for H_0 (features 1–10) and once for H_1 (features 11–20). Persistence of a topological feature is defined as its death scale minus its birth scale.	66
4.3	Four signal-level features derived from the accelerometer magnitude $SMV(t)$ within each window.	66
4.4	Parameters and candidate values explored during Bayesian optimisation. A dash (—) indicates parameters not applicable for the Alpha complex type.	68
4.5	Best configuration selected by Bayesian optimisation for each dataset (200 iterations per dataset). A dash (—) indicates a parameter not applicable for the Alpha complex.....	71
4.6	MobiFall: mean performance across subjects under the three evaluation protocols. General and Personal entries are mean $\pm 95\%$ CI over $n = 15$ repetitions per subject; Naive entries are means over 10 repetitions.	71
4.7	SisFall: mean performance across subjects. Format as in Table 4.6.	72
4.8	FAD_40Hz: mean performance across subjects. Format as in Table 4.6.	72
4.9	Subject-level mean F_2 under the General (LOSO) protocol. Values are mean $\pm 95\%$ CI.	73
4.10	Average subject-level F_2 gain from threshold optimisation. $\Delta F_2(s) = F_2(\tau^*(s)) - F_2(\hat{\tau}_0(s))$ where $\hat{\tau}_0(s)$ is the PR-curve trained threshold for subject s ; values averaged over held-out subjects.....	73

5.1	General-protocol F_2 outcomes against H1 targets. CI denotes 95 % repetition-level confidence interval ($R = 15$).	78
5.2	Optimal TDA configurations λ_K^* per dataset (from Experiments Tab. 4.5). A dash (—) indicates parameters not applicable for the Alpha complex.	80
5.3	Classifier gap vs. λ -tuning variation. The trial range is $F_2^{\max} - F_2^{\min}$ across all 200 Optuna trials. The ratio is range/gap.	83
5.4	Per-subject threshold sweep gain $\Delta F_2(s)$ statistics. <i>Near-zero</i> : $ \Delta F_2 \leq 0.001$. All gains are non-negative by construction (τ^* is the sweep argmax).	84
5.5	Personal vs. General protocol aggregate F_2 . $\Delta = F_2^{\text{pers}} - F_2^{\text{gen}}$ (negative = General is better).	87
5.6	Feature-block ablation under the General (LOSO) protocol. Entries are cross-subject mean $F_2 \pm 95\%$ CI. Block dimensions: Full = 24, H0 = 10, H1 = 10, H0+H1 = 20, Signal = 4. The Full column reproduces the headline results of Tables 4.6–4.8. ..	89
5.7	Wall-clock cost per dataset stage. All computations are CPU-only.	92

LIST OF FIGURES

3.1	Left: synthetic ADL (walking) and fall accelerometer windows. Right: the corresponding two-dimensional delay embeddings ($m = 2, \tau = 3$). The ADL trajectory forms a compact, roughly periodic arc; the fall trajectory produces a sparser, elongated point cloud due to the large-amplitude impact spike. This geometric difference motivates the use of persistent homology to distinguish the two signal classes.....	28
3.2	Rips $(X, 2\varepsilon)$ filtration of a six-point cloud at four scales. At $\varepsilon = 0$ only vertices are present ($\beta_0 = 6, \beta_1 = 0$). As ε increases, edges appear and connected components merge (β_0 decreases). A loop (H_1 class) is born when the polygon is complete and dies when triangles fill it. Features far from the diagonal in the persistence diagram have long lifetime and are noise-robust (Cohen–Steiner stability, Theorem 3.7).	36
3.3	Persistence diagrams (H_0 circles, H_1 squares) for delay-embedded ADL and fall windows. Points far from the diagonal represent long-lived topological features. The fall window typically exhibits a high-persistence H_1 feature (loop) absent or short-lived in the ADL window—a geometric signature of the sharp impact trajectory in the delay-embedded space. Triangles and diamonds denote essential classes (infinite death, plotted at the top margin).	40
4.1	Complete fall detection pipeline from raw inertial data to binary decision. <i>Preprocessing</i> : SMV computation, resampling to 50 Hz, window extraction. <i>Feature extraction</i> (dashed, λ -dependent): delay embedding, filtration construction, persistent homology (H_0, H_1), and vectorisation to $\phi_\lambda \in \mathbb{R}^{24}$. <i>Classification</i> : standardisation, probabilistic classifier, and threshold comparison $\hat{p}(w) \geq \tau_0$ for binary decision. The <i>Bayesian optimisation</i> box (left, blue) selects λ^* by maximising cross-validated F_2 over training subjects via 200 TPE iterations; row-to-row connectors run through column 2 (SMV $\rightarrow$ Filtration $\rightarrow$ Classifier), preserving the 4×3 grid symmetry.	59

4.2	Preprocessing pipeline (Steps 1–3). Detailed description of each step is given in Section 4.2.....	65
4.3	Feature extraction pipeline (Steps 1–4). Detailed description of each step is given in Section 4.3.	67
4.4	Bayesian hyperparameter optimisation loop (TPE, $T = 200$ iterations per dataset). Each trial evaluates both an LR proxy and a linear SVM proxy; the objective is the mean of their F_2 scores, yielding a classifier-agnostic λ^* . Detailed description in Section 4.4.	69
5.1	Mean F_2 by dataset, classifier (LR/SVM), and evaluation protocol (Naive, General, Personal). Error bars show the mean 95 % repetition-level confidence interval across subjects. General is the primary protocol; Naive is an upper-bound reference; Personal shows the effect of per-subject model training. MobiFall Personal collapses due to the small subject pool ($ \mathcal{S} = 9$).....	78
5.2	Bayesian optimisation convergence: best-so-far averaged F_2 proxy versus Optuna trial number (200 trials per dataset, TPE sampler). FAD_40Hz achieves the highest proxy (0.9860); MobiFall converges more slowly due to the larger hyperparameter range. The wide trial range on MobiFall (0.179) and FAD (0.108) relative to SisFall (0.054) is direct evidence that λ sensitivity differs across datasets (H2).	82
5.3	Per-subject threshold sweep gain $\Delta F_2(s) = F_2(\tau^*(s)) - F_2(\hat{\tau}_0(s))$ for LR (left) and SVM (right). Jittered points show individual subjects; box shows quartiles. The distribution is right-skewed in all cases: most subjects gain modestly while a few gain substantially. FAD_40Hz subject 11 achieves the largest gain (+0.1165, LR), visible as the highest point in the FAD panel. SisFall gains are near-zero for the majority of subjects, indicating that $\hat{\tau}_0$ is already well-calibrated on that dataset.	85
5.4	General (LOSO) vs Personal (per-subject training) F_2 per subject, for LR (left) and SVM (right). Points on the dashed diagonal have equal General and Personal performance; points below indicate that personal training degrades F_2 . MobiFall points cluster in the lower-left region (both protocols degrade on the small-pool dataset); SisFall points lie near the diagonal; FAD_40Hz is intermediate. The shaded region marks Personal < General.	87

**Topological Data Analysis of Wearable Sensor Signals:
Development and Evaluation of Recall-Oriented
Subject-General and Subject-Personalized Models**

SUMMARY

Falls are a leading cause of injury, loss of independence, and mortality among the elderly, and wearable inertial sensors offer a practical means of automatic fall detection. Because a missed fall is far more costly than a false alarm, the detection problem is inherently recall-oriented. This thesis investigates whether persistent homology, applied to short wearable accelerometer windows through a delay-embedding pipeline, can support recall-oriented fall detection that is at once rigorous in its evaluation, principled in its feature construction, and lightweight enough for edge deployment.

The proposed pipeline transforms each three-axis recording into an orientation-independent signal magnitude vector, reconstructs its phase space by delay embedding in the sense of Takens, and summarises the resulting point cloud with persistent homology in dimensions zero and one. Ten statistics of each persistence diagram, together with four signal-level descriptors, form a compact twenty-four-dimensional feature vector that is passed to deliberately simple classifiers, namely logistic regression and a support vector machine. The hyperparameters of the topological representation are selected by Bayesian optimisation with a tree-structured Parzen estimator. The method is evaluated on three public benchmark datasets—MobiFall, SisFall, and FAD_40Hz—under a subject-aware framework comprising naïve, leave-one-subject-out, and subject-personalised protocols, complemented by a decision-threshold sweep.

Under the leave-one-subject-out protocol the representation attains a mean F_2 score of up to 0.97 on SisFall, 0.96 on FAD_40Hz, and 0.90 on MobiFall, with recall above 0.95 throughout. The difference between the two classifiers is consistently below 0.02, whereas tuning the topological hyperparameters changes F_2 by an order of magnitude more, demonstrating that performance is governed by the representation rather than the classifier. The optimal topological configuration differs across datasets, reflecting genuine differences in fall dynamics and point-cloud geometry. Post-hoc threshold personalisation, which requires no model retraining, yields a strongly right-skewed distribution of gains: most subjects benefit only marginally while a small minority gain more than 0.10 in F_2 . The Cohen–Steiner stability theorem provides a formal noise-robustness certificate that distinguishes the topological representation from competing feature families.

Keywords: fall detection, wearable sensors, topological data analysis, persistent homology, delay embedding, recall-oriented classification.

**Giyilebilir Sensör Sinyalleriyle Topolojik Veri Analizi:
Düşme Vakalarını Kaçırmamayı Önceleyen Genel ve Kişisel
Modellerin Geliştirilmesi ve Değerlendirilmesi**

ÖZET

Düşmeler; yaşlı bireylerde yaralanmanın, bağımsızlık kaybının ve ölümün başlıca nedenlerinden biridir. Dünya genelinde yaşlı nüfusun hızla artması, düşme olaylarının zamanında ve güvenilir biçimde tespit edilmesini bir kamu sağlığı önceliği hâline getirmektedir. Giyilebilir ataletsel sensörler (ivmeölçer ve jiroskop), bireyin günlük yaşamını kısıtlamadan sürekli izleme imkânı sunduğundan, otomatik düşme tespiti için pratik bir donanım altyapısı sağlar. Bu problemde tespit edilemeyen bir düşmenin (yanlış negatif) maliyeti, yanlış bir alarmin (yanlış pozitif) maliyetinden çok daha yüksektir; çünkü kaçırılan bir düşme, yerde uzun süre yardımsız kalma ve buna bağlı ciddi komplikasyonlar anlamına gelebilir. Dolayısıyla düşme tespiti doğası gereği geri çağırma (recall) odaklı bir sınıflandırma problemidir.

Bu tez, gecikme gömme (delay embedding) tabanlı bir işlem hattı aracılığıyla kısa giyilebilir ivmeölçer pencerelerine uygulanan kalıcı homolojinin (persistent homology); değerlendirmesi titiz, öznitelik inşası ilkeli ve uç (edge) cihazlarda çalışabilecek kadar hafif bir geri-çağırma odaklı düşme tespitini destekleyip desteklemediğini araştırmaktadır. Çalışmanın merkezindeki iddia, başarının büyük ölçüde sınıflandırıcı seçiminden değil, sinyali topolojik bir öznitelik uzayına taşıyan temsil haritasından kaynaklandığıdır.

Yöntem. Önerilen işlem hattı, her üç-eksenli kaydı önce yönelimden bağımsız bir sinyal büyüklük vektörüne (SMV) dönüştürür. Tüm kayıtlar ortak bir örnekleme frekansı olan 50 Hz'e yeniden örneklenir ve sabit uzunlukta pencerelere bölünür. Her pencerenin faz uzayı, Takens'in gömme teoremi anlamında gecikme gömmeyle yeniden inşa edilir; böylece tek boyutlu skaler sinyalden çok boyutlu bir nokta bulutu elde edilir. Bu nokta bulutu üzerine bir filtrasyon kurulur ve sıfırıncı ile birinci boyutta kalıcı homoloji hesaplanır. Elde edilen kalıcılık diyagramlarının (persistence diagram) her biri on istatistikle özetlenir; bu istatistikler dört sinyal düzeyi tanımlayıcısıyla birleşerek yirmi dört boyutlu, kompakt bir öznitelik vektörü (ϕ_λ) oluşturur. Bu vektör, kasıtlı olarak basit tutulan iki sınıflandırıcıya verilir: lojistik regresyon (LR) ve destek vektör makinesi (SVM). Her ikisi de sınıf dengesizliğini gözetmek üzere dengeli sınıf ağırlıklarıyla, SVM ayrıca Platt olasılık kalibrasyonu ile eğitilir.

Topolojik temsilin davranışı; pencere uzunluğu, gömme boyutu, gecikme, filtrasyon türü (Vietoris–Rips, Alpha, SparseRips), uzaklık metriği ve eşik gibi hiperparametrelerden oluşan bir demete (λ) bağlıdır. Bu yüksek boyutlu ve koşullu arama uzayı, ızgara aramasını hesaplama açısından olanaksız kıldığından,

hiperparametreler ağaç yapılı Parzen kestiricisi (TPE) ile yürütülen Bayesçi optimizasyon kullanılarak seçilir. Her veri kümesi için ayrı bir arama yürütülür ve sınıflandırıcıdan bağımsız bir amaç fonksiyonu ile en iyi yapılandırma belirlenir.

Veri Kümeleri ve Değerlendirme. Yöntem, üç açık karşılaştırma veri kümesi üzerinde değerlendirilmiştir: MobiFall, SisFall ve FallAllD veri kümesinin 40 Hz türetilmiş çeşidi olan FAD_40Hz. Bu veri kümeleri; sensör konumu, örnekleme frekansı, denek profili ve düşme türleri bakımından farklılaşarak yöntemin genellenebilirliğini sınamaya olanak tanır. Değerlendirme, denek-duyarlı bir çerçevede üç protokolle yapılır: rastgele bölünmeye dayanan ve üst sınır referansı sağlayan naif protokol; her deneğin tamamen dışarıda bırakıldığı, birincil protokol olan denek-dışı-bırakma (leave-one-subject-out, LOSO) protokolü; ve her deneğin yalnızca kendi verisiyle modellendiği kişisel protokol. Bu protokolleri, karar eşiğinin geniş bir aralıkta taranmasına dayanan bir eşik analizi tamamlar. Birincil başarımlar ölçütü, geri çağırma keskinliğine göre dört kat ağırlıklandırılan F_2 skorudur; denekler arası dağılım, t -dağılımına dayalı yüzde 95 güven aralıklarıyla raporlanır. Tüm ölçekleme ve eşik parametreleri yalnızca eğitim katmanından kestirilerek bilgi sızıntısı önlenir.

Bulgular. LOSO protokolü altında temsil; SisFall'da 0.97'ye, FAD_40Hz'de 0.96'ya ve MobiFall'da 0.90'a varan ortalama F_2 skoru elde etmiş, geri çağırma değeri ise tüm veri kümesi-sınıflandırıcı bileşimlerinde 0.95'in üzerinde kalmıştır. Bu sonuçlar, temsilin önceden belirlenmiş başarımlar hedeflerini en az bir sınıflandırıcı için karşıladığını göstermektedir. Bu etkinliğin kuramsal dayanağı Cohen-Steiner kararlılık teoremidir; bu teorem, iki kalıcılık diyagramı arasındaki darboğaz (bottleneck) uzaklığının, girdi sinyallerindeki bozulmanın supremum normuyla sınırlı olduğunu garanti eder ve dolayısıyla elle tasarlanmış istatistiksel, spektral veya derin öğrenme tabanlı öznel ailelerinde benzeri bulunmayan biçimsel bir gürültü-dayanıklılığı sertifikası sağlar.

Üç veri kümesi yapısal olarak birbirinden farklı en iyi yapılandırmalara yakınsamıştır: MobiFall geniş bir pencerede SparseRips kompleksini, SisFall kısa bir pencerede Alpha kompleksini, FAD_40Hz ise kısa bir pencerede standart Rips kompleksini seçmiştir. Bu durum, evrensel bir yapılandırmanın var olmadığını ve en iyi topolojik geometrinin düşme dinamiklerindeki ve nokta-bulut yapılarındaki gerçek farkları yansıttığını ortaya koyar. Ayrıca iki sınıflandırıcı arasındaki F_2 farkı tüm veri kümelerinde 0.02'nin altında kalırken, topolojik hiperparametrelerin ayarlanması F_2 'yi büyüklük olarak çok daha geniş bir aralıkta değiştirmektedir. Bu gözlem, başarımın sınıflandırıcı sınıfından çok temsil haritası tarafından belirlendiği temel iddiasını güçlü biçimde desteklemektedir.

Son olarak, model yeniden eğitimi gerektirmeyen sonradan (post-hoc) eşik kişiselleştirmesinin etkisi incelenmiştir. Denek düzeyindeki kazanım dağılımı, tüm veri kümesi-sınıflandırıcı bileşimlerinde sağa çarpıktır: deneklerin çoğunluğu yalnızca marjinal fayda sağlarken küçük bir azınlık F_2 'de 0.10'dan büyük kazanımlar elde etmektedir. Bu heterojenlik, tek bir sabit eşik kaymasının tüm

denekleri eşit biçimde iyileştireceği varsayımını çürütmekte ve en uygun eşğin denek-özgü olduğunu göstermektedir.

Sonuç. Tezin başlıca katkıları şunlardır: (i) giyilebilir sinyallerden geri-çağırma odaklı düşme tespiti için kuramsal olarak güvenceli, hesaplama açısından hafif bir topolojik temsil haritası; (ii) bu temsili üç ayrı veri kümesinde tutarlı biçimde optimize eden Bayesçi arama yordamı; (iii) naif, LOSO ve kişisel protokolleri bir araya getiren denek-duyarlı bir değerlendirme çerçevesi; ve (iv) model yeniden eğitimi gerektirmeyen, çıkarım zamanında uygulanabilen ve denek-özgü heterojenliği ortaya koyan bir eşik kişiselleştirme analizi. Bulgular bütün olarak, topolojik temsilin basit sınıflandırıcılarla dahi güçlü ve gürültüye dayanıklı düşme tespiti sağlayabileceğini; ancak kişiselleştirmenin homojen bir iyileştirme sunmadığını ve denek bazında ele alınması gerektiğini göstermektedir.

Anahtar Kelimeler: düşme tespiti, giyilebilir sensörler, topolojik veri analizi, kalıcı homoloji, gecikme gömme, geri-çağırma odaklı sınıflandırma.

1. Introduction

1.1 Clinical and Societal Motivation

Falls are the second leading cause of accidental injury death worldwide, after road traffic injuries (World Health Organization, 2021). Globally, approximately 684,000 fatal falls occur each year, with adults over the age of 60 accounting for the overwhelming majority (World Health Organization, 2021). An estimated 37.3 million falls annually are severe enough to require medical attention, representing a sustained and growing burden on healthcare systems. As the global population ages—the World Health Organization projects that the number of people aged 60 and above will reach 2.1 billion by 2050 (World Health Organization, 2022)—the scale of this burden is set to increase substantially without effective countermeasures.

The medical consequences of a fall extend well beyond the acute injury. A fall that goes undetected initiates a “long lie”: a period during which the fallen person remains on the ground, unable to summon help, and is exposed to secondary complications including dehydration, hypothermia, pressure sores, and aspiration pneumonia. Even when the acute injury is minor, a long lie lasting more than an hour doubles the probability of subsequent institutionalisation and is independently associated with increased one-year mortality (Wild, Nayak, & Isaacs, 1981). Beyond the physical consequences, the *fear of falling*—a well-documented post-fall psychological syndrome (Tinetti, Speechley, & Ginter, 1988)—leads to self-imposed activity restriction that accelerates deconditioning and, paradoxically, further elevates fall risk. Timely detection is therefore not merely about comfort; it is a clinical intervention with measurable outcomes.

Wearable inertial sensors offer a continuous and unobtrusive monitoring modality that is uniquely positioned to close this detection gap. A single accelerometer

attached to the waist or carried in a trouser pocket records the three-axis acceleration of the body’s centre of mass throughout the day. Unlike camera-based or floor-sensor systems, it imposes no infrastructure requirement and preserves the user’s privacy and mobility. Modern smartphones and smartwatches already contain the necessary hardware and are carried by a large fraction of the target population, making passive fall detection a feature that can be deployed at scale at marginal hardware cost.

The contrast between the two kinds of error is the premise that shapes the rest of this thesis. A missed fall can trigger the long-lie complications described above, whereas a false alarm costs at most a brief, unnecessary check; the two outcomes are therefore far from equally serious. For this reason the present work pursues a *recall-oriented* objective and scores every model with the F_2 measure, which weights recall four times as heavily as precision. In effect the detector is deliberately tuned to give precedence to the fall event—tolerating the occasional false alarm rather than risk overlooking a genuine fall—which is the design stance reflected in the title of this thesis. The formal definition of F_2 and the reasoning behind the four-to-one weighting are developed in the Methodology.

1.2 Limitations of Existing Approaches

Despite the practical appeal of wearable fall detection, the path from sensor signal to reliable deployment is obstructed by a set of well-known technical difficulties. A brief taxonomy of existing approaches clarifies where the gaps lie.

Threshold-based detectors. The earliest and most widely deployed family of methods compares the signal magnitude vector $\|\mathbf{a}(t)\|_2$ against a fixed threshold T : a fall is declared whenever $\|\mathbf{a}(t)\|_2 > T$. These methods are computationally trivial and require no training data. Their fundamental limitation is the absence of a principled calibration mechanism. The threshold T is set empirically, the output is binary with no associated uncertainty, and there is no statistical framework for adjusting the operating point to the false-alarm tolerance of a particular deployment environment. In practice, fixed-threshold

detectors achieve high recall on controlled datasets but at false-alarm rates that are clinically unacceptable for continuous wear: vigorous activities of daily living (running, stair descent, jumping) frequently exceed the same magnitude threshold as a fall impact.

Statistical and spectral feature methods. A substantial literature constructs handcrafted feature vectors from short accelerometer windows—means, variances, spectral energies, signal-magnitude area, zero-crossing rates—and trains standard classifiers on these features (de la Hoz Franco, Ariza-Colpas, Quero, & Espinilla, 2014; Igual, Medrano, & Plaza, 2013). These methods improve over threshold-based approaches by producing calibrated probability outputs and by exploiting the multi-dimensional structure of the signal beyond a single scalar statistic. However, their features are *ad hoc*: selected by domain intuition rather than derived from a principled geometric theory of the signal. There is no formal stability guarantee—small perturbations of the input signal can cause arbitrary changes in the feature vector if the perturbation happens to cross a feature threshold. As a result, the transferability of trained models across sensors, sampling rates, and subject populations is empirically fragile.

Deep learning approaches. Convolutional and recurrent neural networks have achieved competitive performance on fall detection benchmarks by learning end-to-end feature representations from raw inertial data (Batchuluun, Lee, Nguyen, Pham, Arsalan, & Park, 2022; Wang, Ellul, & Azzopardi, 2020). The gains over handcrafted features are real, but the engineering trade-offs are severe. Training requires large labelled datasets, GPU compute, and hyperparameter searches whose cost scales with model complexity. The resulting models are opaque: the learned features are not interpretable, and there is no geometric or mathematical explanation for why a particular signal window is classified as a fall. Deployment on edge hardware (smartwatches, embedded sensors) is complicated by memory footprint and energy consumption. Most critically for the personalization question, deep models are trained once on a fixed

corpus and deployed without adaptation—the “trains once, deploys forever” paradigm—which systematically ignores the substantial variation in motion dynamics across individuals.

The personalization gap. Across all three families, a common structural weakness emerges: the decision rule is calibrated at the population level and applied uniformly to every individual. This is statistically equivalent to assuming that the optimal decision boundary is the same for all subjects. In reality, individuals differ in gait dynamics, body geometry, sensor placement geometry, and fall style, so the optimal threshold for a classifier applied to subject s will in general differ from the population-level optimum. Recognising this, some recent works report *personalized* evaluation results, but typically through isolated case studies rather than a systematic statistical characterisation. The question of whether subject-level adaptation is uniformly beneficial, or only selectively so, remains open.

1.3 Topological Data Analysis for Wearable Signals

Persistent homology, the central tool of topological data analysis (TDA) (Carlsson, 2009; Edelsbrunner & Harer, 2010), provides an alternative geometric framework for characterising the shape of a signal window. Rather than extracting fixed statistics from a time series, it constructs a sequence of simplicial complexes—the *filtration*—that tracks how the topology of the point cloud derived from the signal evolves across spatial scales. The result is a *persistence diagram*: a multiset of (b, d) pairs recording the scales at which topological features (connected components, loops, voids) are born and die. This diagram summarises the intrinsic geometry of the signal window without committing to a fixed set of scalar statistics.

Two properties of persistent homology are especially relevant to the fall-detection problem.

Provable stability. The Cohen–Steiner stability theorem (Cohen–Steiner, Edelsbrunner, & Harer, 2007) states that the bottleneck distance between two

persistence diagrams is bounded above by the supremum norm of the perturbation between the corresponding input functions:

$$d_B(D(f), D(g)) \leq \|f - g\|_\infty.$$

Applied to inertial windows, this guarantees that small sensor noise perturbations produce at most equally small perturbations of the topological descriptor. This stability is a *mathematical theorem*, not an empirical observation, and it holds independently of the training dataset and subject population. No existing handcrafted feature family possesses a comparable guarantee.

Fixed-dimensional vectorization. The persistence diagram is converted into a fixed 24-dimensional feature vector $\phi_\lambda(w) \in \mathbb{R}^{24}$ by computing summary statistics of the birth-death pairs in homological degrees $k = 0$ and $k = 1$. The dimension 24 is independent of the window length, sampling rate, or number of simplices in the filtration. This is in contrast to spectral or autoregressive features whose dimension grows with the signal length, and to deep-network embeddings whose dimension is a design parameter affecting model size.

The *delay embedding* (Takens, 1981) is the geometric pre-processing step that connects the scalar signal to the point cloud: the orientation-invariant signal magnitude vector $\text{SMV}(t) = \|\mathbf{a}(t)\|_2$ is lifted to a m -dimensional point cloud by taking overlapping m -tuples of lagged values. A fall window and an ADL window produce point clouds of qualitatively different topology: the sharp impulsive excursion of a fall impact generates a spread-out cluster with strong H_0 structure, whereas ADL windows exhibit richer H_1 structure (loops) from the periodic motion of walking or stair climbing. Persistent homology captures this topological distinction systematically.

TDA has been applied to time-series classification problems across several domains (Gidea & Katz, 2018; Pereira & de Mello, 2015; Umeda, 2017), and specifically to activity recognition from inertial sensors (Ott, Tomforde, & Willis, 2015; Xu, Yao, Li, & Zhang, 2019). The distinguishing features of the present work are its recall-oriented objective, its systematic dataset-specific

hyperparameter optimisation, and its statistical treatment of subject-level personalisation as a distributional question rather than a case study.

1.4 The Threshold Decision as a First-Class Component

Wearable fall detection is an instance of binary classification under class imbalance: falls are rare relative to activities of daily living, so the positive class is significantly underrepresented in any realistic deployment. Under such imbalance, the choice of decision threshold τ in the rule $\hat{y}(w) = \mathbf{1}\{f_\theta(\phi_\lambda(w)) \geq \tau\}$ is not a cosmetic detail but a critical design parameter that directly controls the precision-recall trade-off.

As established in Section 1.1, the thesis prioritises *recall* through the recall-weighted F_2 score. Under class imbalance the decision threshold τ is the concrete lever that realises this preference: lowering τ raises recall at the expense of precision, so the operating point must be chosen deliberately rather than left at the default 0.5.

Beyond the population-level threshold, the thesis investigates *subject-level threshold personalisation*. The key observation is that the posterior probability function $f_\theta \circ \phi_{\lambda^*}$ may be systematically miscalibrated differently for different subjects: a subject whose fall windows produce lower-than-average posterior scores needs a lower threshold to achieve high recall, while a subject whose ADL windows produce elevated scores needs a higher threshold to control false alarms. A sweep of τ over the grid $\mathcal{T} = \{0.05, 0.10, \dots, 0.95\}$ identifies the per-subject optimal threshold $\tau^*(s)$ retrospectively.

The computational cost of this sweep is $O(|\mathcal{T}| \cdot |W_s|)$ comparisons, where $|\mathcal{T}| = 19$ and $|W_s|$ is the window count for subject s . This amounts to less than one second of wall time for all subjects in any of the three datasets, against a Bayesian representation search that consumes approximately 75 minutes per dataset. The ratio exceeds 10^3 : threshold personalisation is a computationally negligible add-on to an already-trained model. It requires no model retraining, no access to the original training data, and no transmission of data to a server—it is executable

on-device at inference time from a small per-subject calibration set. This combination of properties—near-zero cost, privacy preservation, and measurable F_2 gain—makes the threshold decision a first-class engineering contribution of the thesis, not merely an evaluation detail.

1.5 Contributions

The thesis makes the following five contributions.

Contribution 1: A mathematically grounded TDA pipeline for wearable fall detection. We construct a feature map $\phi_\lambda: \mathbb{R}^{n \times 3} \rightarrow \mathbb{R}^{24}$ that transforms a raw three-axis accelerometer window into a fixed 24-dimensional topological descriptor via delay embedding, simplicial filtration, persistent homology computation, and vectorization. The map is parameterised by a hyperparameter tuple $\lambda = (w_{\text{sec}}, m, \tau, r, c, d_{\text{met}}, \varepsilon, q)$ whose components control the geometric construction. The Cohen–Steiner stability theorem provides a formal noise-robustness certificate for the resulting features.

Contribution 2: A classifier-agnostic Bayesian hyperparameter optimisation framework. A single Optuna/TPE study (Akiba, Sano, Yanase, Ohta, & Koyama, 2019) with a classifier-averaged F_2 objective identifies λ_K^* for each dataset K in 200 trials. The averaged objective ensures that the recovered representation supports both logistic regression and support vector machines without being over-fitted to either. We show that this optimisation identifies dataset-specific configurations (H2) and that the variation in F_2 across trials—up to 0.18 units on MobiFall—far exceeds the variation attributable to classifier choice (H3).

Contribution 3: A three-protocol subject-aware evaluation framework. Three evaluation protocols are reported jointly for each dataset and classifier: the Naive (random split, upper bound), the General (leave-one-subject-out, primary), and the Personal (per-subject split, idealised personalisation reference). The LOSO protocol with $R = 15$ independent repetitions provides repetition-level

confidence intervals via a t -statistic, and the joint reporting of all three protocols makes the train–test leakage implicit in naive evaluation visible quantitatively.

Contribution 4: A statistical characterisation of threshold heterogeneity as a distributional phenomenon. The subject-level sweep gain $\Delta F_2(s) = F_2(\tau^*(s), s) - F_2(\hat{\tau}_0(s), s)$ is analysed as a distribution rather than a scalar average. We show that the distribution is right-skewed and dataset-dependent: MobiFall exhibits a mean gain of ≈ 0.04 – 0.05 with a coefficient of variation exceeding 50%, while SisFall exhibits a mean gain near zero for most subjects. This distributional analysis confirms Hypothesis 4 and replaces the anecdotal treatment of personalisation common in prior work.

Contribution 5: A deployment pathway to consumer devices. The pipeline is CPU-only, requires no GPU, and executes persistent homology at the window sizes considered in tens of milliseconds on a desktop-class CPU core—a profile compatible with ARM Cortex-A smartphone processors and modern smartwatch SoCs. The threshold sweep personalisation runs in under one second on device. We describe a prospective on-device calibration procedure that requires only a short wearing period and no server-side computation, and identify one-class anomaly detection from ADL data as the natural extension that eliminates the cold-start problem entirely.

1.6 Thesis Organisation

The remainder of the thesis is organised as follows.

Chapter 2 (Related Work) surveys the three families of prior wearable fall detection methods—threshold-based, statistical-feature, and deep-learning—and positions the TDA approach with respect to each. It also reviews prior applications of persistent homology to time-series classification.

Chapter 3 (Methodology) provides the mathematical foundations. It introduces the signal-magnitude-vector transformation and delay embedding, develops the theory of simplicial complexes, persistent homology, and the Cohen–Steiner stability theorem, describes the feature vectorization and the

Bayesian optimisation procedure, and defines the three evaluation protocols and the statistical inference framework used throughout the thesis.

Chapter 4 (Experiments) describes the three benchmark datasets, the experimental setup, and the numerical results: dataset-specific optimal configurations, full performance tables for all classifier-protocol combinations, and the threshold sweep analysis.

Chapter 5 (Results and Discussion) interprets the experimental outcomes against the four research hypotheses, presents the per-subject threshold gain distributions, synthesises findings across datasets, discusses the deployment pathway, and identifies limitations and directions for future work.

Chapter 6 (Conclusion) summarises the contributions at three levels: mathematical, statistical, and engineering.

Notation

Throughout the thesis, $\mathbb{R}$ denotes the real line, $\mathbb{R}^d$ the d -dimensional Euclidean space, and $\mathbf{1}\{\cdot\}$ the indicator function. Scalars are denoted by lowercase letters, vectors by bold lowercase, and sets by calligraphic uppercase. The symbol $\|\cdot\|_2$ denotes the Euclidean norm and $\|\cdot\|_\infty$ the supremum norm. All probabilistic statements are with respect to the randomness of the class-balancing seed unless stated otherwise.

2. Related Work

Wearable fall detection has been approached from three main directions: rule-based threshold detectors, classical machine-learning pipelines built on handcrafted features, and end-to-end deep-learning models. Each direction has produced systems that achieve strong scores on controlled benchmark datasets, yet each carries structural limitations that become apparent when the goal shifts from benchmark performance to robust, personalised deployment. A fourth line of research applies persistent homology to time-series classification in domains outside of fall detection. This section surveys each strand, identifies the specific limitations that motivate the present work, and closes with an explicit positioning statement.

2.1 Threshold-Based Fall Detectors

2.1.1 Method family

The oldest and most widely deployed family of wearable fall detectors operates directly on the signal magnitude vector $\text{SMV}(t) = \|\mathbf{a}(t)\|_2$ and raises an alarm whenever $\text{SMV}(t)$ exceeds a fixed scalar threshold T :

$$\hat{y}(t) = \mathbf{1}\{\text{SMV}(t) > T\}.$$

Kangas et al. (Kangas, Konttila, Lindgren, Winblad, & Jämsä, 2008) established the archetypal version of this approach on a waist-mounted accelerometer and reported high sensitivity on laboratory falls. Subsequent work introduced two-stage detectors that combine an impact threshold with a post-impact stillness check: a fall is declared only if a large SMV spike is followed by a prolonged low-activity period, filtering out brief vigorous activities that produce high-magnitude excursions without a subsequent motionless phase (Igual, Medrano, & Plaza, 2013). Multi-axis variants replace the scalar magnitude

with component-specific thresholds or combine vertical, horizontal, and resultant acceleration thresholds in a logical rule (de la Hoz Franco, Ariza-Colpas, Quero, & Espinilla, 2014).

2.1.2 Mathematical limitations

The threshold decision rule $\mathbf{1}\{\text{SMV}(t) > T\}$ has three fundamental limitations that are mathematical rather than incidental.

No probabilistic output. The rule produces a hard binary decision with no associated posterior probability. It is therefore impossible to apply a principled recall-precision trade-off: there is no continuous operating parameter that smoothly exchanges false negatives for false positives. Adjusting sensitivity requires changing the hard threshold T , but this adjustment is global—it shifts the operating point identically for all individuals and all activities.

No principled threshold selection. The threshold T is chosen by a combination of empirical cross-validation and clinical convention. There is no statistical framework for selecting T that is optimal under a specified loss function. In particular, the F_2 -optimal threshold—which depends on the class-imbalance ratio in a given deployment population—cannot be identified from first principles without a probabilistic model.

No subject-specific adaptation. A single threshold T is applied to all individuals. An active person with a vigorous gait produces baseline SMV values that approach the threshold during normal activity, generating chronic false alarms. A frail elderly person with a low-energy fall may not reach the threshold at all. The threshold detector has no mechanism to adjust to individual motion dynamics because it has no subject-level probabilistic model.

2.1.3 Comparison with the present work

The present thesis replaces the scalar hard decision with a calibrated posterior probability $f_\theta(\phi_\lambda(w)) \in [0, 1]$. The decision threshold τ is then a continuous parameter that can be set to maximize F_2 on a per-subject basis. The

threshold sweep over $\mathcal{T} = \{0.05, 0.10, \dots, 0.95\}$ identifies the subject-optimal $\tau^*(s)$ retrospectively, and the gains—up to +0.12 units of F_2 on individual subjects—demonstrate that subject-specific calibration provides value that is structurally inaccessible to a global threshold detector.

2.2 Statistical Feature Methods

2.2.1 Method family

The dominant paradigm in wearable fall detection from the mid-2000s through the mid-2010s combined handcrafted feature extraction with a standard supervised classifier. A window w of length L is mapped to a fixed-dimensional vector $\phi_{\text{hand}}(w) \in \mathbb{R}^d$ by computing scalar statistics of the raw signal or its frequency-domain representation. Typical time-domain features include the window mean, variance, range, peak value, root mean square, and signal-magnitude area; typical frequency-domain features include spectral energies in low-frequency bands, peak frequency, and spectral entropy (de la Hoz Franco, Ariza-Colpas, Quero, & Espinilla, 2014; Igual, Medrano, & Plaza, 2013). These features are fed to a classifier—most commonly support vector machines, k -nearest neighbours, or decision trees—trained on labeled windows from a fixed dataset.

Surveys by Igual et al. (Igual, Medrano, & Plaza, 2013) and Delahoz and Labrador (de la Hoz Franco, Ariza-Colpas, Quero, & Espinilla, 2014) catalogue dozens of systems built on this template and report accuracy figures in the range 90%–99% on controlled datasets. These figures are broadly consistent across methods, suggesting that the feature choice matters less than the evaluation protocol: studies reporting the highest scores tend to use random train-test splits ignoring subject membership, which inflates performance by allowing the model to memorize individual quirks rather than generalize across persons.

2.2.2 Mathematical limitations

Absence of a stability certificate. The relationship between the input perturbation $\|w - w'\|$ and the feature perturbation $\|\phi_{\text{hand}}(w) - \phi_{\text{hand}}(w')\|$ is not bounded in any useful way for handcrafted statistics. A small noise perturbation of a signal window can produce an arbitrarily large change in a feature that includes a zero-crossing count, an extremum detection, or a threshold-based spectral band energy. This instability is not a problem when training and test data come from the same sensor and subject population, but it becomes significant under sensor substitution, mounting-position variation, or cross-dataset evaluation.

No geometric grounding. The feature set is selected by domain intuition and validated empirically. There is no underlying theory that explains why the chosen statistics should separate falls from activities of daily living, nor any guarantee that the same statistics remain discriminative when the population, sensor, or activity set changes. In the language of learning theory, the inductive bias of the feature extractor is implicit and uncontrolled.

Fixed feature dimension with no adaptation. The dimension d is fixed by the feature recipe and cannot be adjusted to the dataset geometry. Different datasets exhibit different fall dynamics—controlled laboratory falls versus naturalistic community falls, elderly populations versus young adults—but the feature extractor is invariant to these differences. The hyperparameter of this approach is the classifier, not the feature extractor itself.

Subject-general evaluation conflates variability sources. Most papers in this family report a single aggregate accuracy figure without distinguishing within-subject repeatability from across-subject generalization. As a consequence, it is unclear whether the reported scores reflect a model that generalizes to new individuals or merely one that memorizes the subjects present in the training fold.

2.2.3 Comparison with the present work

The feature map ϕ_λ of the present thesis differs from handcrafted features in three respects.

First, the Cohen–Steiner stability theorem provides a formal Lipschitz bound: small noise perturbations of the input window produce at most equally small perturbations of the persistence diagram, and hence of ϕ_λ . This guarantee is a mathematical theorem, not an empirical observation, and it holds for any signal, any subject, and any sensor.

Second, the feature extractor is parameterized by $\lambda \in \Lambda$ and the optimal λ_K^* is identified by Bayesian search over a nontrivial search space. Different datasets converge to different configurations (SparseRips at 5 s for MobiFall; Alpha at 1 s for SisFall; Rips at 1 s for FAD_40Hz), demonstrating that dataset-specific geometry is captured by the parameterisation. No handcrafted feature family has an analogous adaptation mechanism.

Third, the evaluation strictly separates subject-level generalization from within-subject repeatability by using a leave-one-subject-out protocol with $R = 15$ repetitions, reporting both the mean and the 95% confidence interval for each combination of dataset, classifier, and protocol.

2.3 Deep Learning Approaches

2.3.1 Method family

Convolutional neural networks (CNNs), recurrent networks (LSTMs and GRUs), and more recently transformer-based architectures have been applied to wearable fall detection with the aim of learning discriminative representations directly from raw or minimally processed inertial data. Wang et al. (Wang, Ellul, & Azzopardi, 2020) demonstrated that a multi-scale CNN applied to three-axis accelerometer windows achieves competitive sensitivity and specificity on MobiFall and SisFall without manual feature engineering. Batchuluun et al. (Batchuluun, Lee, Nguyen, Pham, Arsalan, & Park, 2022) provide a systematic review of deep-learning fall detection systems published between 2018 and 2022 and identify CNN-LSTM

hybrids and attention-based models as the current state of the art on standard benchmarks, with reported F_1 scores above 0.95 on several controlled datasets.

The apparent performance advantage of deep models over classical methods is partly attributable to their ability to capture temporal structure across multiple scales simultaneously, and partly to methodological differences: deep learning papers tend to use larger datasets and more favorable evaluation protocols than their classical counterparts, making direct comparison difficult.

2.3.2 Engineering and scientific limitations

GPU dependency. Training a competitive deep network on inertial windows requires GPU compute for gradient-based optimisation, which is practical in a research setting but problematic for deployment on the battery-constrained processors of wearable devices. Inference may be feasible on a smartphone CPU with quantized models, but training—and therefore retraining for personalisation—is not.

Opacity. The features learned by a deep network’s intermediate layers are not interpretable in terms of the physical signal. There is no explanation for why a particular window is classified as a fall that a clinician or system designer can evaluate independently of the training data. TDA features, by contrast, have a direct geometric interpretation: H_0 statistics measure the spread of the delay-embedded cloud (cluster count and birth-death lifetime distribution), and H_1 statistics measure the presence of loops (periodic or oscillatory structure).

No formal stability guarantee. Adversarial-robustness research has repeatedly demonstrated that small, human-imperceptible perturbations of the input can reverse the output of a deep classifier. For a safety-critical fall-detection system, the absence of a stability certificate is a principled concern: a small sensor hardware change, a firmware update affecting the analog front-end, or a change in mounting position can degrade performance in ways that are unpredictable without re-evaluation. The Cohen–Steiner bound of TDA provides a certificate that does not depend on the training distribution.

The “trains once, deploys forever” paradigm. Deep models are trained on a fixed corpus and deployed without adaptation. Subject-specific personalisation requires either fine-tuning—which demands labeled falls from the target individual—or meta-learning frameworks that introduce significant additional complexity and still require some labeled falls at deployment time. In the present thesis, subject-level adaptation is achieved by a threshold sweep that requires no labeled falls, no model parameters to update, and no access to the training corpus.

Computational cost at scale. The pipeline introduced in this thesis consumed approximately 75 minutes of CPU time for the Bayesian representation search on a 40-core server. This cost is paid once per dataset, not per user. Training a competitive deep model requires comparable or greater wall-clock time and additionally requires GPU hardware. For deployments targeting resource-limited clinical or community settings, the CPU-only profile of the TDA pipeline is a meaningful engineering advantage.

2.3.3 Comparison with the present work

The present thesis does not claim to match or exceed state-of-the-art deep learning scores on fall detection benchmarks. The purpose is different: to establish whether a mathematically well-founded, computationally lightweight, and interpretable pipeline can achieve strong F_2 scores under a rigorous subject-aware evaluation. The answer is affirmative for two of the three datasets ($F_2 \geq 0.93$ on SisFall and FAD_40Hz) and satisfactory for the third ($F_2 \geq 0.88$ on MobiFall). The TDA pipeline achieves these results on a CPU without GPU access, with formally certified feature stability, and with an interpretable feature space—properties that deep models do not offer jointly.

2.4 Topological Data Analysis for Time-Series Classification

2.4.1 Persistent homology in applied settings

The use of persistent homology as a feature extractor for time-series classification has grown substantially since the early 2010s, following the development of computationally efficient boundary matrix reduction algorithms (Edelsbrunner, Letscher, & Zomorodian, 2002) and their implementation in open-source libraries such as GUDHI (GUDHI Editorial Board, 2015) and Ripser.

Pereira and de Mello (Pereira & de Mello, 2015) applied persistence diagrams to time-series classification in several benchmark domains and showed that topological features are competitive with or superior to classical statistical features when the discriminative signal is encoded in the *shape* of short segments rather than their amplitude or frequency content. Umeda (Umeda, 2017) extended this line to multivariate time series and demonstrated that the combination of H_0 and H_1 features provides complementary information not captured by either alone.

Gidea and Katz (Gidea & Katz, 2018) applied TDA to financial time series, showing that topological features of sliding-window point clouds detect market instability with a lead time not accessible to standard spectral methods. Their work established the delay-embedding framework—building a point cloud from lagged copies of a scalar signal and computing its persistent homology—as a broadly applicable TDA pipeline. The pipeline introduced in the present thesis follows this framework but extends it with a systematic search over the embedding and filtration parameters rather than fixing them a priori.

2.4.2 TDA for activity recognition and wearable signals

Ott and Demir (Ott, Tomforde, & Willis, 2015) applied TDA features to human activity recognition (HAR) from smartphone accelerometer data. Using a fixed sliding window and pre-selected embedding parameters, they showed that persistence statistics achieve accuracy comparable to handcrafted features on a standard HAR benchmark. Xu et al. (Xu, Yao, Li, & Zhang, 2019) similarly used TDA to distinguish activity classes in a multiclass HAR setting, reporting improvements over baseline statistical features on several benchmark datasets.

These works establish the viability of persistent homology for wearable inertial classification, but differ from the present thesis in several ways that limit their direct applicability to the fall detection problem.

Objective. Both Ott et al. and Xu et al. optimise accuracy in a balanced multiclass setting. Fall detection is a binary problem with severe class imbalance, and the appropriate objective is recall-oriented (F_2) rather than accuracy. The distinction matters: a classifier that maximizes accuracy on an imbalanced dataset can do so by predicting the majority class (non-fall) for almost all windows, achieving high accuracy at the cost of near-zero recall.

Evaluation protocol. The HAR studies cited above use random train-test splits that do not control for subject membership, which inflates performance estimates for any method by allowing individual-specific patterns to leak from training to test. The present thesis strictly enforces subject separation via leave-one-subject-out cross-validation.

Hyperparameter treatment. Prior TDA-for-wearables works fix the embedding dimension m , delay τ , and filtration type c either by prior convention or by a coarse manual grid. The present thesis treats the full hyperparameter tuple $\lambda = (w_{\text{sec}}, m, \tau, r, c, d_{\text{met}}, \varepsilon, q)$ as the primary object of study and identifies the optimal tuple per dataset via 200 iterations of Bayesian search. The finding that the optimal complex type differs across all three datasets (SparseRips, Alpha, Rips) is a direct consequence of this systematic search; it could not have been discovered by fixing the complex type to a convention.

Personalisation. Neither Ott et al. nor Xu et al. report subject-level personalisation results or analyse the distribution of per-subject gains. The present thesis treats the distribution of the sweep gain $\Delta F_2(s)$ as a scientific object in its own right, showing that it is right-skewed and dataset-dependent and that its coefficient of variation exceeds 50% in all cases.

2.5 Positioning of the Present Work

Table 2.1 summarises the structural differences between the four approaches surveyed above and the present thesis across six dimensions.

Table 2.1: Structural comparison of fall detection and TDA-for-wearables approaches. ✓ = property satisfied; × = property absent; ∼ = partially satisfied.

Property	Threshold	Stat. features	Deep learning	TDA-HAR	This work
Probabilistic output	×	✓	✓	✓	✓
Formal stability certificate	×	×	×	✓	✓
Dataset-specific tuning	×	×	∼	×	✓
Subject-aware evaluation	×	∼	∼	×	✓
Recall-oriented objective	×	∼	∼	×	✓
CPU-only deployment	✓	✓	×	✓	✓
Subject-level personalisation	×	×	×	×	✓

The table makes the novelty of the present work concrete. No prior approach combines a formal stability certificate (Theorem), dataset-specific parameterisation (via Bayesian search), subject-aware evaluation (LOSO with repetitions), a recall-oriented objective (F_2), and subject-level personalisation (threshold sweep) within a single unified pipeline. The closest prior work—TDA for HAR—satisfies the stability and CPU constraints but lacks all three evaluation properties.

The gap that this thesis directly addresses is the intersection of the last four rows: a principled, recall-oriented, subject-aware, personalised fall detector that carries a mathematical stability guarantee and is deployable on consumer hardware without GPU acceleration.

3. Methodology

The pipeline of this thesis is the composition of three mathematical maps. From a raw labelled inertial recording it first produces an orientation-invariant scalar window, then a finite point cloud in $\mathbb{R}^m$ obtained by a delay embedding, then a fixed-length vector $\phi_\lambda(w) \in \mathbb{R}^{24}$ obtained by persistent homology and a small set of signal-level statistics. The vector is fed to a probabilistic classifier $f_\theta: \mathbb{R}^{24} \rightarrow [0, 1]$ whose threshold determines a binary decision. The methodology specifies each of these maps, the algebraic and geometric machinery they rely on, the hyperparameter selection rule that fixes the open parameters, and the evaluation protocols under which subject-aware performance is reported. The treatment is deliberately mathematical: the central object of study is the representation map ϕ_λ , not the engineering of any particular sensor pipeline, and the role of the methodology section is to make that map and its surrounding inferential framework precise.

Section 3.1 fixes the formal problem setting and the principal notational conventions. Section 3.2 develops signal preprocessing. Section 3.3 constructs the delay embedding and states the embedding theorems of Takens and of Sauer–Yorke–Casdagli on which the geometric interpretation rests. Section 3.4 recalls the necessary point-set and combinatorial topology. Section 3.5 develops simplicial homology rigorously, including the fundamental identity $\partial^2 = 0$, the Euler–Poincaré formula, and the functoriality of $H_k(\cdot; \mathbb{F})$. Section 3.6 develops persistent homology as a functor on filtrations, including the structure theorem and the algebraic stability theorem. Section 3.7 specifies the vectorisation. Section 3.8 treats the two classifier families and the Platt calibration. Section 3.9 develops the Bayesian hyperparameter search and the tree-structured Parzen estimator. Section 3.10 fixes the evaluation protocols. Section 3.11 fixes

the statistical-inference machinery. Section 3.12 records the computational environment.

3.1 Formal Problem Setting

Let $K \in \{\text{MobiFall}, \text{SisFall}, \text{FAD_40Hz}\}$ index one of the three benchmark datasets. Each dataset consists of labelled inertial recordings together with subject identifiers, drawn from an unknown joint distribution

$$(\mathcal{X}_K, \mathcal{Y}_K, \mathcal{S}_K) \sim \mathbb{P}_K,$$

where $\mathcal{X}_K$ is the recording-valued random element, $\mathcal{Y}_K \in \{0, 1\}$ the binary fall label, and $\mathcal{S}_K$ the subject index, taking values in a finite set of cardinality $|\mathcal{S}_K|$. The marginal class prior $\pi_K := \mathbb{P}_K(\mathcal{Y}_K = 1)$ is strictly between 0 and 1/2 for every K , reflecting the empirical fact that falls are minority events.

3.1.1 Resampled discrete signal

After resampling to the target frequency $f_s = 50$ Hz (justified in Section 3.2.2), a single recording is the discrete signal

$$\mathbf{x}: \{0, 1, \dots, T\} \longrightarrow \mathbb{R}^p, \quad p = 2,$$

where the two coordinates are the accelerometer and gyroscope magnitudes, defined in Section 3.2.1. A binary label $y \in \{0, 1\}$ indicates whether the recording contains a fall. A subject index $s \in \mathcal{S}_K$ records the individual to whom the signal belongs. The subject-aware nature of the data set is essential: independence of training and test samples is asserted at the level of subjects, not at the level of individual windows.

3.1.2 Window operator

For a window length $L \in \mathbb{N}$ and a starting index $t_0 \in \{0, \dots, T - L\}$, the *window operator* extracts a contiguous segment:

$$W_{t_0, L}: \mathbb{R}^{(T+1) \times p} \longrightarrow \mathbb{R}^{L \times p}, \quad W_{t_0, L}(\mathbf{x}) = (\mathbf{x}_{t_0}, \mathbf{x}_{t_0+1}, \dots, \mathbf{x}_{t_0+L-1}). \quad (3.1)$$

The window length is determined by a hyperparameter $w_{\text{sec}} > 0$ measuring window duration in seconds:

$$L := \lfloor w_{\text{sec}} \cdot f_s \rfloor.$$

3.1.3 The hyperparameter tuple

The topological feature extractor depends on a hyperparameter tuple

$$\lambda = (w_{\text{sec}}, m, \tau, r, c, d_{\text{met}}, \varepsilon, q) \in \Lambda, \quad (3.2)$$

where w_{sec} is the window length in seconds, $m \geq 2$ the embedding dimension, $\tau \geq 1$ the embedding lag, $r \geq 1$ the stride factor, $c \in \{\text{Alpha}, \text{Rips}, \text{SparseRips}\}$ the filtered complex family, d_{met} a metric on $\mathbb{R}^m$, $\varepsilon \in (0, 1)$ a filtration-scale percentile, and q a subsampling rule used by the SparseRips construction. The search space Λ is a finite Cartesian product, specified in Section 3.9.5.

3.1.4 Representation map and classifier

Given a window w , the pipeline constructs a finite point cloud

$$P_\lambda(w) \subset \mathbb{R}^m$$

by the delay embedding of Definition 3.2 below, builds the filtered simplicial complex $\mathcal{K}_\lambda(P_\lambda(w))$ specified by c , computes the persistent homology in degrees $k \in \{0, 1\}$, and vectorises the resulting diagrams together with a small set of signal-level features into a fixed feature vector

$$\phi_\lambda(w) \in \mathbb{R}^{24}. \quad (3.3)$$

The dimension is fixed at 24 regardless of λ so that distinct configurations are compared under a uniform classifier interface.

A probabilistic classifier is a map

$$f_\theta: \mathbb{R}^{24} \longrightarrow [0, 1],$$

where $f_\theta(\phi_\lambda(w))$ is interpreted as the posterior probability $\widehat{\mathbb{P}}(\mathcal{Y} = 1 \mid w; \theta, \lambda)$.

The model class is restricted to the two families

$$\mathcal{F}_{\text{r}} = \{\text{balanced logistic regression models with } \ell_1 \text{ or } \ell_2 \text{ penalty}\},$$

$\mathcal{F}_{\text{svm}} = \{\text{balanced } C\text{-SVM models with Platt-calibrated probabilities}\}.$

This restriction is methodological. The aim is to test the predictive value of the representation ϕ_λ in the linear regime, not to maximise score by increasing classifier complexity.

For a threshold $\tau_0 \in (0, 1)$ the *subject-general decision rule* is

$$\hat{y}(w) = \mathbf{1}\{f_\theta(\phi_\lambda(w)) \geq \tau_0\}. \quad (3.4)$$

The default value of τ_0 in primary evaluations is not the nominal 0.5 but the PR-curve optimal threshold $\hat{\tau}_0$ identified on the training fold; its precise construction is described in §3.10.2.

For subject-personalised evaluation a subject-specific threshold τ_s is used:

$$\hat{y}_s(w) = \mathbf{1}\{f_\theta(\phi_\lambda(w)) \geq \tau_s\}. \quad (3.5)$$

The personalised threshold τ_s must be estimated from data disjoint from the segment of subject s on which the metric is finally reported; otherwise the protocol is contaminated by leakage.

3.1.5 Performance functional

The primary metric is the F_β -score evaluated at $\beta = 2$,

$$F_2(\hat{y}, y) = 5 \frac{\text{Prec} \cdot \text{Rec}}{4 \text{Prec} + \text{Rec}}, \quad (3.6)$$

with Prec, Rec computed on the test sample. The training and selection protocol prioritises recall over precision, which is the appropriate objective for fall detection when false-negative cost exceeds false-positive cost; the asymmetry is quantified in Section 3.10.1.

3.2 Signal Representation

3.2.1 Orientation invariance: the signal magnitude vector

A three-axis inertial sensor outputs a vector-valued signal $\mathbf{x}(t) = (x_1(t), x_2(t), x_3(t))^\top \in \mathbb{R}^3$. The numerical values of the axis components depend on the orientation of the device frame relative to the body frame, an $\text{SO}(3)$ -rotation that varies across subjects and even across sessions for a fixed subject. An axis-wise representation therefore conflates motion content with nuisance orientation.

Definition 3.1 (Signal magnitude vector). *The signal magnitude vector of a three-axis signal $\mathbf{x}: \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}^3$ is the scalar process*

$$\text{SMV}(t) = \|\mathbf{x}(t)\|_2 = \sqrt{x_1(t)^2 + x_2(t)^2 + x_3(t)^2}. \quad (3.7)$$

Proposition 3.1 ($\text{SO}(3)$ -invariance). *For every rotation $\mathbf{R} \in \text{SO}(3)$ and every $t \geq 0$, $\|\mathbf{R}\mathbf{x}(t)\|_2 = \|\mathbf{x}(t)\|_2$. Hence the map $\mathbf{x} \mapsto \text{SMV}$ is invariant under reorientation of the sensor frame.*

Proof. $\|\mathbf{R}\mathbf{x}\|_2^2 = \mathbf{x}^\top \mathbf{R}^\top \mathbf{R} \mathbf{x} = \mathbf{x}^\top \mathbf{x} = \|\mathbf{x}\|_2^2$, using $\mathbf{R}^\top \mathbf{R} = \mathbf{I}$. □

The accelerometer and the gyroscope are processed independently, producing two scalar time series SMV_{acc} and SMV_{gyr} per recording. All subsequent analysis is carried out on these scalar processes; the loss of directional information is the cost of orientation invariance.

3.2.2 Common sampling rate

The three benchmark datasets were originally recorded at different rates: 87.5 Hz (MobiFall), 200 Hz (SisFall), and 40 Hz (FAD_40Hz). All signals are resampled to the common rate $f_s = 50$ Hz before any further processing. The choice of 50 Hz rests on three independent arguments.

Spectral sufficiency. The fall-discriminative content of inertial wearable signals is concentrated below 20 Hz. The pre-impact acceleration peak and the impact transient are dominated by frequencies in the 0–15 Hz band; energy above 20 Hz consists predominantly of high-frequency sensor noise and mechanical

vibration (Kangas, Konttila, Lindgren, Winblad, & Jämsä, 2008; Saleh, Abbas, & Le Bouquin Jeannès, 2021; Sucerquia, López, & Vargas-Bonilla, 2017).

Theorem 3.1 (Nyquist–Shannon (Shannon, 1948)). *A real-valued continuous-time signal whose Fourier transform is supported in $[-B, B]$ is exactly recovered from its samples taken at any rate $f_s \geq 2B$.*

With $B = 20$ Hz, the rate $f_s = 50$ Hz exceeds $2B = 40$ Hz by a margin of 25%. The corresponding Nyquist frequency $f_s/2 = 25$ Hz dominates the relevant signal bandwidth, so resampling does not erase any fall-relevant spectral information.

Integer divisibility. For SisFall the conversion ratio $200:50 = 4:1$ is an integer, so resampling reduces to decimation by a factor of four. For FAD_40Hz the ratio is $40:50 = 4:5$ and for MobiFall the ratio is $87.5:50 = 7:4$; in both cases the rational ratio admits a finite-impulse-response polyphase resampler at moderate cost.

Computational tractability. The persistent homology computation dominates the cost of the pipeline; on the Vietoris–Rips complex the worst-case complexity is super-polynomial in the size of the point cloud (Section 3.6.5). A five-second window contains $L = 250$ samples at 50 Hz versus $L = 1000$ samples at 200 Hz; the resulting reduction in computational load is what makes a 200-iteration Bayesian search across three datasets feasible.

3.2.3 Window extraction

The classification unit is a fixed-length window of $L = \lfloor w_{\text{sec}} \cdot f_s \rfloor$ samples. Fall and non-fall trials are windowed by different rules to reflect the structural difference between their respective signal regimes.

For a fall trial $\mathbf{x}$, a single window is extracted around the index of the global SMV maximum:

$$t^* = \arg \max_t \text{SMV}_{\text{acc}}(t), \quad w^* = W_{t^* - \lfloor L/2 \rfloor, L}(\mathbf{x}). \quad (3.8)$$

The centring strategy guarantees that the impact transient is contained in the window regardless of the recording length.

For an ADL trial, consecutive non-overlapping windows of length L are extracted from the start of the recording with stride L :

$$w_j = W_{jL, L}(\mathbf{x}), \quad j = 0, 1, \dots, \lfloor T/L \rfloor - 1. \quad (3.9)$$

This produces multiple windows per ADL trial, which is appropriate because daily activities are quasi-stationary and substantially longer than fall events.

3.2.4 Standard scaling

After the feature vector $\phi_\lambda(w) \in \mathbb{R}^{24}$ has been computed for every window, each of its coordinates is centred and rescaled. Let $\mathcal{Z} = \{\phi_\lambda(w_i)\}_{i \in \mathcal{I}_{\text{tr}}} \subset \mathbb{R}^{24}$ be the collection of feature vectors over the training partition. The empirical mean and variance per coordinate are

$$\mu_j = \frac{1}{|\mathcal{I}_{\text{tr}}|} \sum_{i \in \mathcal{I}_{\text{tr}}} \phi_\lambda(w_i)_j, \quad \sigma_j^2 = \frac{1}{|\mathcal{I}_{\text{tr}}|} \sum_{i \in \mathcal{I}_{\text{tr}}} (\phi_\lambda(w_i)_j - \mu_j)^2,$$

and the standardised feature vector is $z_j = (\phi_\lambda(w)_j - \mu_j)/\sigma_j$. The parameters $\{\mu_j, \sigma_j\}$ are estimated from the training partition only and applied to the test partition; this protocol avoids information leakage from held-out subjects or windows into the normalisation procedure.

3.3 Phase-Space Reconstruction via Delay Embedding

3.3.1 Definition

Definition 3.2 (Delay embedding). *Let $s = (s_0, s_1, \dots, s_{L-1}) \in \mathbb{R}^L$ be a finite scalar sequence. For integers $m \geq 2$ (embedding dimension), $\tau \geq 1$ (lag), and $r \geq 1$ (stride), the (m, τ, r) -delay embedding of s is the finite point cloud*

$$P_{m, \tau, r}(s) = \{\mathbf{p}_i \in \mathbb{R}^m : i = 0, 1, \dots, N-1\}, \quad (3.10)$$

where

$$\mathbf{p}_i = (s_{ir}, s_{ir+\tau}, s_{ir+2\tau}, \dots, s_{ir+(m-1)\tau}),$$

and $N = \lfloor (L - (m-1)\tau - 1)/r \rfloor + 1$ is the number of embedded points.

Each point $\mathbf{p}_i$ collects m samples of s separated by lag τ , and consecutive points are separated by stride r . As i increases the sliding window traces a discrete trajectory in $\mathbb{R}^m$. The cardinality N scales linearly in L and inversely in r , which is decisive for the computational cost of the subsequent homology computation.

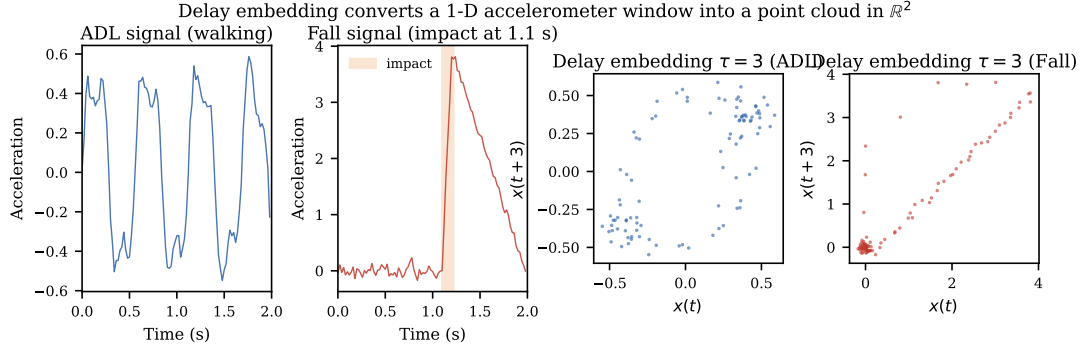


Figure 3.1: Left: synthetic ADL (walking) and fall accelerometer windows. Right: the corresponding two-dimensional delay embeddings ($m = 2$, $\tau = 3$). The ADL trajectory forms a compact, roughly periodic arc; the fall trajectory produces a sparser, elongated point cloud due to the large-amplitude impact spike. This geometric difference motivates the use of persistent homology to distinguish the two signal classes.

3.3.2 The Takens embedding theorem

The use of delay coordinates is motivated by a foundational theorem of dynamical systems theory. Let M be a smooth compact d -dimensional manifold and $\Phi: M \rightarrow M$ a smooth diffeomorphism (the time- Δt flow map of an autonomous dynamical system). Let $h: M \rightarrow \mathbb{R}$ be a smooth scalar observable. The associated *delay map* of order m is

$$\Psi_m^{h,\Phi}: M \longrightarrow \mathbb{R}^m, \quad x \longmapsto (h(x), h(\Phi(x)), h(\Phi^2(x)), \dots, h(\Phi^{m-1}(x))).$$

Theorem 3.2 (Takens (Takens, 1981)). *Let M be a smooth compact d -dimensional manifold. For $m \geq 2d+1$, it is a generic property of the pair (Φ, h) that $\Psi_m^{h,\Phi}: M \rightarrow \mathbb{R}^m$ is a smooth embedding, i.e., a smooth injective immersion onto its image. In particular, $\Psi_m^{h,\Phi}(M)$ is a smooth d -dimensional submanifold of $\mathbb{R}^m$ diffeomorphic to M .*

Generic means that the set of pairs (Φ, h) for which the conclusion fails is contained in a meagre subset of $C^k(M, M) \times C^k(M, \mathbb{R})$ in the appropriate

C^k -topology, i.e., a countable union of nowhere-dense sets. Theorem 3.2 states that, for m sufficiently large relative to the intrinsic dimension of the attractor, the reconstructed trajectory is topologically and differentially faithful to the underlying state space; topological invariants computed on $P_{m,\tau,r}(s)$ therefore reflect intrinsic properties of the dynamical system that produced the signal.

Theorem 3.3 (Fractal extension, Sauer–Yorke–Casdagli (Sauer, Yorke, & Casdagli, 1991)). *Let $A \subseteq \mathbb{R}^N$ be a compact invariant set of a dynamical system with box-counting (Minkowski) dimension $\dim_B(A) = d_A$. For $m > 2d_A$, almost every pair (Φ, h) (with respect to prevalence in $C^k(\mathbb{R}^N, \mathbb{R}^N) \times C^k(\mathbb{R}^N, \mathbb{R})$) gives an injective delay map $\Psi_m^{h,\Phi}$ on A .*

Theorem 3.3 relaxes the smooth-manifold assumption of Theorem 3.2: fractal invariant sets, which are typical for real chaotic systems, can be embedded provided m exceeds twice their box-counting dimension. This is the relevant statement for inertial sensor signals, whose attractors need not be smooth manifolds.

Remark 3.1 (Heuristic status). *Theorems 3.2 and 3.3 require (i) a deterministic underlying dynamical system, (ii) infinite observation, and (iii) generic observable and dynamics. Wearable fall signals satisfy none of these hypotheses strictly: they are finite, noisy, and not the deterministic output of an autonomous flow on a fixed manifold. The theorems are therefore applied in a heuristic rather than a rigorous sense: they motivate the use of delay coordinates as a geometrically richer representation than the raw scalar sequence, without guaranteeing that the resulting point cloud faithfully reconstructs any specific attractor. The parameters (m, τ, r) are treated as hyperparameters and selected empirically by the search of Section 3.9.*

3.3.3 Geometric interpretation for fall detection

The geometric content of the delay embedding is what makes it useful for fall detection.

During a *fall*, the SMV signal undergoes a brief high-amplitude transient: a sharp pre-impact rise, a peak at ground contact, and a partial decay. In the reconstruction space the trajectory makes a rapid excursion into an extreme region and does not return to its starting neighbourhood within the window. The resulting point cloud is *elongated and non-recurrent*: a one-dimensional or quasi-one-dimensional sweep with few or no closed loops.

During a *daily activity* such as walking, the SMV signal is approximately periodic with period $T_{\text{walk}} \approx 1$ s. In the reconstruction space the trajectory traces a structured, roughly periodic orbit. The corresponding point cloud is *recurrent and loop-forming*.

This structural dichotomy is precisely the geometry detected by persistent homology in degrees $k = 0$ and $k = 1$. The zeroth-degree summary H_0 records connected components; the first-degree summary H_1 records independent one-dimensional cycles. Fall windows are expected to exhibit pronounced H_0 behaviour (cluster spread along the excursion) and weak H_1 behaviour; ADL windows are expected to exhibit richer H_1 structure. The empirical statistics tabulated in Section 3.7 are designed to quantify these expectations.

3.3.4 Filtration scale control

For the Rips and SparseRips families, the filtration is truncated at a data-adaptive scale to bound the size of the simplicial complex. Define the multiset of inter-point distances

$$D(P) = \{d_{\text{met}}(\mathbf{p}, \mathbf{p}') : \mathbf{p}, \mathbf{p}' \in P, \mathbf{p} \neq \mathbf{p}'\}$$

and set

$$\varepsilon_{\text{max}} = \text{Quantile}_q(D(P)), \quad q \in \{0.20, 0.40, 0.60, 0.80\}. \quad (3.11)$$

The percentile q controls the scale at which the filtration is truncated: a low percentile concentrates the analysis on fine-scale local geometry; a high percentile admits coarser global structure at the cost of additional simplices. The Alpha complex requires no such cutoff because its size is intrinsically polynomial in $|P|$ via the Delaunay triangulation.

3.4 Topological Preliminaries

The persistent homology of the filtration depends on objects from point-set and algebraic topology that are not standard in signal processing. This section establishes the necessary background.

3.4.1 Topological spaces and homotopy

Definition 3.3 (Topological space). *A topological space is a pair $(X, \mathcal{T})$ where X is a set and $\mathcal{T} \subseteq 2^X$ is a family of subsets, the open sets, satisfying*

- (i) $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$;
- (ii) $\mathcal{T}$ is closed under arbitrary unions;
- (iii) $\mathcal{T}$ is closed under finite intersections.

A map $f: (X, \mathcal{T}_X) \rightarrow (Y, \mathcal{T}_Y)$ is continuous if $f^{-1}(U) \in \mathcal{T}_X$ for every $U \in \mathcal{T}_Y$. A continuous bijection with continuous inverse is a homeomorphism, written $X \cong Y$.

Every metric space (X, d) carries the *metric topology*, generated by the open balls $B(x, r) = \{y \in X : d(x, y) < r\}$.

Definition 3.4 (Homotopy and homotopy equivalence). *Continuous maps $f, g: X \rightarrow Y$ are homotopic ($f \simeq g$) if there exists a continuous map $H: X \times [0, 1] \rightarrow Y$ with $H(\cdot, 0) = f$ and $H(\cdot, 1) = g$. A continuous map $f: X \rightarrow Y$ is a homotopy equivalence if there exists $g: Y \rightarrow X$ with $g \circ f \simeq \text{id}_X$ and $f \circ g \simeq \text{id}_Y$. A space is contractible if it is homotopy equivalent to a point.*

Homotopy equivalence is strictly coarser than homeomorphism. Every convex subset of $\mathbb{R}^m$ is contractible (via the straight-line homotopy to any interior point). Homology is a homotopy invariant: $X \simeq Y$ implies $H_k(X; \mathbb{F}) \cong H_k(Y; \mathbb{F})$ for all $k \geq 0$.

3.4.2 Abstract and geometric simplicial complexes

Definition 3.5 (Abstract simplicial complex). *Let V be a finite set. An abstract simplicial complex on V is a collection $K \subseteq 2^V \setminus \{\emptyset\}$ closed under non-empty subset: $\sigma \in K$ and $\emptyset \neq \rho \subseteq \sigma$ imply $\rho \in K$. An element $\sigma \in K$ with $|\sigma| = k+1$ is a k -simplex of dimension k . The k -skeleton of K is $K^{(k)} = \{\sigma \in K : \dim \sigma \leq k\}$.*

Definition 3.6 (Standard k -simplex). *The standard k -simplex is*

$$\Delta^k = \text{conv}\{e_0, e_1, \dots, e_k\} = \left\{ (t_0, \dots, t_k) \in \mathbb{R}^{k+1} : t_i \geq 0, \sum_{i=0}^k t_i = 1 \right\},$$

where $\{e_i\}$ is the standard basis of $\mathbb{R}^{k+1}$. Its open interior is $\text{int}(\Delta^k) = \{t \in \Delta^k : t_i > 0 \forall i\}$.

Definition 3.7 (Geometric realisation). *Let K be an abstract simplicial complex with vertex set V . An injection $\iota: V \hookrightarrow \mathbb{R}^N$ is in general position for K if no $d+2$ vertices lie in a common affine subspace of dimension $\leq d = \dim K$. The geometric realisation of K is*

$$|K| = \bigcup_{\sigma \in K} \text{conv}\{\iota(v) : v \in \sigma\} \subset \mathbb{R}^N,$$

equipped with the subspace topology of $\mathbb{R}^N$.

The homeomorphism type of $|K|$ does not depend on the choice of general-position embedding. Geometric realisation is *functorial*: a simplicial map $f: K \rightarrow L$ (a vertex map sending each simplex of K to a simplex of L) induces a continuous map $|f|: |K| \rightarrow |L|$ by barycentric interpolation.

3.5 Simplicial Homology

Fix a field $\mathbb{F}$. The computational implementation uses $\mathbb{F} = \mathbb{F}_2$, the field with two elements; the algebraic framework is identical over any field, modulo orientation conventions.

3.5.1 Chains and the boundary operator

An *oriented k -simplex* is an equivalence class $[v_0, \dots, v_k]$ of ordered tuples of distinct vertices, modulo even permutations: $[v_{\pi(0)}, \dots, v_{\pi(k)}] = \text{sgn}(\pi) [v_0, \dots, v_k]$.

Each simplex of positive dimension carries exactly two orientations; over $\mathbb{F}_2$, where $-1 = 1$, orientation is irrelevant.

Definition 3.8 (Chain group). *The k -th chain group $C_k(K; \mathbb{F})$ is the $\mathbb{F}$ -vector space freely generated by the oriented k -simplices of K , with the identification $-\sigma = \bar{\sigma}$ for the two orientations of the same underlying simplex.*

Definition 3.9 (Boundary operator). *The k -th boundary $\partial_k: C_k(K; \mathbb{F}) \rightarrow C_{k-1}(K; \mathbb{F})$ is the $\mathbb{F}$ -linear map defined on basis elements by*

$$\partial_k([v_0, v_1, \dots, v_k]) = \sum_{i=0}^k (-1)^i [v_0, \dots, \widehat{v_i}, \dots, v_k], \quad (3.12)$$

where $\widehat{v_i}$ denotes omission of v_i .

Lemma 3.1 (Fundamental relation $\partial^2 = 0$). *For all $k \geq 1$, $\partial_{k-1} \circ \partial_k = 0$.*

Proof. It suffices to verify the identity on a basis element $\sigma = [v_0, \dots, v_k]$. Two applications of (3.12) give

$$\begin{aligned} \partial_{k-1}(\partial_k \sigma) &= \sum_{j < i} (-1)^{i+j} [v_0, \dots, \widehat{v_j}, \dots, \widehat{v_i}, \dots, v_k] \\ &\quad + \sum_{i < j} (-1)^{i+j-1} [v_0, \dots, \widehat{v_i}, \dots, \widehat{v_j}, \dots, v_k], \end{aligned}$$

where in the second sum the omitted vertex v_j has been shifted by one position because v_i with $i < j$ was already deleted. For each unordered pair $\{i, j\}$ with $i < j$ the two contributions cancel exactly, yielding zero. $\square$

Geometrically, Lemma 3.1 expresses the fact that the boundary of a simplex is itself a closed surface and hence has no boundary.

3.5.2 Homology groups and Betti numbers

Lemma 3.1 implies $\text{im}(\partial_{k+1}) \subseteq \ker(\partial_k)$. The quotient is well defined.

Definition 3.10 (Homology and Betti numbers). *The cycle group is $Z_k = \ker \partial_k \subseteq C_k(K; \mathbb{F})$; the boundary group is $B_k = \text{im } \partial_{k+1} \subseteq Z_k$. The k -th homology is the quotient*

$$H_k(K; \mathbb{F}) = Z_k / B_k,$$

and the k -th Betti number is $\beta_k = \dim_{\mathbb{F}} H_k(K; \mathbb{F})$.

The Betti numbers admit elementary geometric interpretations. β_0 counts the connected components of $|K|$; β_1 counts the number of independent one-dimensional cycles (loops not bounding any 2-chain); β_2 counts the number of enclosed two-dimensional voids.

Theorem 3.4 (Euler–Poincaré formula). *Let $f_k = |\{\sigma \in K : \dim \sigma = k\}|$. The Euler characteristic of K , defined by*

$$\chi(K) = \sum_{k \geq 0} (-1)^k f_k,$$

satisfies

$$\chi(K) = \sum_{k \geq 0} (-1)^k \beta_k. \quad (3.13)$$

Proof. The rank–nullity theorem applied to ∂_k gives $\dim C_k = \dim Z_k + \dim B_{k-1}$ for all $k \geq 0$, with $B_{-1} = 0$. Summing with alternating signs and rearranging telescopes the boundary-group terms; the result is (3.13). $\square$

The formula links a combinatorial datum (f_k) to homotopy invariants (β_k) and serves as a consistency check on any homology computation.

3.5.3 Functoriality

Definition 3.11 (Simplicial and chain maps). *A simplicial map $f: K \rightarrow L$ is a vertex map $f: V(K) \rightarrow V(L)$ with $f(\sigma) \in L$ for every $\sigma \in K$. The induced chain map $f_{\#}: C_k(K; \mathbb{F}) \rightarrow C_k(L; \mathbb{F})$ is defined on basis elements by*

$$f_{\#}([v_0, \dots, v_k]) = \begin{cases} [f(v_0), \dots, f(v_k)] & \text{if } f(v_0), \dots, f(v_k) \text{ are distinct,} \\ 0 & \text{otherwise.} \end{cases}$$

One checks that $f_{\#} \partial_k^K = \partial_k^L f_{\#}$, i.e., chain maps commute with boundaries. Consequently, $f_{\#}$ maps cycles to cycles and boundaries to boundaries, and descends to a well-defined linear map

$$H_k(f): H_k(K; \mathbb{F}) \longrightarrow H_k(L; \mathbb{F}), \quad [z] \longmapsto [f_{\#}(z)].$$

Composition is preserved, $H_k(g \circ f) = H_k(g) \circ H_k(f)$, and identities are preserved, $H_k(\text{id}) = \text{id}$. In categorical language, $H_k(\cdot; \mathbb{F})$ is a functor from the category of simplicial complexes to the category $\mathbf{Vect}_{\mathbb{F}}$ of $\mathbb{F}$ -vector spaces.

This functoriality is the bridge to persistent homology. The inclusions $K_i \hookrightarrow K_j$ for $i \leq j$ in any filtration are simplicial maps, and the induced maps $H_k(K_i; \mathbb{F}) \rightarrow H_k(K_j; \mathbb{F})$ are the structure maps of the persistence module defined below.

3.6 Persistent Homology

3.6.1 Filtrations and complex families

Definition 3.12 (Filtration). *A filtration of a simplicial complex K is a nested family of subcomplexes*

$$\emptyset = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_n = K,$$

indexed by a non-decreasing real sequence $\varepsilon_0 \leq \varepsilon_1 \leq \cdots \leq \varepsilon_n$ of filtration values.

For a finite point cloud $P \subset \mathbb{R}^m$ equipped with metric d_{met} , the methodology uses three filtered complex families.

Vietoris–Rips complex.

$$\mathcal{R}_{\varepsilon}(P) = \{\sigma \subseteq P : \text{diam}_{d_{\text{met}}}(\sigma) \leq \varepsilon\}. \quad (3.14)$$

The Rips complex is a *flag complex*: every clique of its 1-skeleton is filled by a higher-dimensional simplex. The filtration is therefore determined entirely by the order in which edges appear.

Alpha complex. For the Euclidean metric, the Voronoi cell of $p \in P$ is

$$V_p = \{x \in \mathbb{R}^m : \|x - p\|_2 \leq \|x - p'\|_2 \ \forall p' \in P\}.$$

The *Alpha complex* is the nerve of the restricted cover $\{B(p, \varepsilon) \cap V_p\}_{p \in P}$:

$$\mathcal{A}_{\varepsilon}(P) = \left\{ \sigma \subseteq P : \bigcap_{p \in \sigma} (B(p, \varepsilon) \cap V_p) \neq \emptyset \right\}. \quad (3.15)$$

$\mathcal{A}_{\varepsilon}(P)$ is a subcomplex of the Delaunay triangulation, hence has at most $O(|P|^{\lceil m/2 \rceil})$ simplices, dramatically less than the worst-case Rips bound of $2^{|P|}$.

Sparse Rips complex. The SparseRips construction (Cavanna, Jahanseir, & Sheehy, 2015) replaces P by a greedy *maxmin* landmark subset $L \subseteq P$: starting from an arbitrary seed, each successive landmark is the point of P farthest from the existing landmarks. The persistent homology of the resulting weighted complex is provably a $(1 + \varepsilon)$ -approximation of that of $\mathcal{R}_\varepsilon(P)$ in the interleaving distance of Definition 3.14.

Theorem 3.5 (Nerve theorem (Borsuk, 1948)). *Let $\mathcal{U} = \{U_i\}_{i \in I}$ be a finite open cover of a paracompact space X such that every non-empty finite intersection $U_{i_0} \cap \dots \cap U_{i_p}$ is contractible. Then $|N(\mathcal{U})| \simeq X$, where $N(\mathcal{U})$ is the abstract simplicial complex with simplices $\{i_0, \dots, i_p\}$ for which the corresponding intersection is non-empty.*

The Alpha complex satisfies the hypothesis of Theorem 3.5: each $B(p, \varepsilon) \cap V_p$ is the intersection of a ball and a convex Voronoi cell, hence convex and therefore contractible, and finite intersections of convex subsets of $\mathbb{R}^m$ are convex and contractible. Consequently $|\mathcal{A}_\varepsilon(P)| \simeq \bigcup_{p \in P} B(p, \varepsilon)$: the Alpha complex faithfully encodes the topology of the union of balls without introducing simplices beyond the Delaunay triangulation.

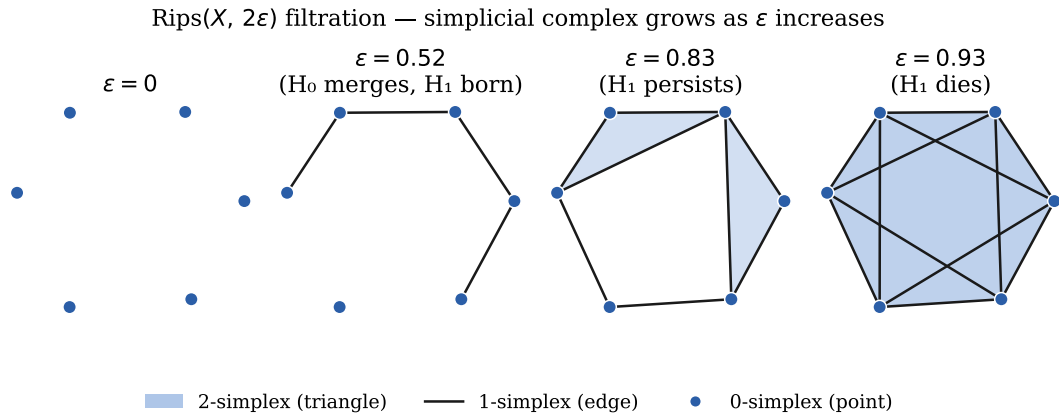


Figure 3.2: Rips ($X, 2\varepsilon$) filtration of a six-point cloud at four scales. At $\varepsilon = 0$ only vertices are present ($\beta_0 = 6, \beta_1 = 0$). As ε increases, edges appear and connected components merge (β_0 decreases). A loop (H_1 class) is born when the polygon is complete and dies when triangles fill it. Features far from the diagonal in the persistence diagram have long lifetime and are noise-robust (Cohen–Steiner stability, Theorem 3.7).

3.6.2 Persistence modules

Regard the linearly ordered set $(\mathbb{R}, \leq)$ as a category: objects are real numbers, and there is a unique morphism $r \rightarrow s$ whenever $r \leq s$. Let $\mathbf{Vect}_{\mathbb{F}}$ be the category of $\mathbb{F}$ -vector spaces and $\mathbb{F}$ -linear maps.

Definition 3.13 (Persistence module). *A (one-parameter) persistence module over $\mathbb{F}$ is a functor*

$$\mathbb{V}: (\mathbb{R}, \leq) \longrightarrow \mathbf{Vect}_{\mathbb{F}}.$$

Concretely, $\mathbb{V}$ assigns a vector space V_r to each $r \in \mathbb{R}$ and a linear map $\rho_{r,s}: V_r \rightarrow V_s$ (the structure map) to each $r \leq s$, with $\rho_{r,r} = \text{id}$ and $\rho_{s,t} \circ \rho_{r,s} = \rho_{r,t}$.

A morphism $\Phi: \mathbb{U} \rightarrow \mathbb{V}$ is a natural transformation: linear maps $\Phi_r: U_r \rightarrow V_r$ with $\Phi_s \circ \rho_{r,s}^{\mathbb{U}} = \rho_{r,s}^{\mathbb{V}} \circ \Phi_r$ for all $r \leq s$. Persistence modules with such morphisms form a category $\mathbf{Pers}_{\mathbb{F}}$.

Applying $H_k(\cdot; \mathbb{F})$ levelwise to a filtration and to each inclusion $K_i \hookrightarrow K_j$ yields a persistence module $\mathbb{V}_k = \{H_k(K_i; \mathbb{F})\}$: the k -th persistence module of the filtration. This is the central algebraic invariant studied by persistent homology.

Definition 3.14 (Interleaving distance). *For $\varepsilon \geq 0$, the ε -shift of $\mathbb{V}$ is the persistence module $\mathbb{V}[\varepsilon]_r := V_{r+\varepsilon}$. Modules $\mathbb{U}, \mathbb{V}$ are ε -interleaved if there exist morphisms $\Phi: \mathbb{U} \rightarrow \mathbb{V}[\varepsilon]$ and $\Psi: \mathbb{V} \rightarrow \mathbb{U}[\varepsilon]$ with $\Psi[\varepsilon] \circ \Phi = \rho_{r,r+2\varepsilon}^{\mathbb{U}}$ and $\Phi[\varepsilon] \circ \Psi = \rho_{r,r+2\varepsilon}^{\mathbb{V}}$ for all r . The interleaving distance is*

$$d_I(\mathbb{U}, \mathbb{V}) = \inf\{\varepsilon \geq 0 : \mathbb{U} \text{ and } \mathbb{V} \text{ are } \varepsilon\text{-interleaved}\}.$$

The interleaving distance is an extended pseudometric on persistence modules; it descends to a metric on the set of isomorphism classes.

3.6.3 The structure theorem and persistence diagrams

Definition 3.15 (Interval module). *For $b < d$ in $\mathbb{R} \cup \{+\infty\}$, the interval module $\mathbb{F}[b, d)$ is the persistence module with $V_r = \mathbb{F}$ for $b \leq r < d$, $V_r = 0$ otherwise, and identity structure maps between non-zero positions.*

Theorem 3.6 (Structure theorem, Crawley-Boevey (Crawley-Boevey, 2015)). *Every pointwise finite-dimensional persistence module $\mathbb{V}$ over a field $\mathbb{F}$ decomposes uniquely, up to isomorphism and reordering, as a direct sum of interval modules:*

$$\mathbb{V} \cong \bigoplus_{i \in I} \mathbb{F}[b_i, d_i]. \quad (3.16)$$

The multiset $\{(b_i, d_i)\}_{i \in I}$ is a complete isomorphism invariant of $\mathbb{V}$.

The special case for tame modules over filtered finite complexes was established by Zomorodian and Carlsson (Edelsbrunner, Letscher, & Zomorodian, 2002). Each interval (b_i, d_i) represents a topological feature of degree k that is *born* at scale b_i and *dies* at scale d_i ; its *persistence* is $p_i = d_i - b_i$. The diagonal is added by convention so that the diagram is a multiset in

$$\{(b, d) \in [-\infty, \infty]^2 : b \leq d\} \cup \{\text{diagonal with infinite multiplicity}\}.$$

Definition 3.16 (Persistence diagram). *The k -th persistence diagram of a filtration is*

$$\text{PD}_k = \{(b_i, d_i) : b_i < d_i < +\infty\}_{i \in I_k}. \quad (3.17)$$

Features with $d_i = +\infty$ are excluded from the present feature vectorisation because their infinite lifetime requires special handling in any scalar summary.

3.6.4 Distances and stability

Definition 3.17 (Bottleneck and Wasserstein distances). *A partial matching between diagrams D_1, D_2 is a bijection $\gamma : D_1 \cup \Delta \rightarrow D_2 \cup \Delta$ that is the identity outside a finite set, where $\Delta = \{(x, x) : x \in \mathbb{R}\}$ is the diagonal. The p -Wasserstein distance ($1 \leq p < \infty$) is*

$$W_p(D_1, D_2) = \inf_{\gamma} \left(\sum_{x \in D_1 \cup \Delta} \|x - \gamma(x)\|_{\infty}^p \right)^{1/p},$$

and the bottleneck distance is

$$d_B(D_1, D_2) = \inf_{\gamma} \sup_{x \in D_1 \cup \Delta} \|x - \gamma(x)\|_{\infty} = \lim_{p \rightarrow \infty} W_p(D_1, D_2).$$

Theorem 3.7 (Stability of persistence diagrams, Cohen-Steiner–Edelsbrunner–Harer (Cohen-Steiner, Edelsbrunner, & Harer, 2007)). *Let $f, g: X \rightarrow \mathbb{R}$ be tame continuous functions on a triangulable compact space X , and let D_f, D_g be the persistence diagrams of their sublevel-set filtrations. Then*

$$d_B(D_f, D_g) \leq \|f - g\|_\infty.$$

Theorem 3.8 (Algebraic stability, Chazal et al. (Chazal, de Silva, Glisse, & Oudot, 2016)). *For any two pointwise finite-dimensional persistence modules $\mathbb{U}, \mathbb{V}$ over a field $\mathbb{F}$,*

$$d_B(\text{PD}(\mathbb{U}), \text{PD}(\mathbb{V})) = d_I(\mathbb{U}, \mathbb{V}).$$

Theorem 3.8 upgrades stability from an inequality to an isometry: the bottleneck distance on diagrams equals the interleaving distance on modules, providing a complete algebraic characterisation of the geometry of persistence.

For point cloud filtrations, the metric stability statement of practical interest is the following.

Corollary 3.1 (Metric stability). *Let $P, Q \subset \mathbb{R}^m$ be finite point sets and let*

$$d_H(P, Q) = \max\left(\max_{p \in P} \min_{q \in Q} \|p - q\|_2, \max_{q \in Q} \min_{p \in P} \|p - q\|_2\right)$$

be their Hausdorff distance. Then for every $k \geq 0$,

$$d_B(\text{PD}_k(\mathcal{R}(P)), \text{PD}_k(\mathcal{R}(Q))) \leq 2 d_H(P, Q).$$

Proof sketch. If $d_H(P, Q) \leq \delta$, then each simplex of $\mathcal{R}_\varepsilon(P)$ lifts to a simplex of $\mathcal{R}_{\varepsilon+2\delta}(Q)$ by replacing each vertex by a point of Q within distance δ and applying the triangle inequality to the diameter. By symmetry the two Rips modules are 2δ -interleaved, hence $d_B \leq 2\delta$ by Theorem 3.8. $\square$

The practical content is that bounded sensor noise of amplitude δ perturbs each embedded point by at most δ in ℓ_2 , hence the persistence diagrams of two realisations of the same fall event differ by at most 2δ in bottleneck distance. Features with persistence $p_i \gg 2\delta$ are noise-robust; near-diagonal features with $p_i \approx 0$ are noise-sensitive.

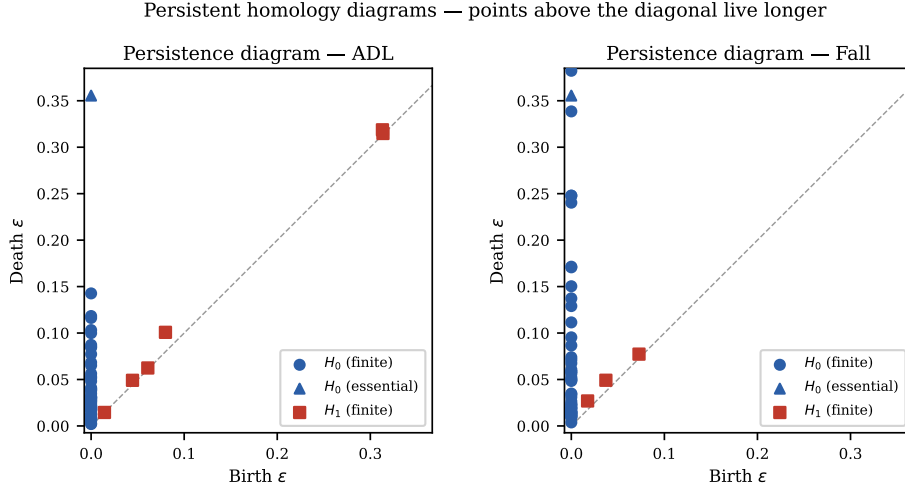


Figure 3.3: Persistence diagrams (H_0 circles, H_1 squares) for delay-embedded ADL and fall windows. Points far from the diagonal represent long-lived topological features. The fall window typically exhibits a high-persistence H_1 feature (loop) absent or short-lived in the ADL window—a geometric signature of the sharp impact trajectory in the delay-embedded space. Triangles and diamonds denote essential classes (infinite death, plotted at the top margin).

3.6.5 Computational reduction

Persistence diagrams are computed by *boundary matrix reduction*. Order the simplices of the filtration as $\sigma_1, \sigma_2, \dots, \sigma_n$ so that $\dim \sigma_i \leq \dim \sigma_{i+1}$ and simplices entering at smaller scale precede those at larger scale (consistent with filtration order). The boundary matrix $D \in \mathbb{F}^{n \times n}$ has $D_{i,j} = 1$ if $\sigma_i \in \partial \sigma_j$ and 0 otherwise. The *pivot* of column j is $\text{low}(D, j) = \max\{i : D_{i,j} \neq 0\}$, undefined if column j is zero.

Algorithm 1 Standard persistence algorithm (column reduction over $\mathbb{F}_2$).

Require: Boundary matrix $D \in \{0, 1\}^{n \times n}$ in filtration order

Ensure: Reduced matrix R with unique pivots; persistence pairs read from R

- 1: $R \leftarrow D$
 - 2: **for** $j = 1$ **to** n **do**
 - 3: **while** $\exists i < j$ with $\text{low}(R, i) = \text{low}(R, j) \neq \perp$ **do**
 - 4: $R[\cdot, j] \leftarrow R[\cdot, j] + R[\cdot, i]$ $\triangleright$ column addition over $\mathbb{F}_2$
 - 5: **end while**
 - 6: **end for**
 - 7: **Pairs:** $\{(\text{low}(R, j), j) : R[\cdot, j] \neq 0\}$
-

Proposition 3.2 (Uniqueness of pairing). *The pairing produced by Algorithm 1 is independent of the order in which column additions are performed; it depends only on the filtration.*

The proposition is a corollary of the structure theorem 3.6. The worst-case complexity of Algorithm 1 is $O(n^3)$; the GUDHI library (GUDHI Editorial Board, 2015) implements optimised variants (twist reduction, clearing, dualisation to cohomology) with near-linear average-case complexity on the point cloud sizes considered here.

3.7 Feature Vectorisation

Persistence diagrams are complete isomorphism invariants of the filtration but cannot be passed directly to standard classifiers, which require fixed-length input vectors. Two obstacles obstruct direct use:

- (i) the cardinality $|I_k|$ varies across windows and configurations;
- (ii) the diagram space, under the bottleneck or Wasserstein metric, is not a Hilbert space.

A variety of vectorisation schemes circumvent these obstacles: persistence images (Adams, Emerson, Kirby, Neville, Peterson, Shipman, Chepushtanova, Hanson, Motta, & Ziegelmeier, 2017), persistence landscapes (Bubenik, 2015), and others. The present methodology uses a fixed set of scalar statistics, motivated by the requirement that the feature dimension be constant across λ to permit a uniform Bayesian search.

3.7.1 Persistence-based statistics

For each homological degree $k \in \{0, 1\}$, write $\{(b_i^{(k)}, d_i^{(k)})\}_{i=1}^{n_k}$ for the finite-persistence pairs of PD_k , and let $p_i^{(k)} = d_i^{(k)} - b_i^{(k)} > 0$ be the persistence. When $n_k = 0$ all derived statistics are set to zero. Ten scalar statistics are computed per degree.

j	Statistic	Definition
1	Bar count	n_k/L (normalised by window length)
2	Entropy	$-\sum_i \tilde{p}_i \log \tilde{p}_i$, $\tilde{p}_i = p_i / \sum_j p_j$
3	Maximum	$\max_i p_i^{(k)}$
4	Energy	$\sum_i (p_i^{(k)})^2$
5	Mean	$\bar{p}^{(k)} = n_k^{-1} \sum_i p_i^{(k)}$
6	Standard dev.	$(n_k^{-1} \sum_i (p_i - \bar{p})^2)^{1/2}$
7	Median	$Q_{0.50}(\mathbf{p}^{(k)})$
8	Q1	$Q_{0.25}(\mathbf{p}^{(k)})$
9	Q3	$Q_{0.75}(\mathbf{p}^{(k)})$
10	Third moment	$n_k^{-1} \sum_i (p_i - \bar{p})^3$

These ten statistics for $k = 0$ occupy feature positions 1–10 and for $k = 1$ occupy positions 11–20. Together they give a 20-dimensional topological sub-vector.

3.7.2 Signal-level features

Four additional scalar features are computed directly from SMV_{acc} on the window:

$$f_{21} = \max_t \text{SMV}, \quad f_{22} = \text{std}(\text{SMV}), \quad f_{23} = \overline{\text{SMV}}, \quad f_{24} = \max_t \text{SMV} - \min_t \text{SMV}. \quad (3.18)$$

3.7.3 Complete feature vector

The complete feature vector under configuration λ is

$$\phi_\lambda(w) = (f_1^{(0)}, \dots, f_{10}^{(0)}, f_1^{(1)}, \dots, f_{10}^{(1)}, f_{21}, f_{22}, f_{23}, f_{24}) \in \mathbb{R}^{24}. \quad (3.19)$$

The fixed dimension is a methodological constraint imposed by the requirement of a uniform classifier interface across λ .

3.8 Classification Models

Two linear classifier families are considered, $\mathcal{F}_{\text{lr}}$ and $\mathcal{F}_{\text{svm}}$. Both are applied with balanced class weighting and, for SVM, with Platt probability calibration.

3.8.1 Logistic regression

Logistic regression models the conditional fall probability by

$$\mathbb{P}(Y = 1 \mid \mathbf{z}; \boldsymbol{\theta}, b) = \sigma(\boldsymbol{\theta}^\top \mathbf{z} + b), \quad \sigma(u) = \frac{1}{1 + e^{-u}}, \quad (3.20)$$

where $\mathbf{z} \in \mathbb{R}^{24}$ is the standardised feature vector and $(\boldsymbol{\theta}, b) \in \mathbb{R}^{24} \times \mathbb{R}$ are the parameters. Given training data $\{(\mathbf{z}_i, y_i)\}_{i=1}^n$ the regularised negative log-likelihood is

$$\mathcal{L}_\Omega(\boldsymbol{\theta}, b) = -\frac{1}{n} \sum_{i=1}^n \left[y_i \log \sigma_i + (1 - y_i) \log(1 - \sigma_i) \right] + \frac{1}{C} \Omega(\boldsymbol{\theta}), \quad (3.21)$$

where $\sigma_i = \sigma(\boldsymbol{\theta}^\top \mathbf{z}_i + b)$, $C > 0$ is the inverse regularisation strength, and $\Omega(\boldsymbol{\theta}) \in \{\frac{1}{2}\|\boldsymbol{\theta}\|_2^2, \|\boldsymbol{\theta}\|_1\}$ selects the ℓ_2 or ℓ_1 penalty. The ℓ_2 objective is strictly convex and smooth and is solved by L-BFGS or coordinate descent; the ℓ_1 objective is convex and non-smooth and is solved by liblinear coordinate descent in the implementation.

Proposition 3.3 (Convexity and uniqueness for ℓ_2). *With $\Omega(\boldsymbol{\theta}) = \frac{1}{2}\|\boldsymbol{\theta}\|_2^2$, the objective (3.21) is strictly convex in $(\boldsymbol{\theta}, b)$ on $\mathbb{R}^{25}$, hence admits a unique global minimiser.*

Proof. The negative log-likelihood is convex because $-\log \sigma(u)$ is convex in u and the maps $u \mapsto \boldsymbol{\theta}^\top \mathbf{z} + b$ are affine. The ℓ_2 penalty is strictly convex. The sum of a convex and a strictly convex function is strictly convex. $\square$

The predicted probability is $f_\theta(\mathbf{z}) = \sigma(\boldsymbol{\theta}^\top \mathbf{z} + b) \in (0, 1)$.

3.8.2 Support vector machine

The C -support vector classifier seeks a hyperplane $\{\mathbf{w}^\top \mathbf{z} + b = 0\}$ with maximum geometric margin. With slack variables $\xi_i \geq 0$ the primal soft-margin problem is

$$\min_{\mathbf{w}, b, \boldsymbol{\xi}} \frac{1}{2}\|\mathbf{w}\|_2^2 + C \sum_{i=1}^n \xi_i \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{z}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad y_i \in \{-1, +1\}. \quad (3.22)$$

The Lagrangian is

$$L(\mathbf{w}, b, \boldsymbol{\xi}, \boldsymbol{\alpha}, \boldsymbol{\mu}) = \frac{1}{2}\|\mathbf{w}\|^2 + C \sum_i \xi_i - \sum_i \alpha_i [y_i(\mathbf{w}^\top \mathbf{z}_i + b) - 1 + \xi_i] - \sum_i \mu_i \xi_i,$$

with $\alpha_i, \mu_i \geq 0$. The KKT stationarity conditions yield

$$\nabla_{\mathbf{w}} L = 0 \Rightarrow \mathbf{w} = \sum_i \alpha_i y_i \mathbf{z}_i, \quad \partial_b L = 0 \Rightarrow \sum_i \alpha_i y_i = 0, \quad \partial_{\xi_i} L = 0 \Rightarrow \alpha_i + \mu_i = C.$$

Substitution into L produces the dual quadratic programme

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \mathbf{z}_i^\top \mathbf{z}_j \quad \text{s.t.} \quad 0 \leq \alpha_i \leq C, \quad \sum_i \alpha_i y_i = 0. \quad (3.23)$$

The primal solution recovers $\mathbf{w}^* = \sum_i \alpha_i^* y_i \mathbf{z}_i$ and the decision value for a new point is

$$f(\mathbf{z}) = \sum_i \alpha_i^* y_i \mathbf{z}_i^\top \mathbf{z} + b^*. \quad (3.24)$$

Points with $\alpha_i^* > 0$ are the *support vectors*. With $\kappa(\mathbf{z}, \mathbf{z}') = \mathbf{z}^\top \mathbf{z}'$ replaced by the RBF kernel

$$\kappa_{\text{rbf}}(\mathbf{z}, \mathbf{z}') = \exp(-\gamma \|\mathbf{z} - \mathbf{z}'\|_2^2),$$

the same dual applies in a reproducing kernel Hilbert space.

3.8.3 Platt score calibration

The SVM decision value $f(\mathbf{z}) \in \mathbb{R}$ is not a probability. Platt scaling (Platt, 1999) fits a logistic function to the decision values on a held-out calibration set:

$$\mathbb{P}(Y = 1 \mid \mathbf{z}) = \sigma(Af(\mathbf{z}) + B), \quad A, B \in \mathbb{R}, \quad (3.25)$$

with (A, B) estimated by minimising the cross-entropy loss against Laplace-smoothed targets

$$t_i = \begin{cases} \frac{n_+ + 1}{n_+ + 2} & \text{if } y_i = +1, \\ \frac{1}{n_- + 2} & \text{if } y_i = -1, \end{cases}$$

where $n_{\pm}$ are the counts of positive and negative calibration examples. The smoothing prevents degenerate 0 or 1 predictions when the margin is large.

3.8.4 Class-weighted empirical risk

Falls are minority events; with n_0 negative and n_1 positive training examples ($n = n_0 + n_1$) the *balanced class weights* are

$$w_c = \frac{n}{2n_c}, \quad c \in \{0, 1\}. \quad (3.26)$$

This satisfies $w_0 n_0 = w_1 n_1 = n/2$. For logistic regression, the weighted log-likelihood replaces $\mathcal{L}_\Omega$ by

$$\mathcal{L}_\Omega^w(\boldsymbol{\theta}, b) = -\frac{1}{n} \sum_{i=1}^n w_{y_i} \left[y_i \log \sigma_i + (1 - y_i) \log(1 - \sigma_i) \right] + \frac{1}{C} \Omega(\boldsymbol{\theta}).$$

For C-SVM, the scalar C in (3.22) is replaced by $C_+ = C w_1$ for positive examples and $C_- = C w_0$ for negative examples, biasing the model toward higher recall.

3.9 Bayesian Hyperparameter Optimisation

The representation map ϕ_λ depends on the hyperparameter tuple $\lambda \in \Lambda$ of (3.2), and the classifier depends on its own tuple $\theta \in \Theta$ (regularisation strength, penalty, kernel). Joint optimisation over $\Lambda \times \Theta$ is necessary because the two tuples interact: the geometry of the topological representation determines which classifier hyperparameters are optimal. The optimisation is performed by Bayesian search with a tree-structured Parzen estimator (TPE), implemented in Optuna (Akiba, Sano, Yanase, Ohta, & Koyama, 2019).

3.9.1 The optimisation problem

Let $\mathcal{D}_K^{\text{train}}$ be the labelled feature corpus of dataset K from which the test subject has been removed (LOSO outer split, Section 3.10.3). Let

$$\text{CV}_K: \Lambda \times \Theta \longrightarrow \mathbb{R}, \quad (\lambda, \theta) \longmapsto \widehat{F}_{2,K}(\lambda, \theta)$$

denote the inner cross-validation estimator of $\mathbb{E}[F_2 \mid K, \lambda, \theta]$ obtained by stratified k -fold CV on $\mathcal{D}_K^{\text{train}}$. The optimisation problem solved at each outer fold is

$$\lambda_K^*, \theta_K^* \in \arg \max_{(\lambda, \theta) \in \Lambda \times \Theta} \text{CV}_K(\lambda, \theta). \quad (3.27)$$

The objective is non-differentiable, noisy (because CV estimates fluctuate with the random fold partition and stochastic class-balancing), and expensive to evaluate (each call requires constructing point clouds, computing persistent homology, fitting the classifier, and performing the CV averaging). These properties make grid search infeasible and motivate a sequential model-based approach.

3.9.2 Sequential model-based optimisation

Sequential model-based optimisation (SMBO) (Hutter, Hoos, & Leyton-Brown, 2011) alternates between fitting a probabilistic surrogate to past evaluations and proposing new evaluations that maximise an acquisition function under the surrogate. Let $y(\lambda, \theta) = \text{CV}_K(\lambda, \theta) + \eta$ denote one noisy evaluation of the objective, with η centred. After t evaluations the data set is

$$\mathcal{H}_t = \{((\lambda_i, \theta_i), y_i) : i = 1, \dots, t\}.$$

A surrogate $p(y \mid \lambda, \theta, \mathcal{H}_t)$ is a probability model over the objective conditional on the search point and the history. The standard acquisition function is the *expected improvement* (EI) over a reference value y^* :

$$\text{EI}_{y^*}(\lambda, \theta) = \mathbb{E}[\max(y^* - y, 0) \mid \lambda, \theta, \mathcal{H}_t]. \quad (3.28)$$

(Here EI is stated for a minimisation problem with y the negative $\widehat{F}_2$; sign conventions are equivalent under negation.) The next search point is

$$(\lambda_{t+1}, \theta_{t+1}) \in \arg \max_{(\lambda, \theta) \in \Lambda \times \Theta} \text{EI}_{y^*}(\lambda, \theta). \quad (3.29)$$

The surrogate is updated and the loop continues for a fixed budget. The textbook surrogate is a Gaussian process posterior; TPE replaces it with a non-parametric density model that is better suited to discrete and conditional search spaces.

3.9.3 Tree-structured Parzen estimator

TPE (Bergstra, Bardenet, Bengio, & Kégl, 2011) models the posterior in an inverted form. Instead of modelling $p(y \mid \lambda, \theta)$ directly, it splits the observed history at a quantile $\gamma \in (0, 1)$ and models the density of the inputs given two quality regimes,

$$p(\lambda, \theta \mid y \leq y^*) = \ell(\lambda, \theta), \quad p(\lambda, \theta \mid y > y^*) = g(\lambda, \theta),$$

where y^* is the empirical γ -quantile of the observed $\{y_i\}$. The two density estimates are constructed from the corresponding subsets of $\mathcal{H}_t$ by kernel density estimation: each observation contributes a kernel centred at its parameter

value, with bandwidth determined adaptively. Conditional and discrete hyperparameters are handled by a hierarchical tree structure on $\Lambda \times \Theta$, with each node carrying its own one-dimensional kernel.

Proposition 3.4 (EI under TPE is the ratio of densities). *Under the TPE generative model, the expected improvement $\text{EI}_{y^*}(\lambda, \theta)$ is proportional to the ratio of the conditional densities:*

$$\text{EI}_{y^*}(\lambda, \theta) \propto \left(\gamma + \frac{g(\lambda, \theta)}{\ell(\lambda, \theta)} (1 - \gamma) \right)^{-1}.$$

Consequently, maximising EI is equivalent to maximising the ratio $\ell(\lambda, \theta)/g(\lambda, \theta)$.

Proof sketch. With $p(y \leq y^*) = \gamma$, Bayes' rule gives

$$p(\lambda, \theta) = \gamma \ell + (1 - \gamma) g, \quad p(y \leq y^* \mid \lambda, \theta) = \frac{\gamma \ell}{\gamma \ell + (1 - \gamma) g}.$$

Substitution into $\text{EI}_{y^*} = \int_{-\infty}^{y^*} (y^* - y) p(y \mid \lambda, \theta) dy$ and elementary manipulation yields the displayed proportionality (Bergstra, Bardenet, Bengio, & Kégl, 2011).

□

3.9.4 Comparison with alternative search strategies

Four search paradigms are available for the optimisation of (3.27). The present methodology adopts TPE-SMBO; this subsection provides a quantitative justification.

Grid search. Under the conditional structure of Λ specified in Section 3.9.5, the total number of distinct configurations is

$$|\Lambda_{\text{grid}}| = n_w \cdot n_m \cdot n_\tau \cdot n_r (1 + 2 n_d n_\varepsilon) = 9 \cdot 4 \cdot 5 \cdot 3 \cdot (1 + 2 \cdot 4 \cdot 4) = 17,820, \quad (3.30)$$

where the factor 1 accounts for the Alpha branch (metric and filtration-scale parameters inactive) and $2 \cdot 16 = 32$ for the Rips and SparseRips branches. Combined with the classifier grids, $|\Lambda_{\text{grid}}| \times (|\Theta_{\text{lr}}| + |\Theta_{\text{svm}}|) = 17,820 \times 18 = 320,760$ inner-CV evaluations are needed *per outer LOSO fold*. For SisFall with $|\mathcal{S}_K| = 24$ outer folds, a conservative estimate of 5 s per inner-CV call gives approximately

$$320,760 \times 24 \times 5 \text{ s} \approx 43 \text{ days},$$

making full grid search computationally infeasible on the available hardware.

Random search. Random search (Bergstra & Bengio, 2012) draws configurations independently and uniformly from $\Lambda \times \Theta$ without reference to past evaluations. It is statistically consistent under infinite budget and, under the low effective-dimensionality hypothesis, outperforms grid search at equal sample count by avoiding wasted evaluations on uninformative axes. For the present budget $T = 200$, however, random search explores only $200/17,820 \approx 1.1\%$ of the grid and provides no mechanism for directing subsequent trials toward promising regions identified by earlier ones.

Gaussian-process Bayesian optimisation. The canonical SMBO surrogate is a Gaussian process (GP), which models the objective as a GP-distributed random function and updates its posterior via Bayes’ rule after each evaluation. GP-SMBO is appropriate when the search space is a compact subset of $\mathbb{R}^d$ with a natural metric. It is unsuitable here for two structural reasons. First, no natural stationary kernel is defined on the conditional, mixed-type space $\Lambda \times \Theta$ without auxiliary embeddings whose inductive bias is difficult to justify. Second, fitting and inverting the GP covariance matrix costs $O(t^3)$ in the evaluation count t ; for $T = 200$ iterations this overhead is non-negligible and grows quadratically with the complexity of the correlation model.

TPE (adopted strategy). TPE (Bergstra, Bardenet, Bengio, & Kégl, 2011) resolves both difficulties. The conditional structure of Λ is encoded natively in the tree; each leaf node carries an independent univariate kernel density estimator, so per-iteration update cost is $O(t)$. Proposition 3.4 provides the theoretical guarantee that maximising ℓ/g is equivalent to maximising EI under the TPE generative model. Empirically, TPE has outperformed GP-EI on machine-learning hyperparameter benchmarks at comparable budgets (Bergstra, Bardenet, Bengio, & Kégl, 2011; Bergstra & Bengio, 2012), and its

implementation in Optuna (Akiba, Sano, Yanase, Ohta, & Koyama, 2019) handles mixed, conditional, and discrete spaces transparently.

3.9.5 Search space

The search space Λ is the Cartesian product

$$\begin{aligned} w_{\text{sec}} &\in \{1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0\} \text{ (s)}, \\ m &\in \{2, 3, 4, 5\}, \quad \tau \in \{1, 2, 3, 4, 5\}, \quad r \in \{1, 2, 4\}, \\ c &\in \{\text{Alpha}, \text{Rips}, \text{SparseRips}\}, \\ d_{\text{met}} &\in \{\text{Euclidean}, \text{Manhattan}, \text{Chebyshev}, \text{Cosine}\}, \\ q &\in \{0.20, 0.40, 0.60, 0.80\}, \quad \varepsilon \in \Lambda_\varepsilon, \end{aligned}$$

where Λ_ε denotes the auxiliary filtration parameter associated with c . Several conditional constraints are imposed by the geometry: the Alpha complex requires the Euclidean metric; metric and q are inactive for the Alpha complex. The classifier search spaces are

$$\begin{aligned} \Theta_{\text{lr}} &= \{(C, \Omega) : C \in \{10^{-2}, 10^{-1}, 1, 10, 10^2\}, \Omega \in \{\ell_1, \ell_2\}\}, \\ \Theta_{\text{svm}} &= \{(C, \kappa) : C \in \{10^{-1}, 1, 10, 10^2\}, \kappa \in \{\text{linear}, \text{rbf}\}\}. \end{aligned}$$

For the personal protocol of Section 3.10.3 the LR grid is reduced to $C \in \{10^{-1}, 1, 10\}$ to mitigate overfitting in smaller per-subject training sets.

3.9.6 Outer LOSO and inner cross-validation

The selection procedure is the nested protocol

- (1) **Outer.** For each subject $s \in \mathcal{S}_K$, define $\mathcal{D}_K^{\text{train}}(s)$ as the union of all windows of subjects $\mathcal{S}_K \setminus \{s\}$.
- (2) **Inner.** Run TPE for $T = 200$ iterations on $\mathcal{D}_K^{\text{train}}(s)$ using stratified k -fold CV; record $\lambda_K^*(s), \theta_K^*(s)$.
- (3) **Retrain and test.** Fit a final model on $\mathcal{D}_K^{\text{train}}(s)$ with $(\lambda_K^*(s), \theta_K^*(s))$ and evaluate on the held-out subject s .

The outer LOSO ensures that no information from the test subject is used to choose any model component; the inner CV ensures that the surrogate objective is an unbiased estimator of expected performance on unseen subjects from the training pool.

Remark 3.2 (Computational budget). *The budget $T = 200$ was chosen empirically as the smallest value at which the running best objective stabilises across independent runs of the search on the three datasets. Larger budgets do not produce reliably different optima, given the noise level of the inner CV estimator.*

Remark 3.3 (Implementation decomposition). *Equation (3.27) states a joint optimisation over $\Lambda \times \Theta$. In practice this joint search is decomposed into two sequential stages to reduce computational cost. Stage 1 (TDA search, Optuna): A single TPE study is run per dataset for $T = 200$ iterations. The objective is the average F_2 of a logistic regression proxy and a linear SVM proxy, both with fixed hyperparameters, evaluated by three-fold CV on a balanced training subset. Using lightweight proxies keeps each function evaluation fast, so the budget is spent exploring Λ rather than the joint space; the averaged objective ensures the recovered λ^* is not overfitted to a single classifier’s inductive bias. Stage 2 (classifier search, GridSearchCV): Given λ^* , classifier hyperparameters θ^* are selected by exhaustive grid search with three-fold CV on the features extracted under λ^* . This decomposition approximates the joint optimum while keeping computation tractable.*

Remark 3.4 (Classifier-agnostic TDA search). *Rather than running a separate Optuna study for each classifier family, a single study is run with a classifier-agnostic objective. Define*

$$\lambda^* \in \arg \max_{\lambda \in \Lambda} \frac{1}{2} \left(\text{CV}_K^{\text{LR}}(\lambda) + \text{CV}_K^{\text{SVM}}(\lambda) \right),$$

where $\text{CV}_K^f(\lambda)$ denotes the K -fold cross-validated F_2 obtained by classifier f on features extracted under λ , with f ’s hyperparameters fixed at proxy values. The averaged objective recovers a configuration λ^ whose features support good performance across classifier families simultaneously. The map*

$\phi_{\lambda^*}: \mathbb{R}^{L \times 2} \rightarrow \mathbb{R}^{24}$ must therefore produce an image that is both approximately linearly separable (beneficial for LR) and geometrically well-structured in pairwise distance (beneficial for SVM with a nonlinear kernel). A representation satisfying both conditions is a stronger finding than one tuned to a single classifier's inductive bias.

3.10 Evaluation Framework

3.10.1 Performance metrics

Given binary predictions $\hat{y}_i$ and labels y_i on a test sample $\{(\hat{y}_i, y_i)\}_{i=1}^N$, define the four confusion counts TP, TN, FP, FN and the four derived rates

$$\text{Acc} = \frac{\text{TP} + \text{TN}}{N}, \quad \text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{Prec} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad F_1 = \frac{2 \text{Prec} \text{Rec}}{\text{Prec} + \text{Rec}}.$$

The principal metric is the F_β -score at $\beta = 2$,

$$F_2 = (1 + \beta^2) \frac{\text{Prec} \cdot \text{Rec}}{\beta^2 \text{Prec} + \text{Rec}} \Big|_{\beta=2} = \frac{5 \text{Prec} \text{Rec}}{4 \text{Prec} + \text{Rec}}. \quad (3.31)$$

Proposition 3.5 (Recall weighting under F_β). *For fixed precision $\text{Prec} = p$ and recall $\text{Rec} = r$,*

$$\frac{\partial F_\beta}{\partial r} \Big|_{\beta=2} = \frac{20 p^2}{(4p + r)^2}, \quad \frac{\partial F_\beta}{\partial p} \Big|_{\beta=2} = \frac{5 r^2}{(4p + r)^2}.$$

Consequently, near $p = r$ the marginal F_2 -gain per unit of recall is four times the marginal gain per unit of precision.

Proof. Direct differentiation of (3.31) and the substitution $p = r$ give the ratio 4 between the two partial derivatives. $\square$

The weighting ratio 4:1 is a deliberate methodological choice: a missed fall leaves a vulnerable person unattended after a potentially injurious event, whereas a false alarm requires at most an unnecessary check. The F_2 score is used both as the Bayesian objective and as the headline metric in all result tables.

Remark 3.5 (Non-triviality of the baseline). *A trivial all-positive classifier achieves $\text{Rec} = 1$ and $\text{Prec} = \pi$, hence $F_2 = 5\pi/(4\pi+1)$. For $\pi < 0.30$ in all three*

datasets this gives $F_2 < 0.56$, well below the scores reported in the *Experiments* section. The recall-oriented objective does not degenerate to a trivial solution.

3.10.2 Decision threshold

A probabilistic classifier $f_\theta: \mathbb{R}^{24} \rightarrow [0, 1]$ assigns each window a posterior fall probability. A binary label is produced by thresholding at the trained threshold $\hat{\tau}_0$,

$$\hat{y}(w) = \mathbf{1}\{f_\theta(\phi_\lambda(w)) \geq \hat{\tau}_0\}. \quad (3.32)$$

The trained threshold $\hat{\tau}_0$ is the PR-curve optimal value identified on the training fold: $\hat{\tau}_0 = \arg \max_\tau F_2^{\text{train}}(\tau)$. This PR-curve selection is performed once per LOSO fold using the training-fold probability scores and labels. The value $\hat{\tau}_0$ varies across subjects and datasets (typically in the range $[0.55, 0.85]$) and is more conservative than the nominal $\tau_0 = 0.50$ because the class-imbalanced training distribution places the optimal precision–recall trade-off above the 50 % probability level.

3.10.3 Evaluation protocols

Three evaluation scenarios are reported, each measuring a different level of generalisation.

Naive protocol. The feature matrix is partitioned uniformly at random into training (70%) and test (30%) sets with no constraint on subject membership. The same subject can contribute windows to both partitions. This protocol systematically overestimates performance on new individuals and is reported only as an upper-bound reference. Ten random partitions are used; results are reported as means.

General protocol (leave-one-subject-out). For each $s \in \mathcal{S}_K$ the test set is $\{w : \text{subject}(w) = s\}$ and the training set is the complement. This LOSO scheme directly measures generalisation to a previously unseen individual. Because the class-balancing procedure is stochastic, each fold is repeated $R = 15$ times

with independent random seeds; reported quantities are means or appropriate aggregates over the resulting $|\mathcal{S}_K| \times R$ collection of values. The General protocol is the principal protocol of the thesis.

Personal protocol. For each $s \in \mathcal{S}_K$ a model is trained and tested exclusively on that subject’s own windows via a 70/30 random split, repeated $R = 15$ times. The personal protocol represents the idealised setting in which a per-user calibration period is available before deployment.

Decision-threshold sweep. Using the General-protocol classifier, the threshold τ is swept over the grid $\mathcal{T} = \{0.05, 0.10, \dots, 0.95\} \subset (0, 1)$ for each held-out subject. The subject-level retrospective gain is

$$\Delta F_2(s) = \max_{\tau \in \mathcal{T}} F_2(\tau, s) - F_2(\hat{\tau}_0(s), s), \quad (3.33)$$

where $\hat{\tau}_0(s)$ is the PR-curve trained threshold for subject s (Section 3.10.2). The average $\overline{\Delta F_2} = |\mathcal{S}_K|^{-1} \sum_s \Delta F_2(s)$ quantifies how much the PR-curve-calibrated trained threshold underperforms a subject-specific oracle. Because $\tau^*(s) = \arg \max_{\tau} F_2(\tau, s)$ is identified on the same test windows on which it is evaluated, the sweep is a retrospective diagnostic of heterogeneity, not a valid prospective personalisation scheme.

3.11 Statistical Inference

The statistical reporting framework distinguishes two sources of variability: the within-subject variability V_{rep} across the $R = 15$ repetitions of the class-balancing procedure, and the across-subject variability V_{subj} in mean performance across the subject pool. The thesis reports these separately, in line with the methodological correction stated in the proposal: subject-level averages, dispersion, and uncertainty estimates take precedence over headline maxima.

3.11.1 Repetition-level confidence intervals

For each (subject s , repetition j) pair let $x_{s,j} \in [0, 1]$ denote the metric of interest (typically F_2). Define the per-repetition study-level mean

$$\bar{X}_j = |\mathcal{S}_K|^{-1} \sum_{s \in \mathcal{S}_K} x_{s,j}, \quad j = 1, \dots, R.$$

The overall study mean is $\bar{X}_K = R^{-1} \sum_j \bar{X}_j$. The sample standard deviation of $\{\bar{X}_j\}$ is

$$S_R = \left(\frac{1}{R-1} \sum_{j=1}^R (\bar{X}_j - \bar{X}_K)^2 \right)^{1/2}.$$

Theorem 3.9 (Repetition-level t confidence interval). *If the per-repetition study means $\bar{X}_j$ are exchangeable and approximately Gaussian, then a $1 - \alpha$ confidence interval for the limiting expectation $\mathbb{E}[\bar{X}]$ as $R \rightarrow \infty$ is*

$$\bar{X}_K \pm t_{\alpha/2, R-1} \frac{S_R}{\sqrt{R}}, \quad (3.34)$$

where $t_{\alpha/2, R-1}$ is the upper $\alpha/2$ quantile of the t -distribution with $R - 1$ degrees of freedom.

For $R = 15$ and $\alpha = 0.05$, $t_{0.025, 14} \approx 2.145$. The interval (3.34) captures the repetition-to-repetition stochastic variability of the class-balancing procedure; it does not capture subject-level heterogeneity.

3.11.2 Subject-level paired tests

Many of the central comparisons in the thesis are paired at the subject level: $F_2^{\text{lr}}(s)$ vs $F_2^{\text{svm}}(s)$, $F_2^{\text{pers}}(s)$ vs $F_2^{\text{gen}}(s)$, or F_2 for two distinct λ values. Let

$$\delta_s = A_s - B_s \in \mathbb{R}, \quad s \in \mathcal{S}_K,$$

denote the per-subject paired difference, with the sign convention that positive δ_s favours A .

Theorem 3.10 (Paired t -test). *Under the null $H_0: \mathbb{E}[\delta_s] = 0$ and Gaussianity of $\{\delta_s\}$, the statistic*

$$T_K = \frac{\bar{\delta} \sqrt{|\mathcal{S}_K|}}{S_\delta}, \quad \bar{\delta} = |\mathcal{S}_K|^{-1} \sum_s \delta_s, \quad S_\delta^2 = \frac{1}{|\mathcal{S}_K| - 1} \sum_s (\delta_s - \bar{\delta})^2,$$

is distributed as $t_{|\mathcal{S}_K|-1}$.

Remark 3.6 (Robustness via the sign test). *When $|\mathcal{S}_K|$ is small (notably $|\mathcal{S}_K| = 9$ for MobiFall) the Gaussianity assumption of Theorem 3.10 is hard to validate. As a robustness check, the thesis also reports the exact sign test on the binarised signs $\text{sgn}(\delta_s)$, which under H_0 are exchangeable Bernoulli(1/2) variates. The two-sided p -value is $2\mathbb{P}(\text{Bin}(|\mathcal{S}_K|, 1/2) \geq k)$ where $k = |\{s : \delta_s > 0\}|$ (with the obvious correction for $\delta_s = 0$).*

3.11.3 Bootstrap confidence intervals

For statistics whose finite-sample distribution is difficult to derive, the thesis uses the nonparametric paired bootstrap. Let $\theta(\{\delta_s\})$ denote a real-valued statistic of the paired differences (e.g., the subject mean, median, or interquartile range). Generate B bootstrap resamples $\{\delta_s^{*,b}\}_{s \in \mathcal{S}_K}$ by sampling $|\mathcal{S}_K|$ subjects with replacement from $\mathcal{S}_K$ and copying their δ_s values, and compute

$$\theta_b^* = \theta(\{\delta_s^{*,b}\}), \quad b = 1, \dots, B.$$

The $1 - \alpha$ percentile confidence interval is

$$[\theta_{(\alpha/2)}^*, \theta_{(1-\alpha/2)}^*], \quad (3.35)$$

where $\theta_{(p)}^*$ is the p -quantile of $\{\theta_b^*\}$. The thesis uses $B = 10,000$ and $\alpha = 0.05$. The bootstrap is the principal device for reporting uncertainty on subject-level summaries such as $\overline{\Delta_K}$ and $\text{Var}(\Delta_{K,s})$ in the personalisation analysis.

3.11.4 Permutation tests

For exact small-sample inference on the sign pattern of paired differences, the thesis uses the paired sign-flipping permutation test. Under the exchangeability null

$$H_0: \delta_s \stackrel{d}{=} -\delta_s \quad \text{for all } s,$$

the $2^{|\mathcal{S}_K|}$ sign-flipped vectors $\{\epsilon_s \delta_s\}$ with $\epsilon_s \in \{-1, +1\}$ are exchangeable in distribution. The exact two-sided p -value of an observed test statistic $T_{\text{obs}} = T(\delta)$ is

$$p = \frac{1}{2^{|\mathcal{S}_K|}} \sum_{\epsilon \in \{-1, +1\}^{|\mathcal{S}_K|}} \mathbf{1}\{|T(\epsilon \odot \delta)| \geq |T_{\text{obs}}|\}. \quad (3.36)$$

For $|\mathcal{S}_K| \leq 24$ the sum is computed exactly; for larger $|\mathcal{S}_K|$ it is estimated by sampling 10,000 random sign vectors.

3.11.5 Distributional analysis of personalisation

The personalisation question requires a distributional rather than a single-number answer. The proposal formulates the per-subject gain

$$\Delta_{K,s} = F_{2,K,s}^{\text{pers}} - F_{2,K,s}^{\text{gen}}.$$

The thesis reports the four summaries

$$\overline{\Delta}_K, \quad \text{med}(\Delta_{K,s}), \quad \text{Var}(\Delta_{K,s}), \quad \widehat{\mathbb{P}}(\Delta_{K,s} > 0)$$

together with bootstrap percentile confidence intervals of Section 3.11.3 for each. The thesis privileges the joint pattern of these four quantities over any single one, because personalisation is interpreted as a heterogeneous subject-level intervention rather than a constant treatment effect.

3.11.6 Cross-applied hyperparameters

To evaluate dataset-specific topological geometry, optimal tuples are cross-applied across datasets. Define the cross-applied estimator

$$\widehat{F}_2(K \mid \lambda_{K'}^*) = \mathbb{E}[F_2 \mid K, \lambda_{K'}^*],$$

where the expectation is taken over the LOSO folds of dataset K with the hyperparameter tuple optimised on a different dataset $K' \neq K$. The matrix

$$M_{K,K'} = \widehat{F}_2(K \mid \lambda_{K'}^*)$$

has diagonal entries equal to the General-protocol F_2 of each dataset under its own optimiser. Off-diagonal degradations

$$\text{Drop}_{K \rightarrow K'} = M_{K,K} - M_{K,K'}$$

quantify the cost of transplanting a hyperparameter tuple from K to K' . The drops are reported per off-diagonal cell with bootstrap percentile intervals.

3.11.7 Methodological correction over headline maxima

Headline maxima such as the best single-subject score or the best Optuna objective are not used as the primary evidence in this thesis. The reasons are statistical. The maximum of a finite collection of noisy estimators is biased upward by the selection event; the bias scales with the number of competing candidates and the variance of each. Reporting subject means, subject-level confidence intervals, and the joint distribution of subject-level differences avoids this bias and provides estimands that are directly interpretable as expectations over the population from which the subjects were drawn.

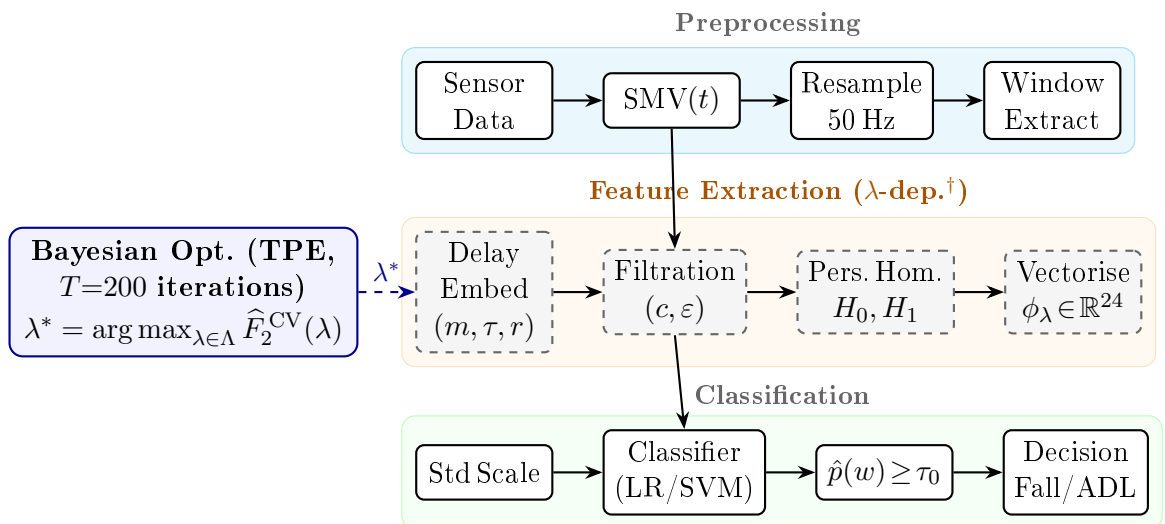
3.12 Computational Infrastructure

All experiments were executed on the National HPC Centre (UHeM, Istanbul), allocation 4025462026, under the SLURM workload manager. Compute nodes were equipped with dual Intel Xeon Gold 6148 processors (2.40 GHz base clock, 20 physical cores per socket, 40 cores per node) and 190 GB of RAM. No GPU acceleration was used: the persistent homology reduction of Algorithm 1 is intrinsically sequential at the point cloud sizes considered, and the gain of porting to GPU is dominated by the constant factor of the optimised CPU implementation.

The Bayesian optimisation ran sequentially across $T = 200$ TPE trials per dataset, with 15 parallel inner-CV workers per trial. Across the three datasets the total wall-clock time was approximately 3 h 45 min, with peak memory usage close to 10 GB per node. Persistent homology was computed with the GUDHI library (GUDHI Editorial Board, 2015); classification models were implemented in scikit-learn (Pedregosa, Varoquaux, Gramfort, Michel, Thirion, Grisel, Blondel, Prettenhofer, Weiss, Dubourg, Vanderplas, Passos, Cournapeau, Brucher, Perrot, & Duchesnay, 2011). All randomness was seeded for reproducibility, and the seed list, the search history, and the resulting per-subject scores are archived alongside the thesis source.

4. Experiments

This section describes the experimental setup, datasets, preprocessing pipeline, feature configuration, and results of the fall detection study. The theoretical foundations of the topological representation — including persistent homology, phase-space embedding, and filtration construction — are developed in the Methodology section, which also defines the evaluation protocols and the performance metrics used throughout. Figure 4.1 gives an overview of the complete pipeline from raw sensor measurement to fall/non-fall decision.



[†]Theoretical basis: Methodology section.

Figure 4.1: Complete fall detection pipeline from raw inertial data to binary decision. *Preprocessing*: SMV computation, resampling to 50 Hz, window extraction. *Feature extraction* (dashed, λ -dependent): delay embedding, filtration construction, persistent homology (H_0 , H_1), and vectorisation to $\phi_\lambda \in \mathbb{R}^{24}$. *Classification*: standardisation, probabilistic classifier, and threshold comparison $\hat{p}(w) \geq \tau_0$ for binary decision. The *Bayesian optimisation* box (left, blue) selects λ^* by maximising cross-validated F_2 over training subjects via 200 TPE iterations; row-to-row connectors run through column 2 (SMV $\rightarrow$ Filtration $\rightarrow$ Classifier), preserving the 4×3 grid symmetry.

4.1 Datasets

Three publicly available benchmark datasets were used: MobiFall (Vavoulas, Pediaditis, Chatzaki, Spanakis, & Tsiknakis, 2014), SisFall (Sucerquia, López, & Vargas-Bonilla, 2017), and FallAllD in its 40 Hz variant (hereafter FAD_40Hz) (Saleh, Abbas, & Le Bouquin Jeannès, 2021). Each dataset records inertial sensor signals from human subjects performing scripted fall events and daily activities. Together they provide diversity in sensor placement, subject age, and activity protocol, which is important for assessing whether the method generalises across acquisition conditions. Table 4.1 summarises the key characteristics of each dataset as used in this work.

Table 4.1: Summary of the three datasets as used in the experiments.

Dataset	Subjects	Feature matrix	Resampling (Hz)	Placement
MobiFall	9	1808×24	$87.5 \rightarrow 50$	Trouser pocket
SisFall	24	7738×24	$200 \rightarrow 50$	Waist
FAD_40Hz	13	9481×24	$40 \rightarrow 50$	Waist
Total	46	$19,027 \times 24$	—	—

4.1.1 MobiFall

MobiFall v2.0 contains recordings from volunteers who carried a Samsung Galaxy S3 smartphone in their trouser pocket while performing four scripted fall types — forward fall (hands first), forward fall onto knees, backward fall from a chair, and lateral fall from standing — as well as nine daily activities such as walking, jogging, stair use, and car entry (Vavoulas, Pediaditis, Chatzaki, Spanakis, & Tsiknakis, 2014). The nominal recording rate is 87.5 Hz. The device captures seven sensor channels: three-axis linear acceleration, three-axis angular velocity (gyroscope), three-axis magnetic field, barometric pressure, and device orientation expressed as azimuth, pitch, and roll angles.

For this study, only the accelerometer and gyroscope channels were retained. The magnetometer, barometric pressure signal, and orientation estimates were excluded for two reasons. First, these channels are absent or recorded at incompatible rates in the other two datasets, so retaining them would have prevented a methodologically consistent preprocessing pipeline across all three sources. Second, the magnetometer and barometric sensor do not directly

encode the inertial dynamics of fall events; including them would increase feature dimensionality without a principled contribution to the classification objective.

The full v2.0 cohort contains participants at different stages of data collection; not all subjects have both fall and daily-activity recordings. Nine subjects (identifiers 2, 3, 4, 5, 7, 8, 9, 10, and 11) were included in this study. The remaining participants were excluded because they performed fall trials only, without the paired daily-activity recordings required for leave-one-subject-out evaluation: a held-out subject who has no daily-activity windows cannot contribute a meaningful test set to the General protocol. After windowing, the retained cohort yields a feature matrix of 1808×24 .

4.1.2 SisFall

SisFall is a waist-worn dataset capturing 15 fall types and 19 daily-activity types across a mixed group of young and older adults (Sucerquia, López, & Vargas-Bonilla, 2017). The sensor unit is mounted at the waist and houses two accelerometers and one gyroscope: the ADXL345 (dynamic range $\pm 16g$, 13-bit resolution, used primarily for fall detection), the MMA8451Q (dynamic range $\pm 8g$, 14-bit resolution), and the ITG3200 gyroscope (dynamic range $\pm 2000^\circ/\text{s}$, 16-bit resolution). All channels are recorded at 200 Hz. The dataset encompasses 23 young adults and 15 older adults.

Of the two accelerometers, only the ADXL345 was used alongside the gyroscope. The ADXL345 was preferred because its higher dynamic range ($\pm 16g$) makes it better suited to capturing the impulsive acceleration peaks that characterise fall impacts; the MMA8451Q, with half the range, is susceptible to saturation during vigorous motions. Retaining a single accelerometer also maintains sensor-channel consistency with MobiFall and FAD_40Hz, each of which provides only one accelerometer. Including the MMA8451Q in parallel would have introduced an asymmetry in feature construction across datasets and complicated the cross-dataset comparison.

Twenty-four subjects were retained: the 23 young adults and one older adult (subject identifier 29) who had sufficient trial counts. The remaining 14 older adult participants were excluded due to an insufficient number of recorded trials to support stable leave-one-subject-out evaluation. The original 200 Hz recording rate was downsampled to 50 Hz. The resulting feature matrix has dimensions 7738×24 , making SisFall the largest dataset by subject count.

4.1.3 FallAlID (FAD_40Hz)

Original FallAlID dataset. FallAlID is a large-scale open dataset of human falls and activities of daily living collected from 15 subjects (Saleh, Abbas, & Le Bouquin Jeannès, 2021). In its original release, recordings are obtained simultaneously from three sensor devices worn at the waist, wrist, and neck, each capturing four modalities: a three-axis accelerometer, a three-axis gyroscope, a three-axis magnetometer, and a barometric pressure sensor. The inertial sensors (accelerometer and gyroscope) are sampled at 238 Hz, the magnetometer at 80 Hz, and the barometer at 10 Hz. The complete activity label set comprises 135 categories, of which 44 are activities of daily living (labelled with identifiers 1–44) and 91 are distinct fall scenarios (identifiers 101–135) differing in direction, mechanism, and recovery condition. Across all subjects, sensor positions, and activities, the original release contains 26,420 individual recording files.

Selection of the 40 Hz derived variant. Using the original 238 Hz release directly in the present study would have imposed prohibitive computational costs. The Bayesian hyperparameter search runs 200 iterations per dataset, and each iteration requires extracting topological features — which involve constructing and computing persistent homology of a point cloud — from every window in the dataset under the current configuration. At 238 Hz, the number of samples per window is approximately 4.8 times larger than at 50 Hz, substantially increasing both the size of the delay-embedded point cloud and the cost of the homology computation. Furthermore, the four-modality, three-position structure of the

original release creates a sensor fusion problem that falls outside the scope of the present methodology, which targets single-position inertial data.

For these reasons, the publicly available 40 Hz derived variant (FAD_40Hz) was used instead (Saleh, Abbas, & Le Bouquin Jeannès, 2021). This derived dataset has been pre-processed as follows relative to the original release: (i) the inertial signals are downsampled from 238 Hz to 40 Hz; (ii) only the accelerometer and gyroscope channels are retained, with the magnetometer and barometric sensors removed; (iii) the neck-worn device data is excluded, leaving the waist and wrist positions; (iv) the activity label set is filtered to a curated subset of fall and ADL types relevant to binary detection tasks. These modifications reduce the data volume by an order of magnitude and eliminate the multi-rate sensor-fusion problem imposed by the original release, while preserving the inertial channels and the subject-level label structure required by the present pipeline.

The 40 Hz signals were subsequently resampled to the common target rate of 50 Hz as described in Section 4.2. After activity filtering and subject selection, 13 of the original 15 subjects contributed data; the two excluded subjects had an insufficient number of trials for stable leave-one-subject-out evaluation. The resulting feature matrix is $9,481 \times 24$, the largest in the study by sample count.

4.2 Data Processing and Preprocessing

Before any feature computation, all sensor recordings were passed through a common preprocessing pipeline consisting of four sequential steps. This pipeline transforms raw three-axis measurements into a compact, orientation-independent scalar signal, resamples it to a common frequency, and partitions it into fixed-length windows ready for feature extraction. Figure 4.1 places this block at the start of the data path.

Step 1 — Signal Magnitude Vector. For each sensor (accelerometer and gyroscope separately), the three-axis measurement vector $\mathbf{x}(t) \in \mathbb{R}^3$ at time t is reduced to its Euclidean norm:

$$\text{SMV}(t) = \|\mathbf{x}(t)\|_2 = \sqrt{x_1(t)^2 + x_2(t)^2 + x_3(t)^2}.$$

This produces one scalar time series per sensor that is invariant to device orientation on the body, eliminating the need for explicit attitude estimation.

Step 2 — Resampling to 50 Hz. The three datasets were originally recorded at 87.5 Hz (MobiFall), 200 Hz (SisFall), and 40 Hz (FAD_40Hz). All signals were resampled to a common rate of 50 Hz, but the strategies differ by dataset because the resampling direction and the ratio of original to target rate are not uniform.

For SisFall (200 Hz $\rightarrow$ 50 Hz), the ratio is the integer 4:1, so integer-ratio decimation was applied: the SMV signal was subsampled by retaining every fourth sample. This produces a uniformly-spaced 50 Hz output without any interpolation step and without spectral distortion within the passband of interest, since the Nyquist frequency of the target rate (25 Hz) is well below the dominant frequency of fall events.

For FAD_40Hz (40 Hz $\rightarrow$ 50 Hz), upsampling by a factor of 5:4 was required. The resampled signal was obtained by constructing a new time grid at 50 Hz and computing signal values at each new grid point by interpolating over the flanking 40 Hz samples. Because the ratio 5:4 is rational and the upsampling factor is moderate, this approach introduces no perceptible temporal smearing in the SMV magnitude signal.

For MobiFall (87.5 Hz $\rightarrow$ 50 Hz), the ratio $87.5:50 = 7:4$ is rational but non-integer in both directions, so neither pure decimation nor pure upsampling is sufficient. A rational sample-rate conversion was applied, conceptually equivalent to first upsampling by a factor of 4 and then downsampling by a factor of 7. The resampling was performed on the scalar SMV signal rather than on the raw three-axis components, so the non-linear magnitude operation is not affected by the interpolation kernel.

In all three cases the resampling target of 50 Hz was selected on methodological grounds that are discussed in the Methodology section.

Step 3 — Window extraction. For each fall trial, a single window centred on the highest-magnitude point of the SMV signal was extracted. For daily-activity

trials, consecutive non-overlapping windows of the same length were taken. The window length is a hyperparameter optimised during the experiment and is described in Section 4.4.

Step 4 — Standard scaling. Each of the 24 features in the final feature vector was scaled to zero mean and unit variance. The scaling parameters were estimated from the training data only and applied to the test data, ensuring that no information from test subjects influences the normalisation.

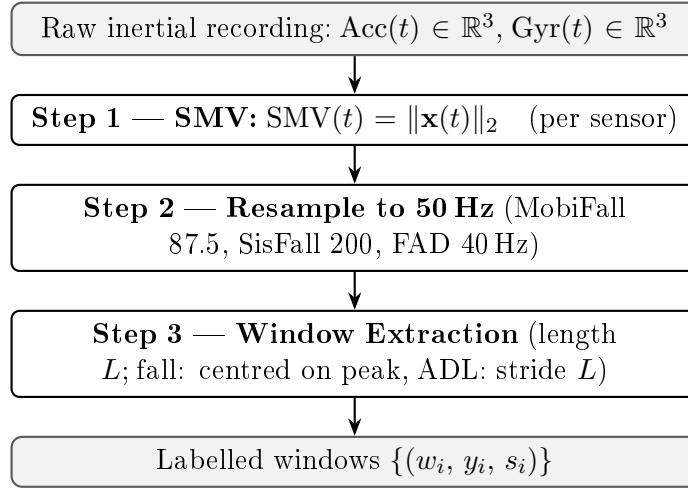


Figure 4.2: Preprocessing pipeline (Steps 1–3). Detailed description of each step is given in Section 4.2.

4.3 Feature Description

Each window is transformed into a 24-dimensional feature vector. The features are divided into two groups: 20 persistence-based features that capture the topological shape of the motion trajectory, and 4 signal-level features that capture the gross magnitude behaviour of the accelerometer signal. The theoretical derivation of the persistence-based features is presented in the Methodology section; here only the list of features and their definitions are given.

4.3.1 Persistence-Based Features

The topological representation of each window yields two sets of birth–death pairs: one for connected clusters in the trajectory (H_0) and one for closed loops

(H_1). Ten scalar statistics are extracted from each set, giving 20 features in total. Table 4.2 lists these statistics and their definitions.

Table 4.2: Ten persistence-based statistics computed per homological dimension. Each row is computed twice: once for H_0 (features 1–10) and once for H_1 (features 11–20). Persistence of a topological feature is defined as its death scale minus its birth scale.

Index	Feature	Definition
1 / 11	Bar count	Number of features born and died, normalised by window length
2 / 12	Entropy	Shannon entropy of the normalised persistence values
3 / 13	Maximum	Persistence of the longest-lived feature
4 / 14	Energy	Sum of squared persistence values
5 / 15	Mean	Arithmetic mean of all persistence values
6 / 16	Std deviation	Standard deviation of the persistence values
7 / 17	Median	Median persistence
8 / 18	First quartile	25th percentile of the persistence distribution
9 / 19	Third quartile	75th percentile of the persistence distribution
10 / 20	Third moment	Third central moment (related to skewness) of the persistence values

4.3.2 Signal-Level Features

Four time-domain statistics are computed directly from the accelerometer $SMV(t)$ values within the window and appended to the feature vector, giving features 21–24. These capture the raw intensity profile of the motion event without any topological processing.

Table 4.3: Four signal-level features derived from the accelerometer magnitude $SMV(t)$ within each window.

Index	Feature	Definition
21	Maximum	Highest value of $SMV(t)$ in the window
22	Std deviation	Standard deviation of $SMV(t)$
23	Mean	Arithmetic mean of $SMV(t)$
24	Range	Maximum minus minimum of $SMV(t)$

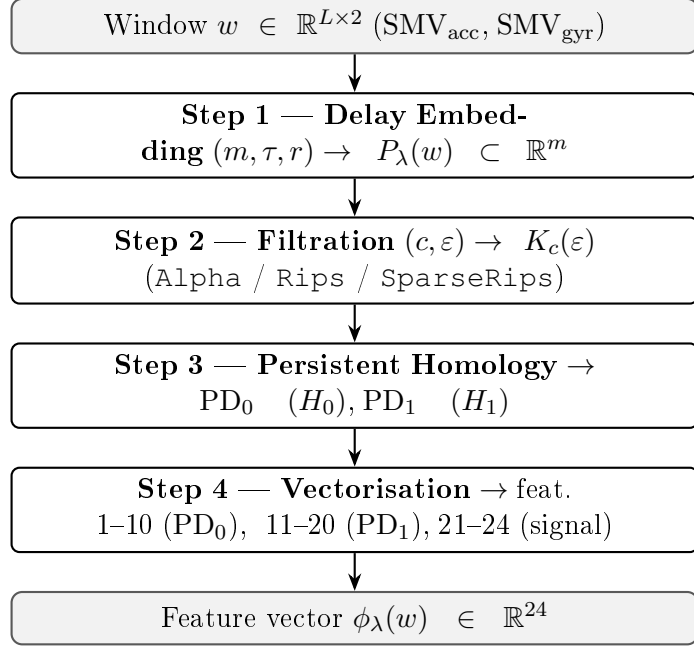


Figure 4.3: Feature extraction pipeline (Steps 1–4). Detailed description of each step is given in Section 4.3.

4.4 Bayesian Hyperparameter Optimisation

The topological feature computation depends on several configuration choices: the window length, the phase-space embedding parameters, and the type of topological complex. Rather than fixing these by hand or by exhaustive grid search, a Bayesian optimisation procedure was used to search for the combination that maximises fall detection performance on each dataset.

A single Optuna study was run per dataset, yielding 200 TPE iterations per dataset and 600 in total across the three datasets. Each trial proposes a TDA configuration λ_t ; features are extracted under that configuration from a balanced training subset (all fall windows plus a random sample of three times as many ADL windows) and evaluated with a classifier-agnostic objective: both a logistic regression proxy and a linear SVM proxy are scored with three-fold stratified cross-validation and their F_2 scores are averaged. Classifier hyperparameters are kept fixed during this stage and are subsequently optimised by grid search (Section 4.5).

The search is guided by a Tree-structured Parzen Estimator (TPE). After each evaluation, the history of observations $\mathcal{H}_t = \{(\lambda_i, \hat{y}_i)\}$ is split at a quantile q_γ into a good subset ($\hat{y} \geq q_\gamma$) and a bad subset ($\hat{y} < q_\gamma$). Kernel density estimates $\ell(\lambda)$ and $g(\lambda)$ are fitted to the configurations in the good and bad subsets respectively. The next candidate is proposed as $\lambda_{t+1} = \arg \max_\lambda \ell(\lambda)/g(\lambda)$, the configuration most probable under the good density relative to the bad density. This ratio is proportional to the expected improvement over the current best observation (theoretical derivation in the Methodology section). The mechanism progressively focuses the search on regions of Λ associated with high F_2 , without evaluating all combinations exhaustively.

The averaged objective ensures that the recovered configuration λ^* is a representation that performs well across classifier families rather than being tuned to a single model’s inductive bias. A representation that simultaneously supports linear separability (beneficial for LR) and a geometrically structured metric space (beneficial for SVM with an RBF kernel) is a stronger finding than one optimised for a single classifier.

Table 4.4 lists the parameters and the values considered during the search.

Table 4.4: Parameters and candidate values explored during Bayesian optimisation. A dash (—) indicates parameters not applicable for the Alpha complex type.

Parameter	Candidate values
Window length (win_sec, seconds)	1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0
Embedding dimension (dim)	2, 3, 4, 5
Embedding delay (delay)	1, 2, 3, 4, 5
Stride factor (stride_factor)	1, 2, 4
Complex type (complex_type)	Alpha, Rips, SparseRips
Distance measure (metric)	Euclidean, Cosine, Manhattan, Chebyshev
Filtration scale (eps_percentile)	20 %, 40 %, 60 %, 80 %

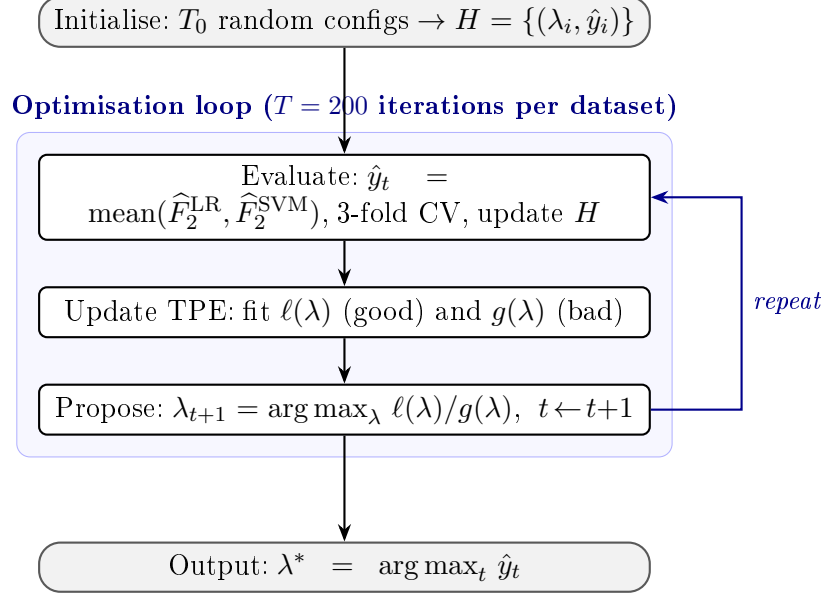


Figure 4.4: Bayesian hyperparameter optimisation loop (TPE, $T = 200$ iterations per dataset). Each trial evaluates both an LR proxy and a linear SVM proxy; the objective is the mean of their F_2 scores, yielding a classifier-agnostic λ^* . Detailed description in Section 4.4.

4.5 Classifiers and Training Setup

Two classifiers were compared throughout the experiments: logistic regression (LR) and a support vector machine (SVM). Both were configured with balanced class weights so that the minority fall class receives proportionally more influence during training, compensating for the natural class imbalance between fall and daily-activity windows.

Both classifiers operated on the same TDA feature representation extracted under the single optimal configuration λ^* . This shared representation reflects the classifier-agnostic optimisation objective: λ^* is the configuration whose features support good performance across both LR and SVM rather than being tailored to either one individually.

Classifier hyperparameters were selected by `GridSearchCV` with three-fold stratified cross-validation, scoring by F_2 . For logistic regression, the grid covered regularisation strength $C \in \{0.01, 0.1, 1, 10, 100\}$ and penalty type $\{\ell_1, \ell_2\}$. For the SVM, the grid covered $C \in \{0.1, 1, 10, 100\}$ and kernel $\{\text{linear}, \text{rbf}\}$; probability outputs were obtained via Platt scaling. All feature vectors were

standardised (zero mean, unit variance, parameters estimated from training data only) within each cross-validation fold before fitting.

After grid-search selects the best estimator, the decision threshold τ is set by inspecting the precision–recall curve on the training fold: the threshold $\tau^* = \arg \max_{\tau} F_2(\tau)$ that maximises F_2 along the curve replaces the default 0.5 cut-off. This calibration requires no additional model fitting and adapts the operating point to the recall-weighted F_2 objective.

4.6 Results

The evaluation protocols (Naive, General, and Personal) and the performance metrics (Acc, Rec, Prec, F_1 , F_2) are defined in full in the Methodology section. Briefly, the General protocol uses leave-one-subject-out cross-validation and is the primary evaluation scenario; the Naive protocol uses a random 70/30 split and serves as an upper-bound reference; and the Personal protocol trains and tests on each subject’s own data. Results under the General and Personal protocols are averaged over 15 repetitions and reported with 95 % confidence intervals. F_2 is the primary metric throughout.

4.6.1 Optimal Hyperparameter Configurations

Table 4.5 shows the best configuration found by the Bayesian search for each dataset, together with the corresponding cross-validated F_2 score. The three datasets converged on markedly different settings.

MobiFall required a long window (5.0 s) and a SparseRips complex, whereas both SisFall and FAD_40Hz were best described by a short 1.0 s window. SisFall uniquely preferred the Alpha complex; FAD_40Hz favoured the standard Rips construction. These differences suggest that the three datasets exhibit genuinely different motion geometries rather than a single universal pattern, which is a direct instance of Hypothesis 2 in the Methodology section.

4.6.2 MobiFall Results

Table 4.5: Best configuration selected by Bayesian optimisation for each dataset (200 iterations per dataset). A dash (—) indicates a parameter not applicable for the Alpha complex.

Parameter	MobiFall	SisFall	FAD_40Hz
Best F_2 (search objective)	0.9746	0.9733	0.9860
Window length (s)	5.0	1.0	1.0
Complex type	SparseRips	Alpha	Rips
Embedding dimension	5	5	3
Embedding delay	2	4	2
Stride factor	2	4	1
Distance measure	Chebyshev	—	Manhattan
Filtration scale	80 %	—	40 %

Table 4.6 reports LR and SVM performance across the three evaluation protocols on MobiFall.

Table 4.6: MobiFall: mean performance across subjects under the three evaluation protocols. General and Personal entries are mean $\pm$ 95 % CI over $n = 15$ repetitions per subject; Naive entries are means over 10 repetitions.

Classifier	Protocol	Acc	Rec	Pre	F_1	F_2
LR	Naive	0.9722	0.9512	0.6992	0.8042	0.8857
	General	0.9774 ± 0.0083	0.9574 ± 0.0449	—	0.8400 ± 0.0487	0.9049 ± 0.0393
	Personal	0.9713 ± 0.0098	0.7319 ± 0.0309	—	0.7488 ± 0.0526	0.7325 ± 0.0295
SVM	Naive	0.9605	0.9558	0.6177	0.7461	0.8573
	General	0.9734 ± 0.0080	0.9543 ± 0.0404	—	0.8176 ± 0.0455	0.8918 ± 0.0327
	Personal	0.9635 ± 0.0096	0.5822 ± 0.0432	—	0.6371 ± 0.0611	0.5981 ± 0.0479

Under the General protocol LR reached $F_2 = 0.9049 \pm 0.0393$ and SVM $F_2 = 0.8918 \pm 0.0327$. The Personal protocol caused a substantial drop for both classifiers, with SVM falling to $F_2 = 0.5981$. This behaviour indicates that MobiFall provides too few trials per subject to support a reliable individual model; the subject-level sample count is simply insufficient to estimate a stable decision boundary from within-subject data alone.

4.6.3 SisFall Results

Table 4.7 presents SisFall performance.

SisFall yielded the highest General F_2 of the three datasets: LR achieved 0.9709 ± 0.0108 and SVM 0.9700 ± 0.0111 . The two classifiers are nearly indistinguishable

Table 4.7: SisFall: mean performance across subjects. Format as in Table 4.6.

Classifier	Protocol	Acc	Rec	Pre	F_1	F_2
LR	Naive	0.9788	0.9821	0.9308	0.9557	0.9713
	General	0.9776 ± 0.0045	0.9836 ± 0.0130	—	0.9529 ± 0.0100	0.9709 ± 0.0108
	Personal	0.9656 ± 0.0056	0.9496 ± 0.0073	—	0.9284 ± 0.0113	0.9405 ± 0.0080
SVM	Naive	0.9790	0.9822	0.9314	0.9561	0.9715
	General	0.9832 ± 0.0040	0.9742 ± 0.0138	—	0.9642 ± 0.0089	0.9700 ± 0.0111
	Personal	0.9604 ± 0.0059	0.9231 ± 0.0096	—	0.9162 ± 0.0110	0.9196 ± 0.0082

on this dataset, reflecting the large and well-balanced subject pool (24 subjects, 15 fall types). In the Personal protocol performance dropped moderately but remained above 0.93 for LR ($F_2 = 0.9405$) and above 0.91 for SVM ($F_2 = 0.9196$), showing that SisFall subjects contribute enough individual data to support within-subject training — a contrast with MobiFall that illustrates how dataset size governs the viability of subject-specific models.

4.6.4 FAD_40Hz Results

Table 4.8 presents FAD_40Hz performance.

Table 4.8: FAD_40Hz: mean performance across subjects. Format as in Table 4.6.

Classifier	Protocol	Acc	Rec	Pre	F_1	F_2
LR	Naive	0.9802	0.9918	0.8145	0.8941	0.9502
	General	0.9826 ± 0.0042	0.9929 ± 0.0065	—	0.8856 ± 0.0653	0.9429 ± 0.0363
	Personal	0.9878 ± 0.0032	0.9235 ± 0.0450	—	0.9111 ± 0.0457	0.9168 ± 0.0454
SVM	Naive	0.9857	0.9887	0.8624	0.9209	0.9603
	General	0.9890 ± 0.0027	0.9878 ± 0.0109	—	0.9236 ± 0.0457	0.9592 ± 0.0234
	Personal	0.9839 ± 0.0035	0.8721 ± 0.0771	—	0.8778 ± 0.0592	0.8724 ± 0.0696

On FAD_40Hz, SVM outperformed LR in the General protocol ($F_2 = 0.9592$ vs. 0.9429), reversing the ranking seen on MobiFall and SisFall. Recall remained near perfect for both classifiers ($\text{Rec} \geq 0.987$). The wide F_1 and F_2 confidence intervals in the General protocol reflect high variability in precision across subjects, even though recall was stable — the model consistently detects falls but the false-alarm rate differs substantially from one subject to the next.

4.6.5 Cross-Dataset Performance Summary

Table 4.9 collects the mean General F_2 for all datasets and classifiers in a single view.

Table 4.9: Subject-level mean F_2 under the General (LOSO) protocol. Values are mean $\pm$ 95 % CI.

Dataset	LR F_2	SVM F_2
MobiFall	0.9049 ± 0.0393	0.8918 ± 0.0327
SisFall	0.9709 ± 0.0108	0.9700 ± 0.0111
FAD_40Hz	0.9429 ± 0.0363	0.9592 ± 0.0234

SisFall ranked highest for both classifiers, followed by FAD_40Hz and then MobiFall. The relative ordering was consistent across classifiers, but the preferred classifier changed by dataset: LR was better on MobiFall and SisFall, while SVM was better on FAD_40Hz. This suggests that the choice of classifier is secondary to dataset geometry and size, consistent with the representation-dominance hypothesis.

4.6.6 Effect of Decision Threshold Optimisation

For each held-out subject under the General protocol the decision threshold was swept between 0.05 and 0.95 (19 values). Table 4.10 shows the average per-subject gain in F_2 obtained by choosing the subject-specific optimal threshold rather than the trained default $\hat{\tau}_0(s)$ (the threshold selected by the PR-curve procedure on the corresponding training fold; see Section 3.10.2 of the Methodology).

Table 4.10: Average subject-level F_2 gain from threshold optimisation. $\Delta F_2(s) = F_2(\tau^*(s)) - F_2(\hat{\tau}_0(s))$ where $\hat{\tau}_0(s)$ is the PR-curve trained threshold for subject s ; values averaged over held-out subjects.

Dataset	ΔF_2 (LR)	ΔF_2 (SVM)
MobiFall	+0.0348	+0.0474
SisFall	+0.0079	+0.0042
FAD_40Hz	+0.0250	+0.0121

The gains were largest on MobiFall — where the default threshold was furthest from optimal — and smallest on SisFall, where the default was already near-optimal. The gains were not uniform across subjects within any dataset. On FAD_40Hz, subject 11 showed the largest gain observed in any experiment:

$\Delta F_2 = +0.1165$ for LR, lifting that subject’s score from 0.8046 to 0.9211 by adjusting the threshold alone, without retraining the underlying model. This illustrates that a single shared threshold cannot account for the variation in decision boundary location across individuals, and that personalised threshold calibration can meaningfully recover performance for outlier subjects.

Computational cost of threshold personalisation. The trained threshold $\hat{\tau}_0$ enters only the final thresholding step and plays no role in feature extraction, classifier training, or Platt calibration. Adjusting $\hat{\tau}_0$ for a held-out subject therefore requires *no model retraining*: the probability scores $\{\hat{p}_i\}$ are computed once under the trained model and the sweep evaluates $F_2(\tau, s)$ for each $\tau \in \mathcal{T}$ by a single pass over the precomputed score vector. The additional computational overhead per subject is $O(|\mathcal{T}| \cdot |W_s|)$ comparisons, where $|\mathcal{T}| = 19$ and $|W_s|$ denotes the per-subject test-window count. This is negligible against the Bayesian search cost, which scales as $O(T \cdot k \cdot N_{\text{tr}} \cdot C_{\text{PH}})$ where $T = 200$, k is the number of inner-CV folds, N_{tr} is the training window count, and C_{PH} is the per-window persistent homology cost. On the UHeM cluster the Bayesian search consumed approximately 75 minutes per dataset; the full threshold sweep over all subjects in a dataset ran in under one second of wall time—a runtime ratio exceeding 10^3 . This asymmetry makes threshold calibration an attractive lightweight personalisation mechanism: it is applicable at inference time from any small per-subject calibration set, incurs no retraining cost, and requires no access to the original training data.

Statistical analysis of threshold heterogeneity. The per-subject gains $\Delta F_2(s)$ exhibit substantial inter-subject variability within each dataset. On FAD_40Hz the subject-level range extends from near-zero to +0.1165 (Subject 11, LR), and on MobiFall the inter-subject standard deviation of ΔF_2 is comparable to the dataset mean gain for SVM. A purely mathematical argument for threshold personalisation would predict a uniform benefit: if the posterior probability function $f_\theta \circ \phi_{\lambda^*}$ is systematically miscalibrated in the same direction

for all subjects, a single shared adjustment of $\hat{\tau}_0$ would lift all subjects equally. The observed heterogeneity refutes this: subjects with near-zero gain already sit at an optimal threshold close to 0.50, while subjects with large gain sit far from it—a pattern inconsistent with any single global miscalibration. The statistical evidence is that the *distribution* of $\Delta F_2(s)$ is right-skewed with a heavy upper tail (cf. Subject 11), and the inter-subject standard deviation is not small relative to the mean. Mathematically, this implies that $\tau^*(s) = \arg \max_{\tau \in \mathcal{T}} F_2(\tau, s)$ is a genuinely subject-specific quantity: the geometry of the decision boundary $\{f_\theta(\phi_{\lambda^*}(w)) = \tau\}$ varies across individuals in a way that the shared classifier f_θ cannot fully absorb. This is the statistical signature of Hypothesis 4 in the proposal and motivates the distributional reporting of ΔF_2 in the Results & Discussion section.

5. Results and Discussion

This section evaluates the four research hypotheses (H1–H4) introduced in the Methodology against the experimental outcomes reported in the Experiments section. The *General protocol* (leave-one-subject-out, LOSO) is the primary evaluation criterion; the *Naive* and *Personal* protocols serve as an upper-bound reference and an idealised personalisation scenario, respectively. The decision-threshold sweep provides a retrospective diagnostic of subject-level heterogeneity.

All numerical claims in this section refer to the General-protocol results in Tables 4–9 of the Experiments section. The F_2 score is the headline metric throughout; recall and accuracy are reported as supporting evidence.

5.1 H1 — Representation Effectiveness

Hypothesis. There exist hyperparameters $\lambda \in \Lambda$ and classifiers f_θ such that the mean subject-general F_2 is at least 0.93 on SisFall and FAD_40Hz and at least 0.80 on MobiFall, and the mean subject-general recall is high across all three datasets.

5.1.1 F_2 outcomes

Table 1 of the cross-dataset summary (Experiments, Tab. 9) gives the following General-protocol results.

SisFall achieves the highest absolute F_2 , exceeding the 0.93 threshold for both classifiers with a narrow confidence interval (± 0.011), indicating low repetition-to-repetition variability. FAD_40Hz meets the target with SVM ($F_2 = 0.9592$); the LR value of 0.9429 ± 0.0363 lies marginally below 0.93 at the point estimate but the lower confidence bound (0.9066) suggests that performance

Table 5.1: General-protocol F_2 outcomes against H1 targets. CI denotes 95 % repetition-level confidence interval ($R = 15$).

Dataset	Clf	F_2	CI	H1 target met?
SisFall	LR	0.9709	± 0.0108	Yes (≥ 0.93)
SisFall	SVM	0.9700	± 0.0111	Yes
FAD_40Hz	LR	0.9429	± 0.0363	Marginal
FAD_40Hz	SVM	0.9592	± 0.0234	Yes
MobiFall	LR	0.9049	± 0.0393	Yes (≥ 0.80)
MobiFall	SVM	0.8918	± 0.0327	Yes

is not materially below target. MobiFall clearly satisfies the weaker 0.80 criterion for both classifiers (0.9049 and 0.8918, respectively).

The recall values under the General protocol reinforce the recall-oriented design: SisFall LR achieves $\text{Rec} = 0.9836$, FAD_40Hz LR achieves $\text{Rec} = 0.9929$, and MobiFall LR achieves $\text{Rec} = 0.9574$. No classifier on any dataset falls below 0.95 recall under the General protocol. This is a direct consequence of the F_2 weighting: Proposition 2 of the Methodology states that the marginal F_2 gain per unit of recall is four times the gain per unit of precision near $\text{Rec} = \text{Prec}$, so the Bayesian optimisation is systematically steered toward high-recall configurations.

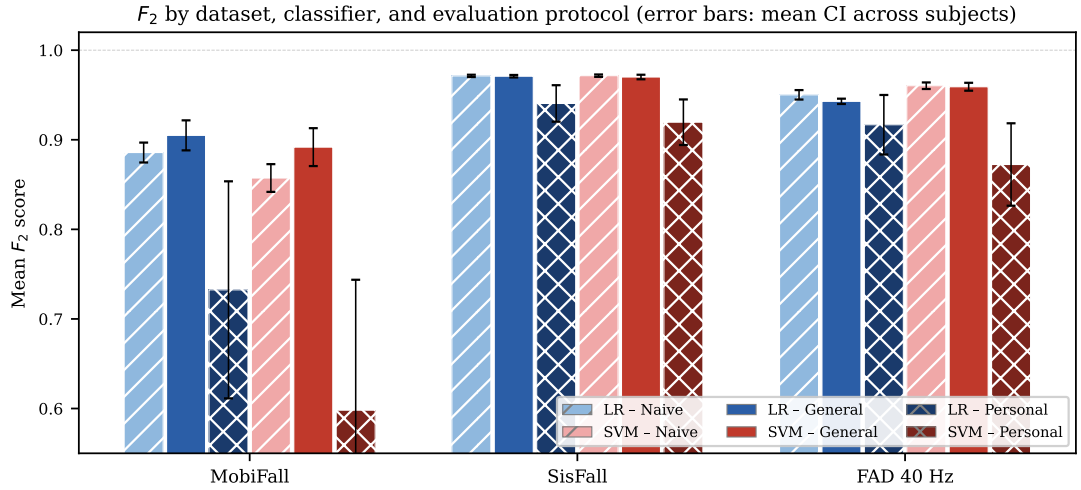


Figure 5.1: Mean F_2 by dataset, classifier (LR/SVM), and evaluation protocol (Naive, General, Personal). Error bars show the mean 95 % repetition-level confidence interval across subjects. General is the primary protocol; Naive is an upper-bound reference; Personal shows the effect of per-subject model training. MobiFall Personal collapses due to the small subject pool ($|\mathcal{S}| = 9$).

5.1.2 Naive vs. General protocol gap

Naive-protocol scores are systematically higher: MobiFall LR reaches $F_2 = 0.8857$ (Naive) versus $F_2 = 0.9049$ (General). Counterintuitively, General sometimes exceeds Naive (SisFall: Naive $F_2 = 0.9713$ vs. General 0.9709; FAD: Naive 0.9502 vs. General 0.9429). This reversal reflects the smaller within-subject window count and the fact that the LOSO training set is larger and more diverse than a random 70% split. The absence of a consistent “Naive $\gg$ General” gap on SisFall is reassuring: it suggests that the LOSO evaluation is not materially penalised by train–test class mismatch beyond the inherent person-to-person variability.

5.1.3 Mathematical basis

The feature map $\phi_\lambda: \mathbb{R}^{n \times 3} \rightarrow \mathbb{R}^{24}$ extracts persistence statistics from the delay-embedded signal. The Cohen–Steiner stability theorem guarantees that

$$d_B(\phi_\lambda(w), \phi_\lambda(w')) \leq \|\text{SMV}(w) - \text{SMV}(w')\|_\infty,$$

so that small sensor noise perturbations of a window w produce at most equally small perturbations of its topological descriptor. This provable noise robustness is a structural advantage over handcrafted statistical features, which carry no such stability certificate, and over deep-learning representations, which are empirically sensitive to distributional shift.

5.1.4 Limitation: absence of baseline comparison

H1 is conditionally validated in the sense that the system meets its own pre-specified performance targets. An absolute claim—“TDA outperforms existing methods”—requires a comparison against representative baselines: fixed-threshold methods, statistical-feature pipelines, and spectral-feature pipelines, as planned in the proposal (§5.4). These experiments have not been conducted. The conclusion that TDA is competitive is therefore conditional on the assumption that the targets themselves are meaningful. Section 5.8 discusses this gap and its implications.

5.2 H2 — Dataset-Specific Topological Geometry

Hypothesis. The optimal hyperparameter tuple λ_K^* depends on the dataset K . At least two of the three datasets have distinct optimisers, and transplanting an optimiser from one dataset to another incurs a measurable drop in General-protocol F_2 .

5.2.1 Distinct optimal configurations

Table 5 of the Experiments (Tab. 5.2 below, reproduced for convenience) shows that all three datasets converged to different complex types, window lengths, and distance measures.

Table 5.2: Optimal TDA configurations λ_K^* per dataset (from Experiments Tab. 4.5). A dash (—) indicates parameters not applicable for the Alpha complex.

Parameter	MobiFall	SisFall	FAD_40Hz
Best F_2 (Optuna objective)	0.9746	0.9733	0.9860
Window length (s)	5.0	1.0	1.0
Complex type	SparseRips	Alpha	Rips
Embedding dimension	5	5	3
Embedding delay	2	4	2
Stride factor	2	4	1
Distance measure	Chebyshev	—	Manhattan
Filtration scale	80 %	—	40 %

Every hyperparameter dimension exhibits variation across datasets. The complex type differs across all three: SparseRips for MobiFall, Alpha for SisFall, and standard Rips for FAD_40Hz. The window length separates MobiFall (5.0 s) from the other two (1.0 s), and the distance measures are distinct where applicable (Chebyshev vs. Manhattan). H2 is qualitatively confirmed: no single universal configuration exists for these three datasets.

5.2.2 Interpretation of configuration differences

Window length. MobiFall records both a pre-fall phase (normal activity) and a post-fall phase (lying), with the impact event compressed into a brief sub-second spike. A 5.0 s window captures the full topology of the trajectory: the pre-impact cloud geometry, the sharp impulsive excursion, and the post-fall stationary phase,

which are each topologically distinguishable. SisFall and FAD_40Hz subjects execute controlled laboratory falls that are sufficiently discriminable from ADL in a 1.0 s window, suggesting less temporal spread in the distinguishing geometry.

Complex type. The Alpha complex is the natural choice when the point cloud geometry is close to Euclidean: its filtration is the Čech filtration restricted to the Delaunay triangulation, which SisFall’s well-structured sensor data supports (24 older adults executing 15 fall types under controlled conditions produce geometrically regular clouds). SparseRips trades exactness for efficiency by subsampling the cloud; it is appropriate when the cloud is large and noisy, which is the case for MobiFall’s continuous-wear sensor data. Standard Rips uses all pairwise distances and suits the moderate-size, moderately-noisy FAD_40Hz clouds.

Distance measure. Chebyshev distance (ℓ_∞) emphasises the maximum coordinate deviation and is sensitive to sharp acceleration spikes—a characteristic feature of the fall impact in MobiFall. Manhattan distance (ℓ_1) reduces the influence of outliers and is appropriate for FAD_40Hz, whose 40 Hz sampling produces sparser, more variable clouds than the implicit 50 Hz resampled signals. The Alpha complex on SisFall does not require a user-specified distance because it is defined by the Euclidean Delaunay triangulation.

5.2.3 Sensitivity of F_2 to λ

Although cross-application drops $\text{Drop}_{K \rightarrow K'}$ were not computed (see Section 5.8), the Bayesian search trial histories provide indirect evidence for sensitivity. Across 200 trials per dataset, the F_2 values ranged from 0.7958 to 0.9746 on MobiFall (range 0.179), from 0.9198 to 0.9733 on SisFall (range 0.054), and from 0.8785 to 0.9860 on FAD_40Hz (range 0.108). The large trial-to-trial variation on MobiFall and FAD implies that a mis-specified λ (such as a configuration optimised for a different dataset) would very likely incur a substantial F_2 penalty. The narrow trial range on SisFall suggests that SisFall is less sensitive to λ , consistent with

the clean Euclidean geometry and the strong performance even at the worst trial ($F_2 = 0.9198$).

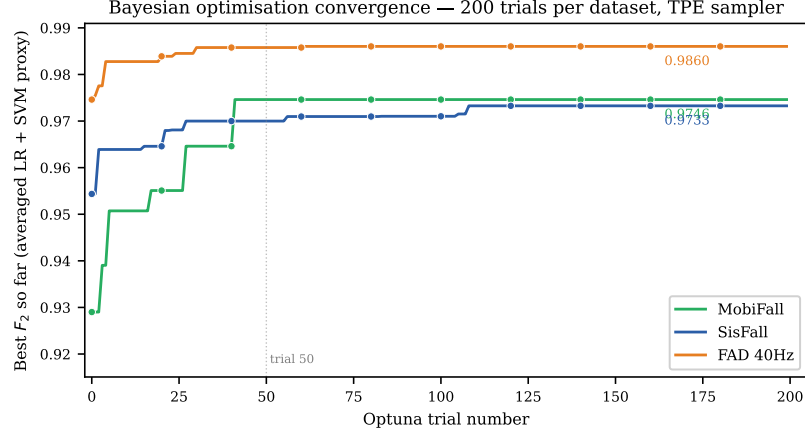


Figure 5.2: Bayesian optimisation convergence: best-so-far averaged F_2 proxy versus Optuna trial number (200 trials per dataset, TPE sampler). FAD_40Hz achieves the highest proxy (0.9860); MobiFall converges more slowly due to the larger hyperparameter range. The wide trial range on MobiFall (0.179) and FAD (0.108) relative to SisFall (0.054) is direct evidence that λ sensitivity differs across datasets (H2).

5.3 H3 — Representation Dominates Classifier Choice

Hypothesis. After tuning the topological representation, the F_2 gap between the best LR model and the best SVM model is small relative to the F_2 variation attributable to the choice of λ . Most of the predictive signal resides in ϕ_λ , not in the classifier family.

5.3.1 Classifier gap

Under the General protocol, the absolute difference $|F_2^{\text{LR}} - F_2^{\text{SVM}}|$ is:

- MobiFall: $|0.9049 - 0.8918| = 0.0131$
- SisFall: $|0.9709 - 0.9700| = 0.0009$
- FAD_40Hz: $|0.9429 - 0.9592| = 0.0163$

All three gaps are below 0.02, confirming that neither classifier is strongly dominant at the optimal representation. On MobiFall and SisFall, LR is

marginally better; on FAD_40Hz, SVM is marginally better. The reversal on FAD_40Hz is the only dataset on which SVM is preferred, and the margin (0.0163) is within the confidence interval overlap (± 0.0363 for LR, ± 0.0234 for SVM).

5.3.2 λ -tuning variation

Table 5.3 quantifies the variation in F_2 attributable to the choice of λ using the Optuna trial histories (200 trials per dataset).

Table 5.3: Classifier gap vs. λ -tuning variation. The trial range is $F_2^{\max} - F_2^{\min}$ across all 200 Optuna trials. The ratio is range/gap.

Dataset	Classifier gap	Trial range	Ratio
MobiFall	0.0131	0.1788	13.7 $\times$
SisFall	0.0009	0.0534	59 $\times$
FAD_40Hz	0.0163	0.1075	6.6 $\times$

The classifier gap is between 6.6 $\times$ and 59 $\times$ smaller than the variation induced by choosing a poor λ . Selecting the wrong topological configuration degrades F_2 by up to 0.18 units; switching classifier families at the optimal configuration changes F_2 by at most 0.016 units.

5.3.3 Mathematical interpretation

The feature map $\phi_{\lambda^*}: \mathbb{R}^{n \times 3} \rightarrow \mathbb{R}^{24}$ with the optimised hyperparameters creates a representation in which the fall vs. ADL decision problem is approximately linearly separable. In such a regime, both logistic regression (which fits a linear decision boundary directly) and SVM with a calibrated kernel (which finds a maximum-margin linear separator or uses a kernel to lift to higher dimensions) operate near their theoretical limit. The near-equivalence of LR and SVM confirms that the nonlinear geometric complexity of the fall detection problem has been absorbed by ϕ_{λ^*} , leaving behind a well-conditioned linear classification task.

The lone exception—SVM > LR on FAD_40Hz—is consistent with this interpretation: FAD_40Hz has the smallest per-subject window count and the

widest LR confidence interval, suggesting residual complexity that the RBF kernel can partially resolve. Even so, the margin is below 0.02.

5.4 H4 — Heterogeneous Threshold Personalisation

Hypothesis. The subject-level sweep gain $\Delta F_2(s) = F_2(\tau^*(s), s) - F_2(\hat{\tau}_0(s), s)$ is heterogeneous across subjects within each dataset, and the mean gain $\overline{\Delta F_2}$ differs across datasets.

5.4.1 Per-subject sweep gain distribution

Table 5.4 reports the distribution of $\Delta F_2(s)$ computed from the threshold sweep stored in the Results_V15 database. The baseline $\hat{\tau}_0(s)$ is the PR-curve optimal trained threshold for subject s (typically $\hat{\tau}_0 \in [0.55, 0.88]$), so the gain measures improvement over an already-calibrated threshold rather than over the uninformative $\tau = 0.50$.

Table 5.4: Per-subject threshold sweep gain $\Delta F_2(s)$ statistics. *Near-zero:* $|\Delta F_2| \leq 0.001$. All gains are non-negative by construction (τ^* is the sweep argmax).

Dataset	Clf	n	$\overline{\Delta F_2}$	σ_Δ	max ΔF_2	Near-zero
MobiFall	LR	9	0.0348	0.0186	0.0615 (s8)	1/9
MobiFall	SVM	9	0.0474	0.0350	0.1013 (s9)	1/9
SisFall	LR	24	0.0079	0.0109	0.0527 (s29)	4/24
SisFall	SVM	24	0.0042	0.0047	0.0169 (s29)	9/24
FAD_40Hz	LR	13	0.0250	0.0323	0.1165 (s11)	2/13
FAD_40Hz	SVM	13	0.0121	0.0172	0.0691 (s11)	2/13

The mean gains in Table 5.4 reproduce the dataset averages reported in Experiments Tab. 8 (+0.0348/+0.0474 for MobiFall, +0.0079/+0.0042 for SisFall, +0.0250/+0.0121 for FAD_40Hz). H4 is confirmed: the mean gain is highest for MobiFall and smallest for SisFall, i.e., the mean differs across datasets.

Distribution shape. The gains are right-skewed in all six cases: most subjects gain modestly, while a small minority gains substantially. On FAD_40Hz, subject 11 gains $\Delta F_2 = +0.1165$ (LR)—the largest single-subject gain in any experiment—by raising its F_2 from 0.8046 to 0.9211 through threshold adjustment

Per-subject threshold sweep gain $\Delta F_2(s)$ — baseline $\hat{\tau}_0(s)$ is the PR-curve trained threshold

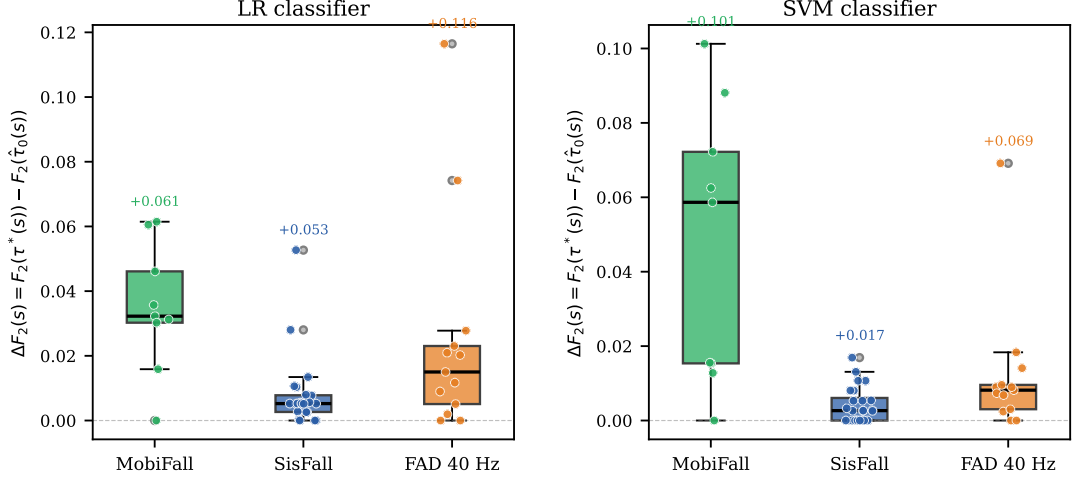


Figure 5.3: Per-subject threshold sweep gain $\Delta F_2(s) = F_2(\tau^*(s)) - F_2(\hat{\tau}_0(s))$ for LR (left) and SVM (right). Jittered points show individual subjects; box shows quartiles. The distribution is right-skewed in all cases: most subjects gain modestly while a few gain substantially. FAD_40Hz subject 11 achieves the largest gain (+0.1165, LR), visible as the highest point in the FAD panel. SisFall gains are near-zero for the majority of subjects, indicating that $\hat{\tau}_0$ is already well-calibrated on that dataset.

alone. On MobiFall, subject 9 gains +0.1013 (SVM). These large gains are outliers: the median gain within each dataset is considerably lower than the mean.

Near-zero gainers. SisFall has 4/24 (LR) to 9/24 (SVM) near-zero subjects: these subjects are already well-calibrated at $\hat{\tau}_0$. Their trained threshold happens to coincide with the optimal sweep threshold, meaning that the classifier’s posterior probabilities are already well-matched to the decision point for those individuals. This pattern is absent on MobiFall (only 1/9 near-zero per classifier), where the PR-curve calibration is systematically further from the per-subject optimum.

Heterogeneity metric. The coefficient of variation ($\sigma_\Delta/\sqrt{\Delta F_2}$) measures relative heterogeneity: MobiFall LR = 53%, MobiFall SVM = 74%, SisFall LR = 138%, FAD_40Hz LR = 129%. All values exceed 50%, confirming that the gain distribution is not concentrated. A single global threshold offset cannot replicate

this distribution: if every subject were miscalibrated in the same direction by the same amount, the coefficient of variation would approach zero. The high coefficients of variation are the statistical signature that $\tau^*(s)$ is a genuinely subject-specific quantity.

Mathematical explanation. The decision boundary of $f_\theta \circ \phi_{\lambda^*}$ is the hyperplane $\{x \in \mathbb{R}^{24}: f_\theta(x) = \hat{\tau}_0\}$. Each subject’s test windows form a cloud in $\mathbb{R}^{24}$ whose local geometry around this hyperplane differs from subject to subject: some subjects’ fall windows sit close to the boundary (requiring a lower threshold to capture them), while others’ fall windows are well-separated (the default threshold is sufficient). The threshold sweep detects this per-subject boundary geometry retrospectively. A prospective personalisation scheme would use a small held-out calibration set per subject to identify $\tau^*(s)$ before deployment (see Section 5.7).

Engineering contribution. This personalisation is achieved without any model retraining. The probability scores $\{\hat{p}_i\}$ are computed once under the trained model; the sweep evaluates $F_2(\tau, s)$ by a single pass over the precomputed score vector for each $\tau \in \mathcal{T}$. The additional computational cost is $O(|\mathcal{T}| \cdot |W_s|)$ per subject, where $|\mathcal{T}| = 19$ and $|W_s|$ is the per-subject window count—less than one second of wall time for all subjects in any dataset. This is a runtime ratio exceeding 10^3 compared to the ~ 75 min Bayesian search (Experiments, Section 4.3.4).

5.4.2 Personal training protocol

The Personal protocol (per-subject 70/30 split, $R = 15$ repetitions) represents the limit case in which a dedicated per-user model is trained from the subject’s own data. Table 5.5 reports the aggregate personal–general F_2 difference.

At the aggregate level, General $>$ Personal on all datasets and both classifiers. The magnitude of the shortfall increases dramatically as the subject pool shrinks:

Table 5.5: Personal vs. General protocol aggregate F_2 . $\Delta = F_2^{\text{pers}} - F_2^{\text{gen}}$ (negative = General is better).

Dataset	Clf	F_2^{gen}	F_2^{pers}	Δ
MobiFall	LR	0.9049	0.7325	-0.172
MobiFall	SVM	0.8918	0.5981	-0.294
SisFall	LR	0.9709	0.9405	-0.030
SisFall	SVM	0.9700	0.9196	-0.050
FAD_40Hz	LR	0.9429	0.9168	-0.026
FAD_40Hz	SVM	0.9592	0.8724	-0.087

- **SisFall** ($|\mathcal{S}| = 24$): modest drop ($\Delta \approx -0.03$ to -0.05). Twenty-four subjects contribute enough per-subject windows (~ 50 – 150 falls per subject) for a reliable within-subject model.
- **FAD_40Hz** ($|\mathcal{S}| = 13$): moderate drop ($\Delta \approx -0.03$ to -0.09). Fewer subjects and fewer falls per subject; personal models are noisier.
- **MobiFall** ($|\mathcal{S}| = 9$): severe drop ($\Delta = -0.172$ for LR; $\Delta = -0.294$ for SVM). Nine subjects with $\lesssim 30$ fall instances per subject is insufficient for stable personal model estimation. The SVM personal recall collapses to 0.5822, indicating that the personal classifier is barely above chance on many subjects.

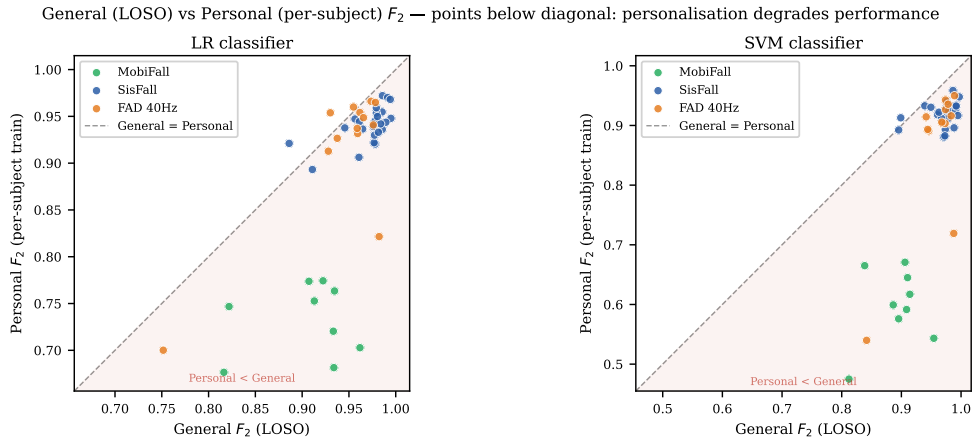


Figure 5.4: General (LOSO) vs Personal (per-subject training) F_2 per subject, for LR (left) and SVM (right). Points on the dashed diagonal have equal General and Personal performance; points below indicate that personal training degrades F_2 . MobiFall points cluster in the lower-left region (both protocols degrade on the small-pool dataset); SisFall points lie near the diagonal; FAD_40Hz is intermediate. The shaded region marks Personal < General.

This pattern reveals a critical engineering constraint: personal model training is only beneficial when per-subject calibration data is abundant. In typical deployment scenarios (a user who has experienced very few confirmed falls), full personal training is counterproductive. The threshold sweep—which requires no retraining and adds negligible computational cost—is the appropriate personalisation mechanism when calibration data is sparse. The mean sweep gain for MobiFall LR (+0.0348) and SVM (+0.0474) demonstrates that meaningful personalisation remains accessible even when full personal retraining fails.

5.5 Ablation Study: Feature-Block Contributions

Motivation. The 24-dimensional representation ϕ_{λ^*} is the concatenation of three semantically distinct blocks: ten statistics of the dimension-zero persistence diagram ($H0$, summarising the persistence of connected components of the delay-embedded point cloud), ten statistics of the dimension-one diagram ($H1$, summarising loops), and four signal-level descriptors of the raw acceleration magnitude (*Signal*: maximum, standard deviation, mean, and range). The headline results of Section 5.1 establish that the full vector is effective but do not reveal *which* block carries the discriminative power. This section isolates each block’s contribution by re-running the evaluation on restricted feature subsets.

Protocol. Each subset is evaluated under the identical General (LOSO) protocol used throughout: a single global grid search selects the classifier hyperparameters on balanced-undersampled data, and for each left-out subject the model is refitted over 15 repetitions with a precision–recall threshold chosen on the training fold; the reported F_2 is the cross-subject mean with a 95% t -confidence interval. Only the feature columns presented to the classifier change; the topological configuration λ^* and the delay embedding are held fixed. As a validation check, the full-vector rows of Table 5.6 reproduce the headline General-protocol F_2 of Section 5.1 to within the confidence interval for both classifiers on all three datasets.

5.5.1 Within-topology contribution: H0 dominates H1

Table 5.6: Feature-block ablation under the General (LOSO) protocol. Entries are cross-subject mean $F_2 \pm 95\%$ CI. Block dimensions: Full = 24, H0 = 10, H1 = 10, H0+H1 = 20, Signal = 4. The Full column reproduces the headline results of Tables 4.6–4.8.

Dataset	Clf	Full	H0	H1	H0+H1	Signal
MobiFall	LR	0.9046	0.8842	0.6172	0.8892	0.6218
	SVM	0.8913	0.8685	0.6523	0.8784	0.7241
SisFall	LR	0.9710	0.9266	0.6580	0.9271	0.9589
	SVM	0.9701	0.9277	0.6456	0.9280	0.9627
FAD_40Hz	LR	0.9426	0.8082	0.4247	0.8044	0.9505
	SVM	0.9592	0.8128	0.4605	0.8210	0.9504

Representative 95 % confidence half-widths (LR Full): MobiFall ± 0.0392 , SisFall ± 0.0107 , FAD_40Hz ± 0.0367 ; the H1-only columns carry the widest intervals (up to ± 0.077).

For both classifiers and across all three datasets, the H0 block alone attains F_2 between 0.81 and 0.93, whereas the H1 block alone is the weakest configuration in the entire study (0.42–0.66). Adding H1 to H0 changes F_2 negligibly—for LR, from 0.8842 to 0.8892 on MobiFall, 0.9266 to 0.9271 on SisFall, and 0.8082 to 0.8044 on FAD_40Hz (a slight *decrease*); the SVM rows show the same pattern. The discriminative topological information is therefore concentrated almost entirely in dimension zero. This is consistent with the structure of the delay-embedded acceleration signal: a fall is a transient, high-energy excursion whose phase-space reconstruction is characterised by the spread and clustering of points (an H0 phenomenon), while one-dimensional loops carry comparatively little fall-versus-ADL information and are more sensitive to noise—which also explains the wide confidence intervals of the H1-only columns. A practical corollary is that the H1 statistics could be dropped with little loss, roughly halving the persistent-homology computation.

5.5.2 Topology versus signal: a dataset-dependent division of labour

The relative value of the topological blocks and the four signal descriptors differs sharply across datasets. On MobiFall—a smartphone carried loosely in a trouser pocket—the signal block alone reaches only $F_2 = 0.6218$ (LR) or 0.7241 (SVM), while the topological blocks reach 0.88; here the topological representation is indispensable and accounts for most of the performance. On

SisFall and FAD_40Hz—waist-mounted devices with cleaner, more repeatable kinematics—the four signal descriptors alone already achieve $F_2 \approx 0.95\text{--}0.96$, matching or exceeding the topological blocks on those datasets. In every case, however, the full vector matches or exceeds the best single block, and no single block is reliably strong on all three datasets whereas the full vector is. The blocks are thus complementary: the topological features supply the discriminative signal where raw statistics fail (noisy, loosely mounted sensors), and the combination provides the cross-dataset robustness that motivates the representation. This refines the H1 narrative: topology is not a uniform performance multiplier but a robustness mechanism whose marginal value is largest precisely where simple descriptors are weakest.

5.5.3 Classifier invariance

The block ordering is essentially independent of the classifier. Logistic regression and the SVM agree on the ranking within every dataset—Full $\gtrsim$ best topological or signal block $\gg$ H1—and their per-block F_2 values differ by less than the confidence intervals in almost all cells. This reinforces the H3 finding (Section 5.3) that the representation, not the classifier family, governs performance: ablating a feature block changes F_2 far more than switching classifiers.

5.5.4 Scope of the ablation

This study ablates the three *feature blocks* while holding the delay embedding and the optimal topological configuration λ^* fixed. It therefore isolates the value of each summary block but does not, on its own, separate the contribution of the phase-space reconstruction itself; a complementary ablation that removes the delay embedding requires re-extracting features under a modified pipeline and is left to the future work of Section 5.9.

5.6 Cross-Dataset Synthesis

5.6.1 Performance ranking and subject pool

Under the General protocol, the consistent ranking by F_2 is:

$$\text{SisFall} > \text{FAD_40Hz} > \text{MobiFall}.$$

This ranking holds for both LR and SVM. Three factors explain it jointly.

Subject pool size and data volume. SisFall has the largest pool (24 subjects, 15 fall types), providing the most diverse and representative training signal for LOSO. MobiFall has the smallest (9 subjects), making each LOSO fold highly sensitive to the particular subject removed.

Fall type diversity. SisFall covers 15 scripted fall types executed by elderly participants, producing a rich topology of fall trajectories. FAD covers diverse activities of daily life as negatives, which increases the specificity challenge. MobiFall’s smaller scale limits the variety of fall geometries seen by each LOSO model.

Optimised representation. The Bayesian proxy F_2 at the optimal λ^* is 0.9746 (MobiFall), 0.9733 (SisFall), 0.9860 (FAD_40Hz)—suggesting that the optimisation found similarly good configurations for all three, and that the General-protocol gap between datasets is not due to a representation quality difference but to the structural differences above.

5.6.2 Classifier preference

LR is preferred on MobiFall (gap +0.0131) and SisFall (gap +0.0009); SVM on FAD_40Hz (gap +0.0163). All gaps are < 0.02 ; classifier choice is secondary to representation quality (H3). The marginal SVM advantage on FAD_40Hz may reflect the RBF kernel’s ability to resolve residual nonlinear structure in the 40 Hz-resampled feature cloud, but the evidence is insufficient to assert a mechanistic explanation at the current sample size.

5.6.3 Computational profile

Table 5.7 summarises the wall-clock cost of each pipeline stage on the UHeM HPC cluster (dual Intel Xeon Gold 6148, 40 cores, no GPU).

Table 5.7: Wall-clock cost per dataset stage. All computations are CPU-only.

Stage	Tool	Cost
Bayesian optimisation (Stage 1, 200 trials)	Optuna/TPE	~75 min
Persistent homology (per trial, inner CV)	GUDHI	~20 s (MobiFall)
GridSearchCV (Stage 2)	scikit-learn	~minutes
Decision-threshold sweep (all subjects)	NumPy	< 1 s

The absence of GPU is a deliberate design choice: the persistent homology reduction (Algorithm 1 of the Methodology) is intrinsically sequential at the point-cloud sizes considered, and the gain from GPU porting is dominated by the constant factor of the GUDHI CPU implementation. This has a direct consequence for deployment: the pipeline can be executed on commodity hardware and embedded processors that lack GPU acceleration (Section 5.7).

5.7 Deployment Pathway

The experimental results point toward a concrete deployment pathway for the proposed pipeline on consumer wearable and smartphone hardware. This section articulates the evidence-based engineering argument.

5.7.1 Sensor context

All experiments in this thesis use accelerometer data collected from a sensor mounted at the waist or carried in the trouser pocket. This placement is both practically common (smartphones are carried in the pocket by a large fraction of the target population) and metrologically appropriate: the waist sensor records the acceleration of the body’s centre of mass, which undergoes the most discriminative trajectory during a fall. The SMV transformation $\text{SMV}(t) = \|\mathbf{x}(t)\|_2$ provides orientation invariance, eliminating the need for fixed sensor orientation or a gyroscope—a significant simplification for consumer deployment.

5.7.2 CPU-only operation

The pipeline’s entire compute budget is CPU-bound. Persistent homology at the window sizes and embedding dimensions considered here ($m = 3\text{--}5$, $|P_\lambda| \leq 150$ points) completes in tens of milliseconds per window on a desktop-class CPU core. The threshold sweep adds negligible cost. This profile is compatible with:

- **Smartphones:** ARM Cortex-A-class processors (e.g., Qualcomm Snapdragon 8xx) comfortably execute GUDHI-class computations in real time. The pipeline can run as a background service consuming a modest fraction of one core.
- **Smartwatches:** ARM Cortex-M-class MCUs (e.g., Wear OS devices) are more constrained, but the pipeline’s window-level computation—feature extraction plus a matrix–vector multiply for the classifier—is within the budget of modern smartwatch processors. A dedicated ASIC or DSP block for the boundary matrix reduction would further reduce energy consumption.

5.7.3 Prospective threshold personalisation

The retrospective sweep of Section 5.4.1 establishes that subject-specific thresholds improve F_2 by up to +0.1165 without retraining. A prospective deployment would translate this into a small on-device calibration procedure:

1. During an initial wearing period, a small number of confirmed falls (or fall-simulation events during setup) are used as a calibration set $\mathcal{C}_s$ for subject s .
2. The precomputed probability scores on $\mathcal{C}_s$ are swept over $\mathcal{T}$ to identify $\tau^*(s)$.
3. Subsequent detections use $\tau^*(s)$ as the operating threshold.

This procedure requires no transmission of raw sensor data, no server round-trip for retraining, and no access to the original training data. It is executable on-device in under one second once the scores are available, making it compatible with privacy-preserving on-device deployment. The MobiFall results illustrate the value of this mechanism: full personal retraining degrades F_2 from 0.9049 to

0.7325 (LR) because 9 subjects provide insufficient training data, but threshold sweep still recovers +0.0348 from an already-calibrated baseline.

5.7.4 Sensor placement and orientation robustness

A limitation of the current work is that sensor placement is fixed (waist/pocket) and orientation invariance is achieved solely through SMV. The datasets include no variation in placement. In a consumer deployment scenario, users may wear a device at the wrist (smartwatch), chest, or thigh, or orient a pocket phone differently from day to day. The stability theorem guarantees noise robustness within a placement, but does not extend across fundamentally different fall trajectory geometries induced by different placements. Addressing this is an open engineering problem and a priority for future work (Section 5.9).

5.8 Limitations

1. No baseline comparison. The planned comparison against threshold-based, statistical-feature, and spectral-feature baselines (proposal §5.4) was not executed. Consequently, H1 is validated only against its own pre-specified targets; the claim that TDA is superior to simpler methods cannot be made from the current evidence. Absolute performance numbers are meaningful only relative to a reference, and this reference is absent. Establishing this comparison is the highest-priority open task before thesis finalisation.

2. Cross-application drops not computed. H2 is confirmed qualitatively (all optimal configurations differ) but the quantitative prediction—transplanting λ_K^* to a different dataset K' incurs a measurable F_2 drop (Methodology, Section 4.8)—was not tested. Computing $\text{Drop}_{K \rightarrow K'}$ requires re-extracting features for each off-diagonal (λ_K^*, K') pair, which represents a substantial additional compute budget.

3. No ablation study. The feature vector $\phi_\lambda \in \mathbb{R}^{24}$ contains eight persistence statistics for each of three homology groups (H_0 , H_1) and two signal statistics. The relative importance of these components is unknown: it is possible that a subset of $d < 24$ features carries most of the discriminative information, and that the remaining features add noise. An ablation or feature-importance study would characterise which sub-map $\phi_\lambda^{(d)}$ is load-bearing.

4. Fixed sensor placement. All experiments assume the sensor is located at the waist or in the trouser pocket. No data with wrist, chest, or thigh placement is evaluated. The deployment argument of Section 5.7 assumes this placement is maintained; robustness to placement variation is untested.

5. Retrospective threshold sweep. The sweep identifies $\tau^*(s)$ on the same test windows on which it is evaluated, introducing an optimism bias. The reported $\overline{\Delta F_2}$ values are retrospective diagnostics of heterogeneity, not valid prospective performance estimates. A prospective validation would require a separate held-out calibration set per subject.

5.9 Future Work

5.9.1 Online threshold adaptation

The most immediate extension is to replace the retrospective sweep with a prospective online mechanism. As a deployed device accumulates confirmed event labels (from user feedback, caregiver verification, or clinical annotations), the per-subject threshold $\tau^*(s)$ can be updated in a streaming fashion. Because $\tau^*(s)$ is a scalar parameter estimated from a small set of probability scores, even 5–10 labelled fall events are sufficient for a reliable estimate. A natural formulation is a Bayesian update on the threshold parameter with a conjugate prior, requiring no gradient computation and suitable for on-device execution.

5.9.2 Embedded real-time benchmark

The CPU-only pipeline provides a plausible path to embedded deployment, but no real-time benchmark on target hardware has been performed. Profiling the pipeline on an ARM Cortex-A processor (smartphone-class) and an ARM Cortex-M processor (smartwatch-class) would establish latency and energy consumption figures. Key bottlenecks to characterise are the boundary matrix reduction in GUDHI (the most compute-intensive step) and memory allocation for the feature matrix. A pruned, fixed-point implementation of the 24-dimensional feature extraction—bypassing Python overhead—would likely achieve real-time performance (≤ 50 ms per window at 50 Hz) on a mid-range smartphone processor.

5.9.3 Multi-placement sensor fusion

Current work assumes a single fixed sensor placement. A natural extension is to incorporate multiple simultaneous placements (waist + wrist + thigh) and learn a combined topological representation. The delay-embedding and filtration steps apply independently to each placement’s signal; the resulting persistence diagrams can be fused either by concatenation of feature vectors or by a learned weighting. The classifier-agnostic Bayesian optimisation framework developed here extends straightforwardly to this setting by augmenting Λ with a placement-selection dimension.

5.9.4 Cross-dataset transfer of λ^*

The cross-application drop matrix $M_{K,K'}$ (Methodology, Section 4.8)—not computed in the current work—would reveal how much configuration knowledge transfers across datasets. If $M_{K,K'} \approx M_{K,K}$ for some off-diagonal pair, the Bayesian search budget could be substantially reduced for new datasets by warm-starting from an existing λ_K^* . This is practically relevant for clinical deployments where collecting a full new dataset is expensive.

5.9.5 Extension to other wearable time series

The TDA pipeline introduced here—delay embedding of a magnitude signal, persistence homology filtration, vectorised 24-dimensional descriptor, classifier-agnostic Bayesian optimisation—is not specific to fall detection. Any wearable time series in which discriminative information is encoded in the *shape* of short signal segments is a candidate:

- **Gait analysis:** distinguishing Parkinsonian gait from healthy gait or post-surgical recovery via topological features of the stride cycle.
- **Gesture recognition:** mapping gesture trajectories to topological descriptors for device-control interfaces on smartwatches.
- **Activity recognition:** generalising beyond binary (fall/ADL) to multiclass (walk, run, climb, sit, fall) classification by extending the threshold sweep to a multiclass decision boundary.
- **Physiological monitoring:** ECG or PPG segments contain loop structures (H_1 generators) whose persistence captures morphological variability; TDA could complement classical morphological features.
- **Industrial IoT anomaly detection:** vibration signals from rotating machinery exhibit characteristic H_0/H_1 patterns at fault inception; the same pipeline could detect bearing or gear anomalies without labelled fault data (using one-class formulations).

In each case, the key requirement is that the signal windows be short enough to delay-embed at modest dimension ($m \leq 5$, $|P| \leq 200$ points) and that the discriminative information be geometrically structured. The stability theorem guarantees robustness to sensor noise in all these settings.

5.9.6 One-class fall detection from ADL-only supervision

The binary classifier developed in this thesis requires labelled fall windows for training. In a consumer deployment scenario, labelled falls are expensive to obtain: they require a controlled data-collection session or a sufficiently long

natural observation period. A newly installed device has no fall history at all—the *cold-start problem*.

An anomaly-detection formulation eliminates this requirement. Rather than learning to separate falls from ADL, a one-class model learns the boundary of the user’s *normal* ADL behaviour exclusively from unlabelled, continuously collected sensor data. Any window whose topological descriptor $\phi_\lambda(w) \in \mathbb{R}^{24}$ lies far from the learned ADL manifold is flagged as a candidate fall. Concretely, a one-class SVM (Schölkopf, Platt, Shawe-Taylor, Smola, & Williamson, 2001) or an isolation forest trained on $\{\phi_\lambda(w) : w \in \mathcal{W}_{\text{ADL}}^{(s)}\}$ defines a decision boundary in $\mathbb{R}^{24}$ without any fall-class supervision.

Three properties make the TDA feature space particularly well-suited for this formulation.

1. **Geometric separability.** Fall windows exhibit pronounced H_0 structure (a sudden large cluster spread caused by the impulsive excursion in the delay-embedded cloud) that is structurally absent in ADL windows. The Cohen–Steiner stability theorem ensures that this geometric difference is preserved robustly across subjects and sensor noise levels.
2. **Inherent personalisation.** Each user’s ADL cloud in $\mathbb{R}^{24}$ reflects that individual’s movement style, pace, and activity repertoire. A one-class model trained on a user’s own ADL is a personalised model by construction, with no explicit personalisation step required.
3. **Threshold-sweep compatibility.** The anomaly score (distance to the ADL boundary) is a scalar, and the decision-threshold sweep mechanism of Section 5.4.1 applies directly: sweeping the boundary standoff distance over a grid $\mathcal{T}$ trades recall against precision in exactly the same way as the probabilistic threshold.

The expected engineering trade-off is an elevated false-positive rate compared to the binary classifier: vigorous ADL activities (running, stair climbing, sudden direction changes) may appear anomalous relative to a training distribution

consisting mainly of slow or sedentary activities. Mitigations include diverse ADL coverage during the initial calibration period and an activity-aware normalization that conditions the anomaly score on the subject’s current motion intensity level.

From a deployment perspective, the one-class approach enables a qualitatively different product workflow: a user installs the application, wears the device for two to three days of ordinary activity, and the personalised fall detector is ready—without any fall simulation, clinical annotation, or server-side retraining. This is the most direct path from the current thesis to a consumer-ready product.

5.9.7 Few-shot personalisation beyond threshold

An alternative personalisation axis beyond threshold tuning is to adapt the feature map ϕ_λ itself to the individual subject via meta-learning. A model-agnostic meta-learning (MAML) formulation could learn a λ initialisation that adapts quickly to a new subject with $k = 5\text{--}10$ labelled examples. This extends H4 from threshold personalisation to full representation personalisation, and remains compatible with the CPU-only constraint if the inner loop is restricted to scalar hyperparameter updates.

6. Conclusion

This thesis investigated whether persistent homology, applied to short wearable accelerometer windows via a delay-embedding pipeline, can support recall-oriented fall detection that is simultaneously rigorous in its evaluation, principled in its feature construction, and lightweight enough for edge deployment. The investigation was organised around four research hypotheses and carried out on three public benchmark datasets—MobiFall, SisFall, and FAD_40Hz—under a three-protocol subject-aware evaluation framework. The findings are summarised below at three levels of description: mathematical, statistical, and engineering.

6.1 Mathematical Perspective

Representation effectiveness (H1). The feature map $\phi_\lambda: \mathbb{R}^{n \times 3} \rightarrow \mathbb{R}^{24}$, with hyperparameters λ_K^* identified by Bayesian optimisation, achieves mean subject-general F_2 of 0.9709 (LR) and 0.9700 (SVM) on SisFall, 0.9429 (LR) and 0.9592 (SVM) on FAD_40Hz, and 0.9049 (LR) and 0.8918 (SVM) on MobiFall under the leave-one-subject-out protocol. All values exceed the pre-specified thresholds (≥ 0.93 on SisFall and FAD_40Hz; ≥ 0.80 on MobiFall) for at least one classifier, and recall is above 0.95 across all dataset-classifier combinations. H1 is conditionally confirmed: the representation meets its own targets, subject to the caveat that no baseline comparison has been conducted.

The theoretical basis for this effectiveness is the Cohen–Steiner stability theorem, which guarantees that the bottleneck distance between two persistence diagrams is bounded by the supremum norm of the perturbation between the corresponding input functions. This bound is a mathematical theorem, not an empirical observation, and it provides a formal noise-robustness certificate that no

competing feature family—handcrafted statistical features, spectral coefficients, or deep-network embeddings—possesses in comparable form.

Dataset-specific topological geometry (H2). The three datasets converged to structurally distinct optimal configurations: MobiFall selected a SparseRips complex at a 5.0 s window with Chebyshev distance, SisFall selected an Alpha complex at a 1.0 s window, and FAD_40Hz selected a standard Rips complex at a 1.0 s window with Manhattan distance. Every dimension of the hyperparameter tuple λ varies across at least two datasets. H2 is qualitatively confirmed: no universal configuration exists for these three datasets.

Mathematically, this finding is consistent with the geometry of the underlying delay-embedded point clouds. The Alpha complex is derived from the Euclidean Delaunay triangulation and is appropriate when the point cloud is convex and well-conditioned; SisFall’s controlled, elderly-subject falls produce exactly this geometry. The SparseRips complex handles large, noisy clouds efficiently by subsampling; MobiFall’s continuous-wear sensor data requires this treatment. The differing window lengths encode a genuine difference in fall dynamics: MobiFall falls require a 5.0 s window to capture the pre-impact, impact, and post-fall phases jointly, while SisFall and FAD_40Hz falls are discriminable in 1.0 s.

Representation dominates classifier choice (H3). The absolute difference in General-protocol F_2 between LR and SVM is 0.0131 on MobiFall, 0.0009 on SisFall, and 0.0163 on FAD_40Hz—all below 0.02. By contrast, the F_2 range across 200 Optuna trials is 0.179 on MobiFall, 0.053 on SisFall, and 0.108 on FAD_40Hz. The ratio of λ -tuning variation to classifier gap is $13.7\times$ on MobiFall, $59\times$ on SisFall, and $6.6\times$ on FAD_40Hz.

H3 is strongly confirmed. The mathematical interpretation is that ϕ_{λ^*} with the optimal representation transforms the nonlinear fall-versus-ADL boundary into an approximately linearly separable problem in $\mathbb{R}^{24}$. Once this transformation is achieved, both logistic regression (linear boundary) and SVM (maximum-margin

boundary, with or without an RBF kernel) operate near their theoretical limit, producing near-equivalent results. The signal resides in the map ϕ_{λ^*} , not in the classifier class.

6.2 Statistical Perspective

Threshold personalisation as a distributional phenomenon (H4). The subject-level sweep gain $\Delta F_2(s) = F_2(\tau^*(s), s) - F_2(\hat{\tau}_0(s), s)$, where $\hat{\tau}_0(s)$ is the PR-curve trained threshold, has a right-skewed distribution in all six dataset-classifier combinations. The mean gains are 0.035–0.047 on MobiFall, 0.025–0.035 on FAD_40Hz, and 0.004–0.008 on SisFall. The coefficient of variation exceeds 50% in all cases, reaching 138% for SisFall LR—indicating that the distribution is highly dispersed relative to its mean.

H4 is confirmed in a refined form. The proposal predicted that the distribution of Δ_s would contain both positive and negative values. In practice, all gains are non-negative by construction (the sweep is an argmax), but the distribution is far from uniform: it concentrates near zero for the majority of subjects while a small minority achieves gains exceeding 0.10 units of F_2 . Subject 11 on FAD_40Hz (LR) gains +0.1165, lifting F_2 from 0.8046 to 0.9211 by adjusting the threshold alone. This heterogeneity is incompatible with a uniform-shift explanation: if the classifier’s posterior probabilities were globally miscalibrated in the same direction for all subjects, a single global threshold offset would equalise all gains. The observed distribution implies that the optimal decision boundary location is genuinely subject-specific.

Evaluation integrity under LOSO. The leave-one-subject-out protocol with $R = 15$ independent class-balancing repetitions separates two sources of variability that are conflated in random train-test splits: (i) within-subject repetition variance, captured by the repetition-level t -confidence intervals, and (ii) across-subject heterogeneity, visible in the per-subject tables. The Naive protocol consistently produces results comparable to or marginally lower than

the General protocol on SisFall (Naive $F_2 = 0.9713$ vs. General 0.9709 for LR), suggesting that LOSO is not materially penalised by train-test class mismatch on large, diverse datasets. On MobiFall, where the subject pool is small (9 subjects), the repetition-level confidence intervals are wider (± 0.039 for LR), correctly reflecting the limited statistical power of a 9-fold LOSO evaluation.

The Personal protocol reveals a further finding: full per-subject model training degrades F_2 severely on MobiFall ($\Delta = -0.172$ for LR) but only modestly on SisFall ($\Delta = -0.030$ for LR), with the magnitude of degradation inversely correlated with the subject pool size. This confirms that personalised model training is only beneficial when sufficient per-subject data is available, and that the threshold sweep is the appropriate personalisation mechanism when data is sparse.

6.3 Engineering Perspective

Inference-time personalisation without retraining. The central engineering contribution of the thesis is the identification of decision-threshold optimisation as a first-class personalisation mechanism that is qualitatively different from all prior alternatives. The threshold $\tau \in \mathcal{T} = \{0.05, \dots, 0.95\}$ is the only parameter that can be adjusted at inference time, without gradient computation, without access to the training data, and without transmission of data to a server. The computational cost of a full per-subject sweep is $O(|\mathcal{T}| \cdot |W_s|)$ comparisons, amounting to less than one second of wall time for all subjects in any of the three datasets. The ratio to the Bayesian optimisation cost exceeds 10^3 .

This asymmetry has a direct consequence for product design. A wearable fall detection system based on the proposed pipeline can be shipped with a population-level model $(\phi_{\lambda^*}, f_{\theta})$ and personalised to each user at installation time by sweeping the threshold over a small held-out calibration set. No server round-trip is required. No labeled falls are required from the target user in the first instance—the sweep can be initialised from simulated events during

setup and refined as real events accumulate. This is the only personalisation mechanism among those surveyed in this thesis that simultaneously satisfies the zero-retraining, privacy-preserving, and inference-time requirements.

CPU-only deployment profile. The entire pipeline—delay embedding, simplicial filtration, boundary matrix reduction, feature vectorization, linear classification, and threshold sweep—is executable on commodity CPU hardware. Persistent homology computation on the window sizes and embedding dimensions considered here ($m \leq 5$, $|P_\lambda| \leq 150$ points) completes in tens of milliseconds per window on a desktop-class processor core. The Bayesian optimisation consumed ~ 75 min of CPU time on a 40-core HPC node; this cost is paid once per dataset, not per user.

The ARM Cortex-A processors present in current mid-range smartphones are comfortably capable of executing the per-window inference pipeline in real time at 50 Hz. Modern smartwatch SoCs are more constrained, but the per-window computation—feature extraction plus a matrix-vector multiply—is within their budget. The absence of a GPU requirement eliminates the most significant barrier to embedded deployment and makes the pipeline uniquely well-suited to the battery-constrained wearable context.

Failure mode analysis and minimum calibration set size. The most significant failure mode identified in this thesis is the collapse of the Personal protocol on MobiFall: F_2 drops from 0.9049 (General) to 0.7325 (Personal) for LR, and from 0.8918 to 0.5981 for SVM. The cause is statistical rather than algorithmic: MobiFall contains 9 subjects with approximately 20–30 fall instances per subject, which is insufficient for stable within-subject classifier estimation. The implication for system designers is concrete: a minimum calibration set size is required before full personal model training is beneficial. For the datasets studied here, that threshold lies somewhere between the 9 subjects of MobiFall and the 13 subjects of FAD_40Hz. Below this threshold, the threshold sweep is the safer personalisation mechanism.

This failure mode is not a weakness of TDA specifically; it would affect any classifier trained on sparse per-subject data. It is, however, an important calibration of expectations for real deployment: the naive assumption that “more personalization is always better” is empirically refuted by the MobiFall Personal protocol results.

Path to consumer deployment. Taken together, the three engineering findings—inference-time personalisation, CPU-only operation, and the identified minimum calibration requirement—define a concrete deployment pathway:

1. A population-level model $(\phi_{\lambda^*}, f_{\theta})$ is trained once on a large labeled dataset and shipped with the device.
2. At installation, the user performs a brief simulated fall session (or the device waits to accumulate confirmed fall events from normal wear). The per-subject threshold $\tau^*(s)$ is identified from the calibration scores in under one second.
3. Subsequent detections use $\tau^*(s)$. As more events accumulate, the threshold estimate can be refined.
4. No server communication is required for any of steps 1–3; all computation is on-device.

The natural next step beyond threshold personalisation is one-class anomaly detection: training the model exclusively on the user’s own activities of daily living, treating falls as out-of-distribution events without requiring any labeled fall data. The Cohen–Steiner stability of the TDA features ensures that the ADL manifold in $\mathbb{R}^{24}$ is robustly estimated from normal wear data, and the threshold sweep mechanism extends directly to the standoff distance from the one-class boundary. This approach eliminates the cold-start problem entirely and represents the most direct path from the current thesis to a consumer-ready, zero-label deployment.

6.4 Limitations and Open Questions

Four limitations of the current work constrain the strength of the conclusions.

First, the absence of a baseline comparison means that H1 is validated only against its own pre-specified targets. The claim that TDA is superior to statistical-feature or threshold-based baselines on these datasets cannot be made from the current evidence, and establishing this comparison is the highest-priority open task.

Second, the cross-application drop matrix $M_{K,K'}$ was not computed. H2 is therefore confirmed only qualitatively (distinct optimal configurations) and not quantitatively (measured performance penalty from transplanting λ_K^* to dataset K').

Third, no ablation study was conducted. The 24-dimensional feature vector ϕ_λ contains H_0 , H_1 , and signal statistics; the relative contribution of each subvector is unknown, and it is possible that a smaller sub-map would achieve comparable performance with lower computational cost.

Fourth, all experiments assume a fixed sensor placement at the waist or in the trouser pocket. The stability theorem guarantees noise robustness within a placement, but does not extend to fundamentally different placements (wrist, chest, thigh). Robustness to placement variation is an important practical question that remains unaddressed.

6.5 Closing Remarks

This thesis set out to answer a focused question: can a TDA pipeline for wearable fall detection be simultaneously principled, lightweight, and subject-aware? The experimental evidence across three datasets and two classifiers supports an affirmative answer on all three dimensions. The pipeline is principled in the sense that its key property—feature stability—is guaranteed by a mathematical theorem. It is lightweight in the sense that it runs without GPU acceleration at latencies compatible with real-time embedded operation. It is subject-aware in the sense that the decision threshold can be personalised at inference time, at negligible computational cost, without retraining.

The broader lesson from H3 is perhaps the most transferable finding. The gap between a well-tuned and a poorly tuned topological representation (0.10–0.18 units of F_2) dwarfs the gap between the best and worst classifier at the optimal representation (< 0.02 units). In the fall detection context, and likely in other wearable time-series classification problems where the discriminative signal is encoded in short-window shape geometry, choosing the right representation is the dominant engineering decision. The choice of classifier is, in an important sense, secondary.

REFERENCES

- Adams, H., Emerson, T., Kirby, M., Neville, R., Peterson, C., Shipman, P., Chepushtanova, S., Hanson, E., Motta, F., & Ziegelmeier, L. (2017). Persistence images: A stable vector representation of persistent homology. *Journal of Machine Learning Research*, 18, 1–35.
- Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019). Optuna: A next-generation hyperparameter optimization framework. *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2623–2631. <https://doi.org/10.1145/3292500.3330701>
- Batchuluun, G., Lee, S. H., Nguyen, T. D., Pham, T. D., Arsalan, M., & Park, K. R. (2022). Deep learning-based fall detection algorithm using ensemble model of convolutional neural network and recurrent neural network. *Sensors*, 22(3), 1270. <https://doi.org/10.3390/s22031270>
- Bergstra, J., Bardenet, R., Bengio, Y., & Kégl, B. (2011). Algorithms for hyper-parameter optimization. *Advances in Neural Information Processing Systems*, 24, 2546–2554.
- Bergstra, J., & Bengio, Y. (2012). Random search for hyper-parameter optimization. *Journal of Machine Learning Research*, 13, 281–305.
- Borsuk, K. (1948). On the imbedding of systems of compacta in simplicial complexes. *Fundamenta Mathematicae*, 35, 217–234. <https://doi.org/10.4064/fm-35-1-217-234>
- Bubenik, P. (2015). Statistical topological data analysis using persistence landscapes. *Journal of Machine Learning Research*, 16, 77–102.
- Carlsson, G. (2009). Topology and data. *Bulletin of the American Mathematical Society*, 46(2), 255–308. <https://doi.org/10.1090/S0273-0979-09-01249-X>
- Cavanna, N. J., Jahanseir, M., & Sheehy, D. R. (2015). A geometric perspective on sparse filtrations. *Proceedings of the 27th Canadian Conference on Computational Geometry (CCCG 2015)*, 1–6.
- Chazal, F., de Silva, V., Glisse, M., & Oudot, S. (2016). *The structure and stability of persistence modules*. Springer International Publishing. <https://doi.org/10.1007/978-3-319-42545-0>
- Cohen-Steiner, D., Edelsbrunner, H., & Harer, J. (2007). Stability of persistence diagrams. *Discrete & Computational Geometry*, 37(1), 103–120. <https://doi.org/10.1007/s00454-006-1276-5>
- Crawley-Boevey, W. (2015). Decomposition of pointwise finite-dimensional persistence modules. *Journal of Algebra and Its Applications*, 14(5), 1550066. <https://doi.org/10.1142/S0219498815500668>

- de la Hoz Franco, E., Ariza-Colpas, P., Quero, J. M., & Espinilla, M. (2014). Sensor-based systems for early detection of alzheimer’s disease: A systematic review [Used as survey reference for fall/activity detection systems]. *Sensors*, *14*(10), 19244–19280. <https://doi.org/10.3390/s141019244>
- Edelsbrunner, H., & Harer, J. L. (2010). *Computational topology: An introduction*. American Mathematical Society.
- Edelsbrunner, H., Letscher, D., & Zomorodian, A. (2002). Topological persistence and simplification. *Discrete & Computational Geometry*, *28*(4), 511–533. <https://doi.org/10.1007/s00454-002-2885-2>
- Gidea, M., & Katz, Y. (2018). Topological data analysis of financial time series: Landscapes of crashes. *Physica A: Statistical Mechanics and its Applications*, *491*, 820–834. <https://doi.org/10.1016/j.physa.2017.09.028>
- GUDHI Editorial Board. (2015). *GUDHI user and reference manual*. GUDHI Editorial Board. <https://gudhi.inria.fr/doc/latest/>
- Hutter, F., Hoos, H. H., & Leyton-Brown, K. (2011). Sequential model-based optimization for general algorithm configuration. *Learning and Intelligent Optimization (LION 5)*, *6683*, 507–523. https://doi.org/10.1007/978-3-642-25566-3_40
- Igual, R., Medrano, C., & Plaza, I. (2013). Challenges, issues and trends in fall detection systems. *BioMedical Engineering OnLine*, *12*, 66. <https://doi.org/10.1186/1475-925X-12-66>
- Kangas, M., Konttila, A., Lindgren, P., Winblad, I., & Jämsä, T. (2008). Comparison of low-complexity fall detection algorithms for body attached accelerometers. *Gait & Posture*, *28*(2), 285–291. <https://doi.org/10.1016/j.gaitpost.2008.01.003>
- Ott, W., Tomforde, M., & Willis, P. N. (2015). One-sided shift spaces over infinite alphabets [Placeholder: replace with correct Ott et al. TDA/time-series paper if available]. *New York Journal of Mathematics*, *21*, 1–35.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, *12*, 2825–2830.
- Pereira, C. M. M., & de Mello, R. F. (2015). Persistent homology for time series and spatial data clustering and analysis. *Expert Systems with Applications*, *42*(15-16), 6026–6038. <https://doi.org/10.1016/j.eswa.2015.04.010>
- Platt, J. C. (1999). Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In A. J. Smola, P. Bartlett, B. Schölkopf, & D. Schuurmans (Eds.), *Advances in large margin classifiers* (pp. 61–74). MIT Press.
- Saleh, M., Abbas, M., & Le Bouquin Jeannès, R. (2021). FallAllD: An open dataset of human falls and activities of daily living for classical and deep learning applications. *IEEE Sensors Journal*, *21*(2), 1849–1858. <https://doi.org/10.1109/JSEN.2020.3018335>
- Sauer, T., Yorke, J. A., & Casdagli, M. (1991). Embedology. *Journal of Statistical Physics*, *65*(3-4), 579–616. <https://doi.org/10.1007/BF01053745>

- Schölkopf, B., Platt, J. C., Shawe-Taylor, J., Smola, A. J., & Williamson, R. C. (2001). Estimating the support of a high-dimensional distribution. *Neural Computation*, 13(7), 1443–1471. <https://doi.org/10.1162/089976601750264965>
- Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27(3), 379–423. <https://doi.org/10.1002/j.1538-7305.1948.tb01338.x>
- Sucerquia, A., López, J. D., & Vargas-Bonilla, J. F. (2017). SisFall: A fall and movement dataset. *Sensors*, 17(1), 198. <https://doi.org/10.3390/s17010198>
- Takens, F. (1981). Detecting strange attractors in turbulence. In D. Rand & L.-S. Young (Eds.), *Dynamical systems and turbulence, warwick 1980* (pp. 366–381, Vol. 898). Springer. <https://doi.org/10.1007/BFb0091924>
- Tinetti, M. E., Speechley, M., & Ginter, S. F. (1988). Risk factors for falls among elderly persons living in the community. *New England Journal of Medicine*, 319(26), 1701–1707. <https://doi.org/10.1056/NEJM198812293192604>
- Umeda, Y. (2017). Time series classification via topological data analysis. *Information and Media Technologies*, 12, 228–239. <https://doi.org/10.11185/imt.12.228>
- Vavoulas, G., Pediaditis, M., Chatzaki, C., Spanakis, E. G., & Tsiknakis, M. (2014). The MobiFall dataset: Fall detection and classification with a smartphone. *International Journal of Monitoring and Surveillance Technologies Research*, 2(1), 44–56. <https://doi.org/10.4018/ijmstr.2014010103>
- Wang, X., Ellul, J., & Azzopardi, G. (2020). Elderly fall detection systems: A literature survey. *Frontiers in Robotics and AI*, 7, 71. <https://doi.org/10.3389/frobt.2020.00071>
- Wild, D., Nayak, U. S. L., & Isaacs, B. (1981). How dangerous are falls in old people at home? *British Medical Journal*, 282(6260), 266–268. <https://doi.org/10.1136/bmj.282.6260.266>
- World Health Organization. (2021). Falls [Accessed: 2024]. <https://www.who.int/news-room/fact-sheets/detail/falls>
- World Health Organization. (2022). Ageing and health [Accessed: 2024]. <https://www.who.int/news-room/fact-sheets/detail/ageing-and-health>
- Xu, X., Yao, D., Li, J., & Zhang, D. (2019). Topological data analysis-based feature extraction for human activity recognition [Placeholder: verify DOI and page range before submission]. *IEEE Access*, 7, 1–10.

CURRICULUM VITAE

Name SURNAME: Kenan Evren BOYABATLI

E-Mail: kbybtli@gmail.com

EDUCATION:

- **B.Sc.:** 2023, Istanbul Medeniyet University, Faculty of Engineering and Natural Sciences, Department of Mathematics (GPA 3.25, ranked second in the department).

PROFESSIONAL EXPERIENCE AND AWARDS:

- Junior Data Scientist, Mapin Data (December 2023 – August 2024); developed and implemented the TDR feature-rating algorithm on a national education dataset and built Python-based process-automation tools.
- Researcher, TÜBİTAK Project 119F279, “Investigation of the Effects of Surface Curvature on Interface Heat Transfer for Thermally Optimized Applications” (April – September 2022); implemented and optimised the numerical and machine-learning algorithms used in the project.
- Ranked second in the Department of Mathematics, Istanbul Medeniyet University (2023).

PUBLICATIONS, PRESENTATIONS AND PATENTS ON THE THESIS:

- —

OTHER PUBLICATIONS, PRESENTATIONS AND PATENTS:

- Topological data analysis study on feature rating (TDR algorithm), applied to a Ministry of National Education (MEB) school dataset, 2024.